SPARQL UniProt.RDF

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Tuesday, November 27, 2012

Everyone has had some introduction slash knowledge of RDF.

A few notes before we begin

- SPARQL 1
  - Some what useful
  - Standardized in 2008
- SPARQL 1.1
  - Very useful
  - Currently in recommended standard

- Still finding incompatibilities
- Or not yet implemented features

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Raise your hand if you have questions



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Tutorial plan

- Set up Topbraid Composer
  - Skipped in talk
  - On VM
- Gather data from uniprot website
  - Already there.
- Learn sparql

You do not need Topbraid Composer to use UniProt RDF data or do sparql queries.

You can use beta.sparql.uniprot.org as well.

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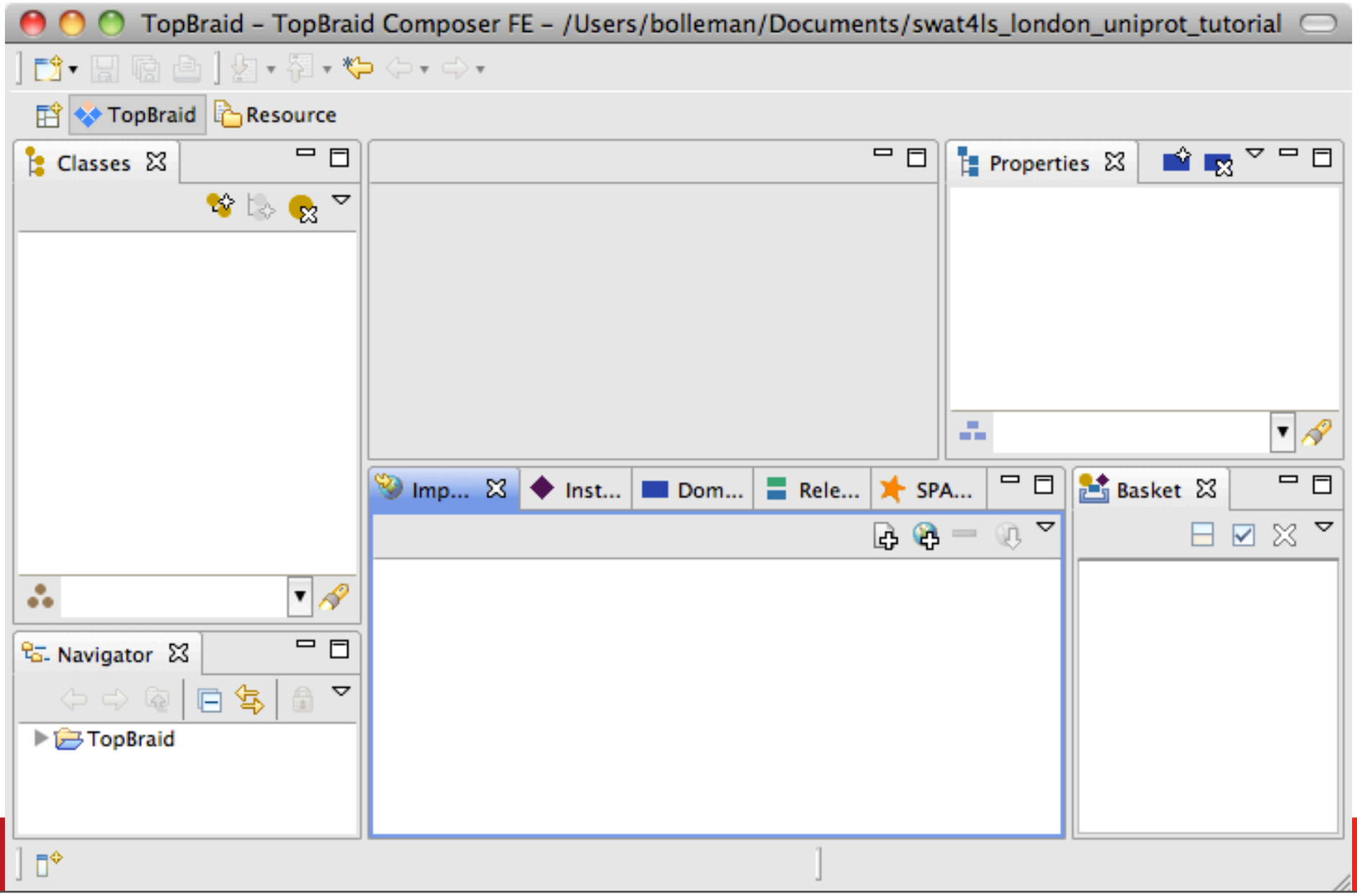
You should have used Topbraid composer in this mornings tutorial.
If not have a look at next few slides.

Download and install Topbraid composer

- Requirements
  - Sun/Oracle JVM
- Go to
  - http://www.topquadrant.com/products/TB_download.html
  - Register
  - Select any edition, free is ok for today

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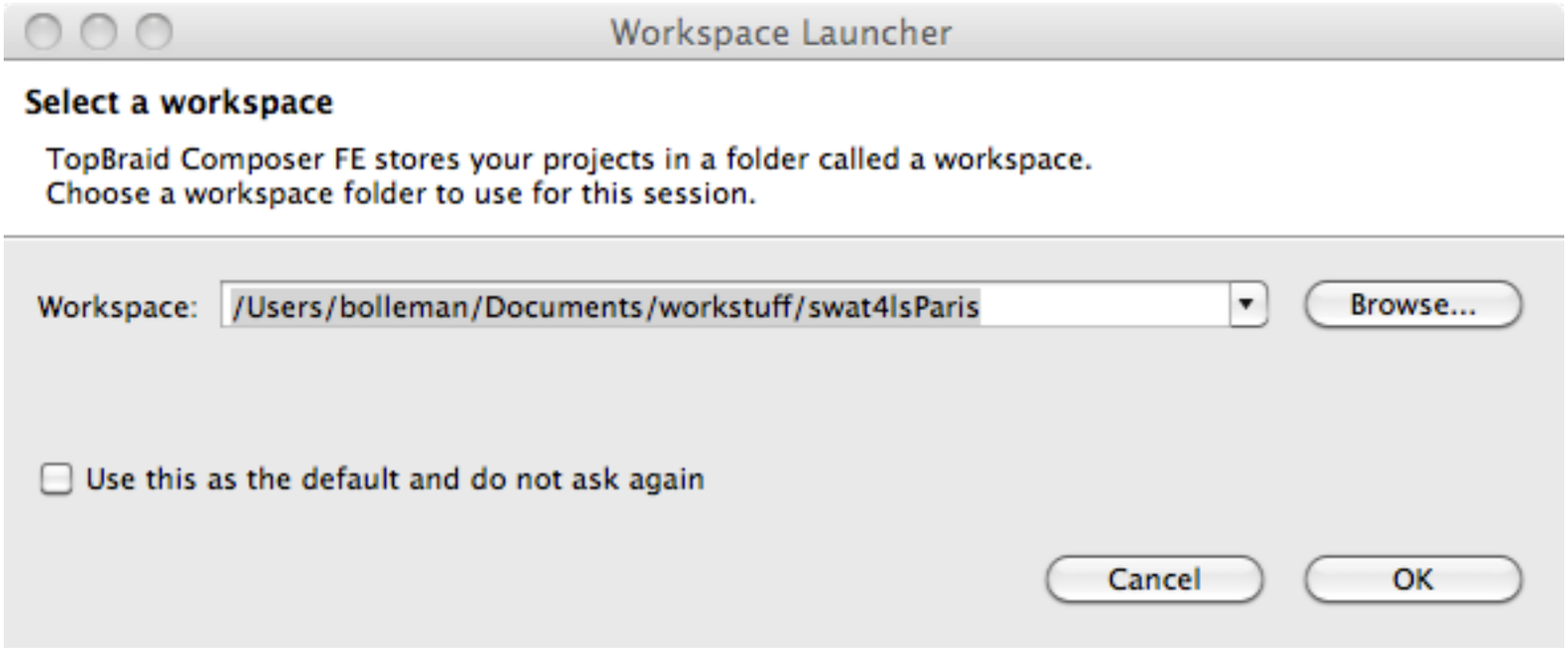
Start Topbraid



If you never used topbraid before you will have an empty workspace.

Setting up a workspace for this tutorial

- http://www.topquadrant.com/products/TB_download.html

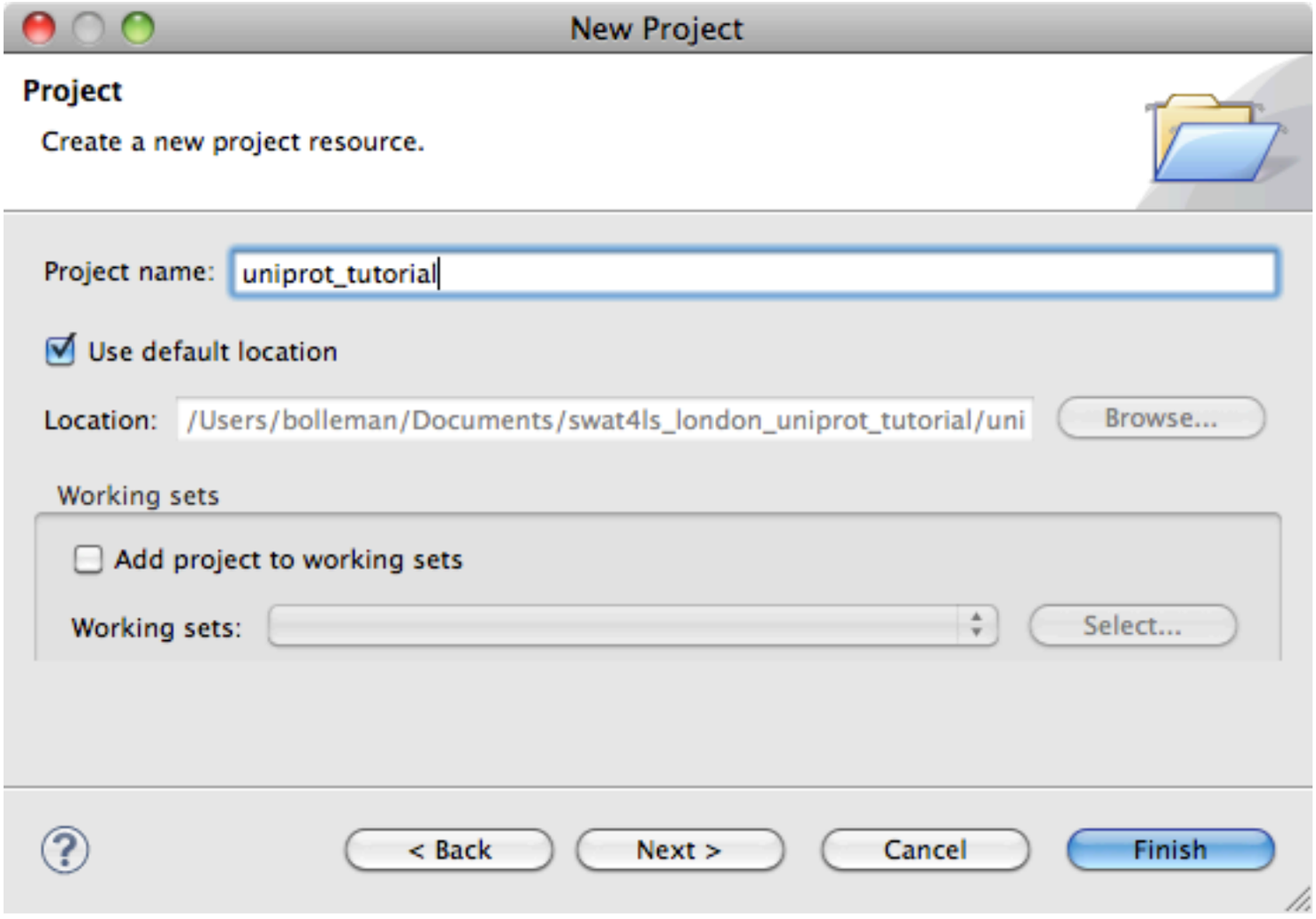


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For today please create a new empty workspace that does not influence your previous work

New project

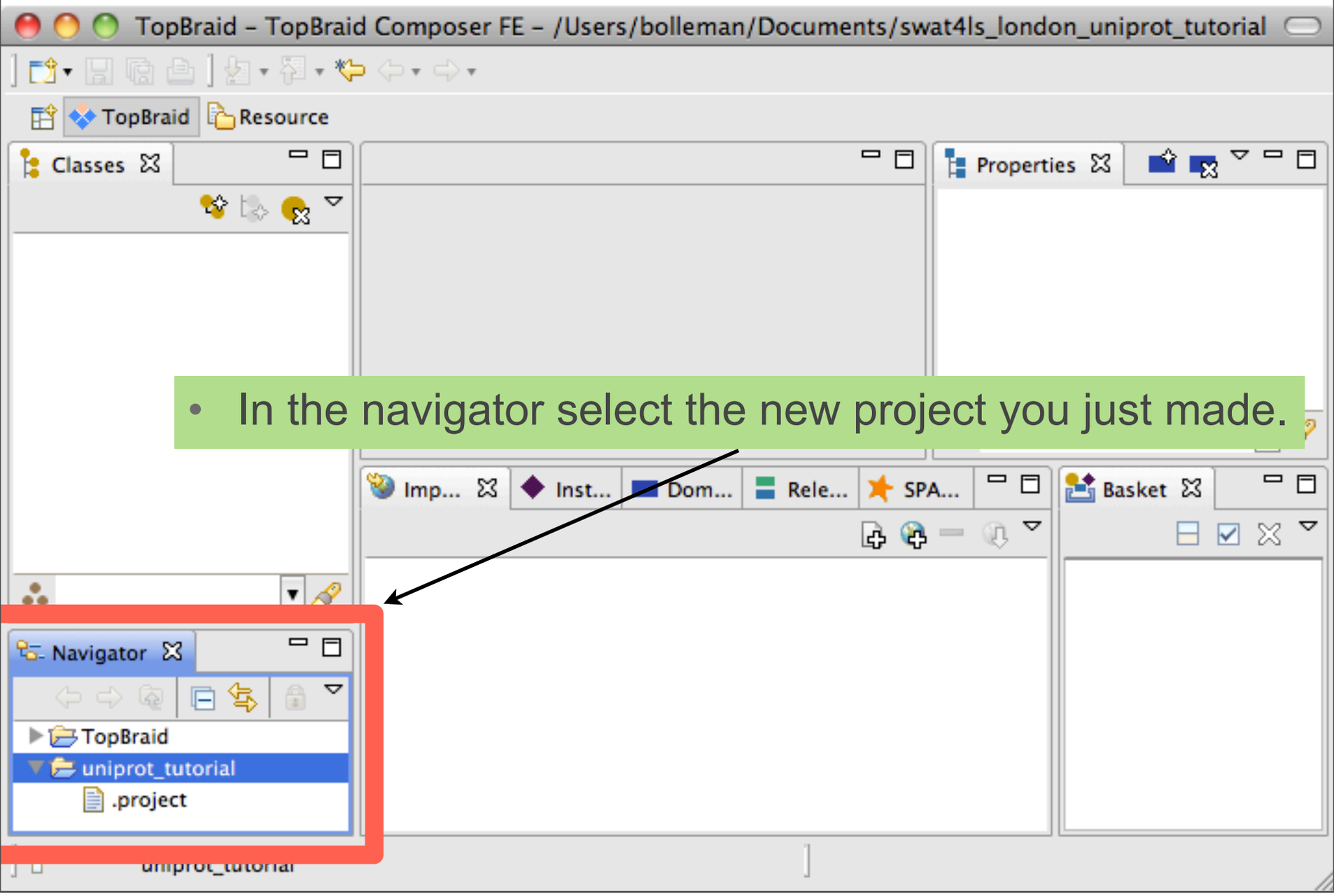
- File > New Project > General



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Give your project a recognizable name

Gather data from uniprot.org website

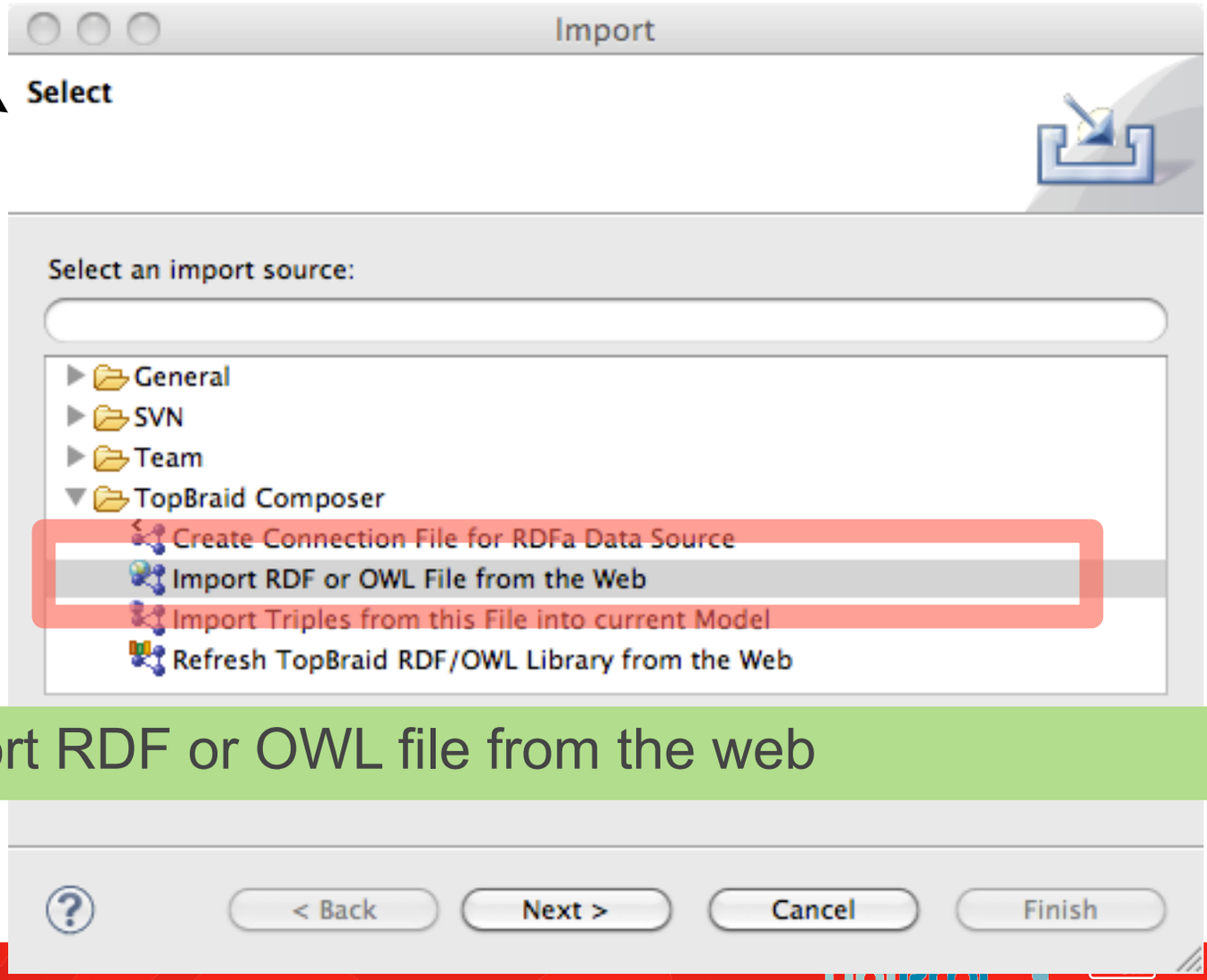


- In the navigator select the new project you just made.

The .project file contains the project details do not delete it!

Gather data from uniprot.org website

Right click on your new project.
Select “Import” in the drop down menu



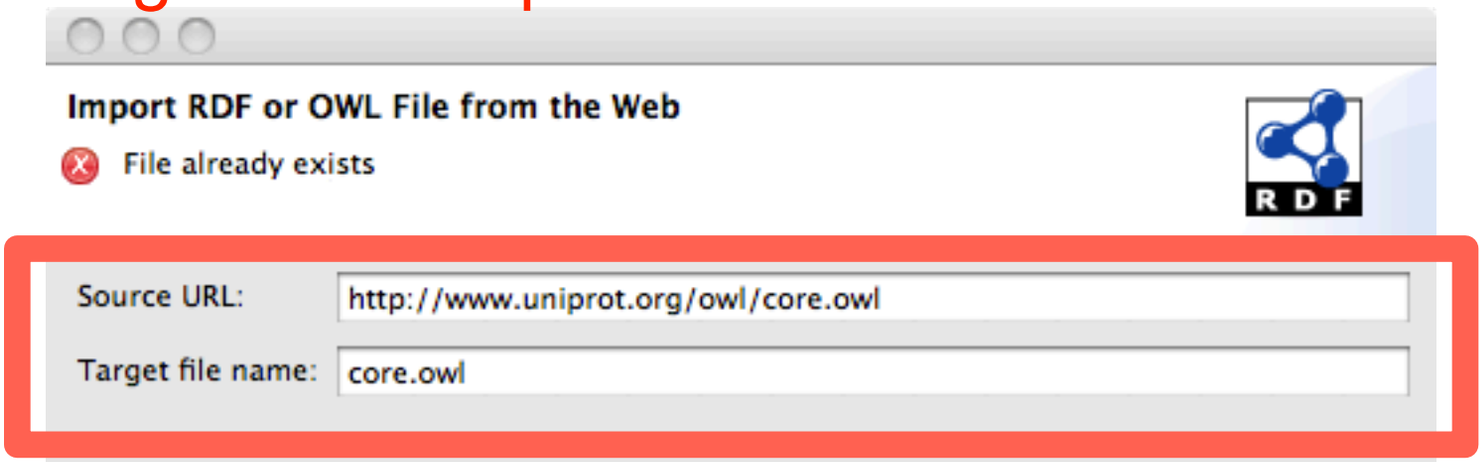
- Import RDF or OWL file from the web

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In this case we are going to use an uniprot entry for our examples.

Using the same process download core.owl

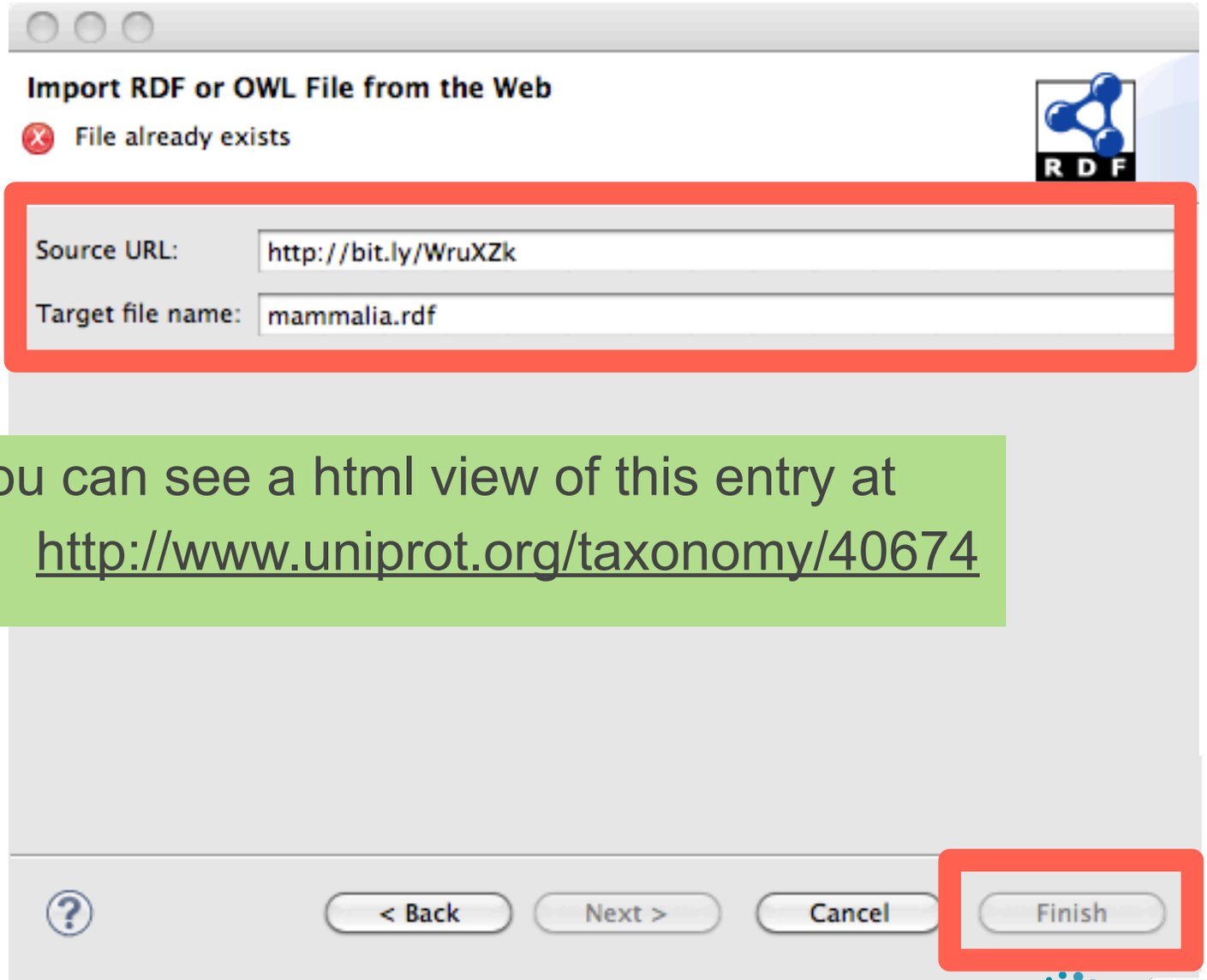


You can see a html view of this schema ontology at

http://www.uniprot.org/core/

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Gather data from uniprot.org website



You can see a html view of this entry at
http://www.uniprot.org/taxonomy/40674

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UniProt

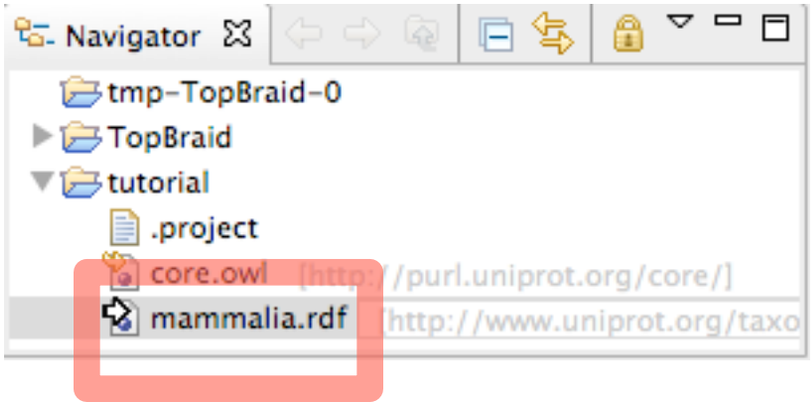
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Fill in the source and target url.

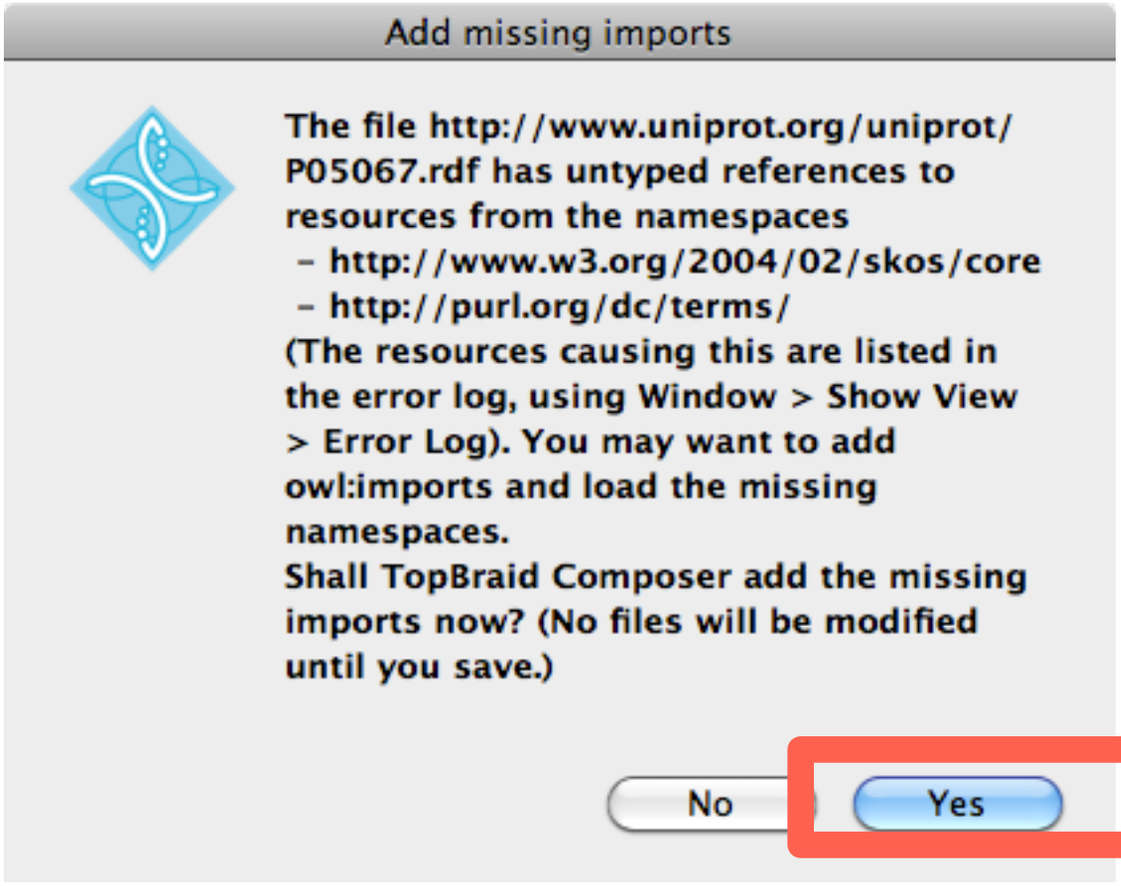
Gather data from uniprot.org website

- Open the mammalia.rdf file by double clicking



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You get a very helpfull dialog.
Hit yes



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This auto imports ontologies used by uniprot that are not inside the core.owl file.
And their imports as well.

Its SPARQLy mammal time !!



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SPECIES Tamias striatus (Eastern chipmunk)

UniProtKB (60) | Branch (1) | Taxonomy help
rdf/xml

Mnemonic	TAMST
Taxon identifier	45474
Scientific name	Tamias striatus
Common name	Eastern chipmunk
Synonym	-
Rank	SPECIES
Lineage	› cellular organisms › Eukaryota › Opisthokonta › Metazoa › Eumetazoa › Bilateria › Coelomata › Deuterostomia › Chordata › Craniata › Vertebrata › Gnathostomata

Taxonomy navigation

- ⇧ › Tamias
- ⇩ › Tamias striatus fisheri

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imagess3.enature.com



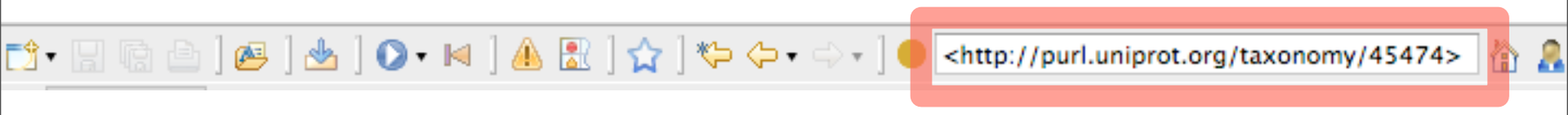
ohioline.osu.edu



www.biolib.cz



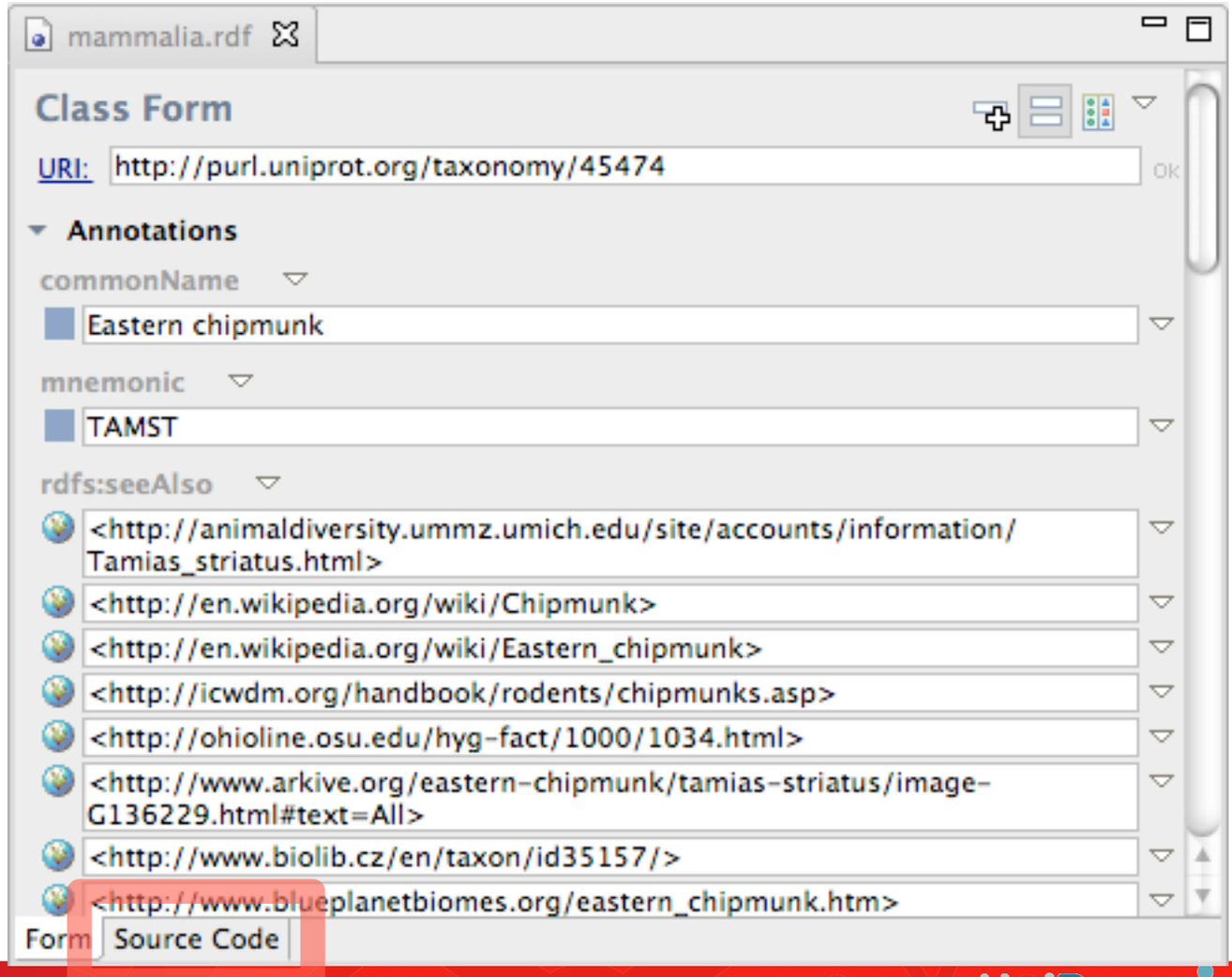
Lets navigate to it in TopBraid



- Type the uri as is with the angle brackets

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Investigate the taxon record

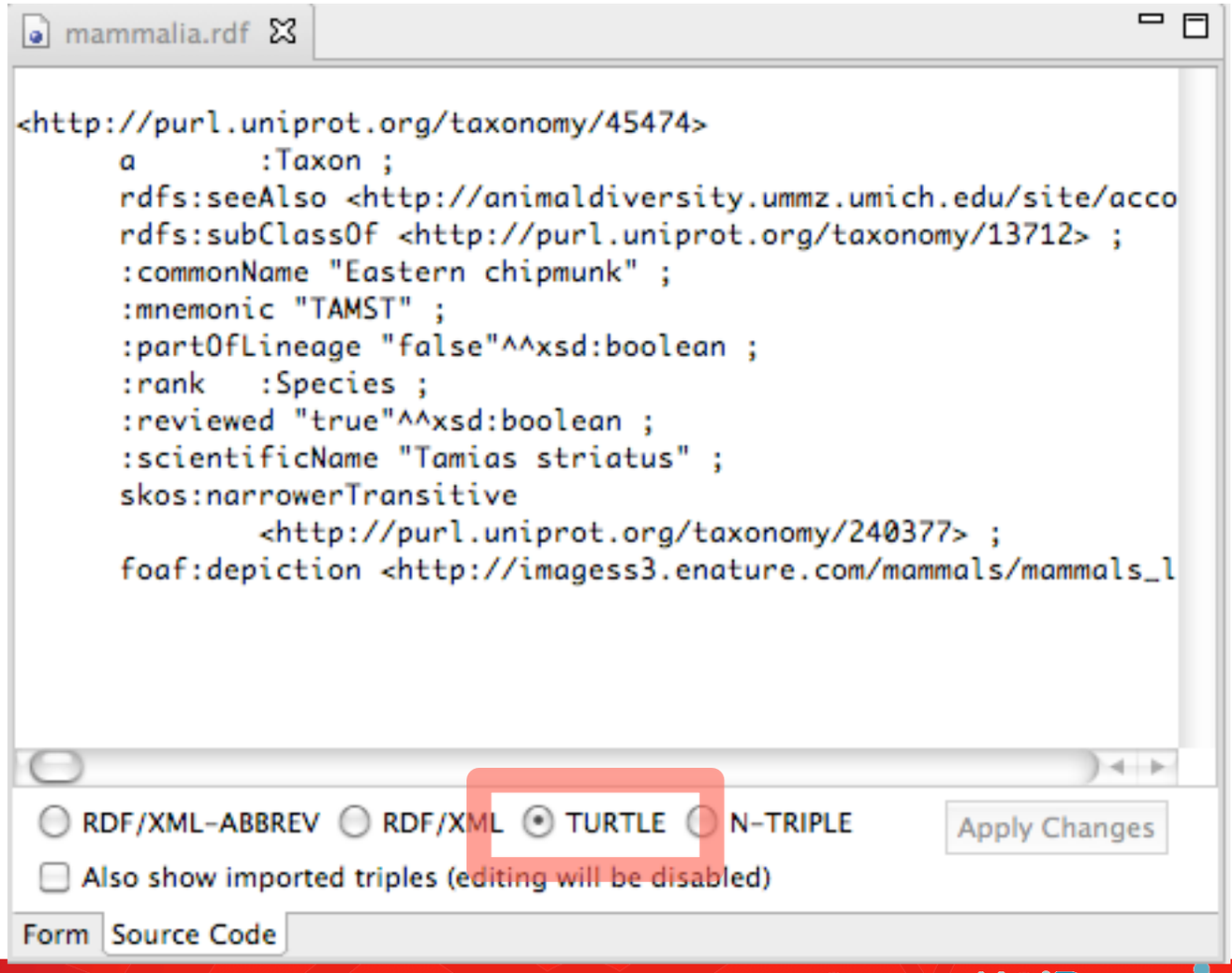


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The “Eastern Chipmunk” in turtle



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Turtle is the RDF serialization aligned with SPARQL

- Shorthand to avoid typing so much
  - . ‘dot’ is end statement
  - ; ‘semi-colon’ repeat subject
  - , ‘comma’ is repeat subject and predicate
- prefix
  - before ‘:’ is abbreviation of uri



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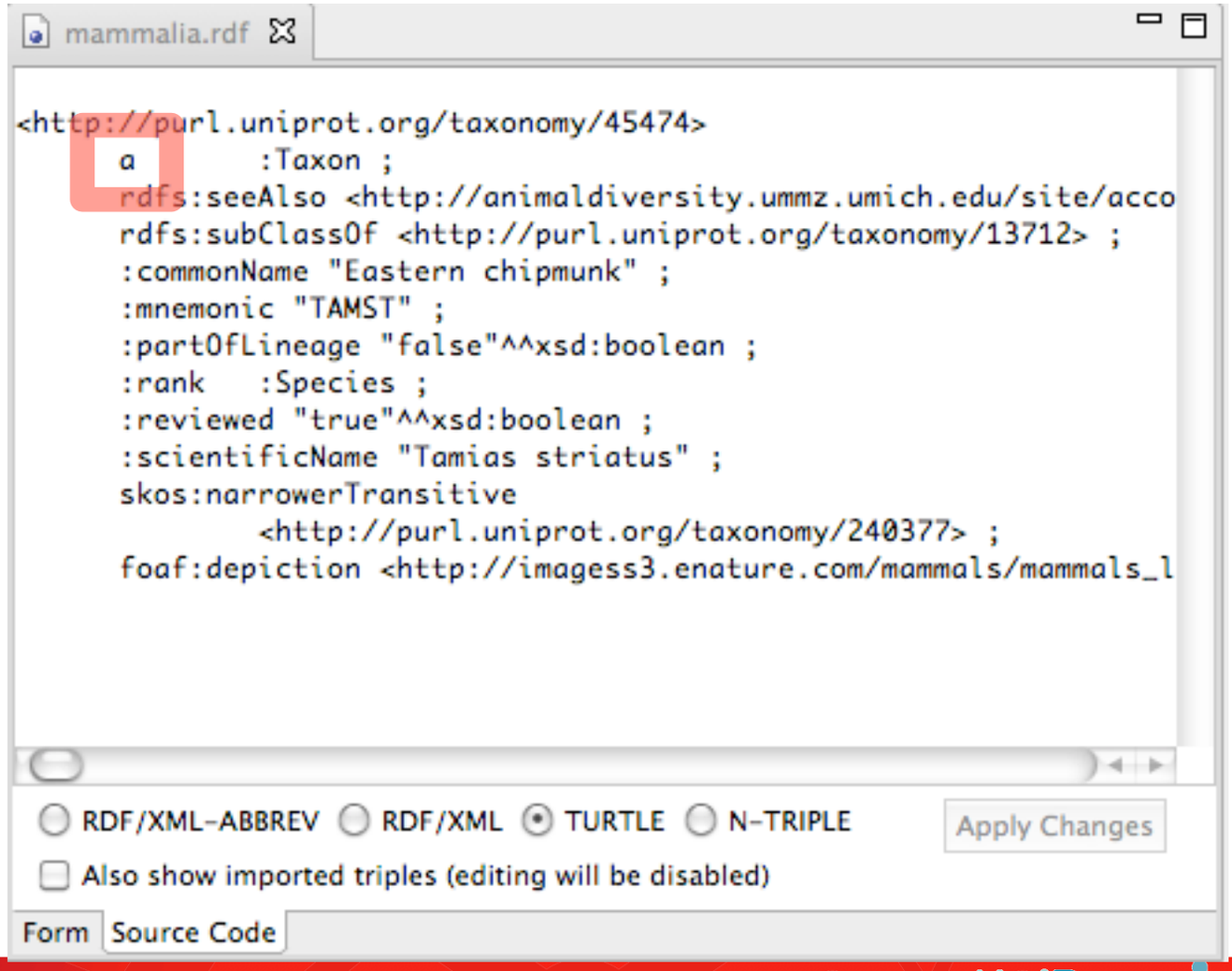
Why don’t these queries work on the web?

- PREFIX
  - Topbraid composer uses the prefixes defined in the files “overview” tab.
  - On the web you often have to add these.

```
PREFIX :<http://purl.uniprot.org/core/>
SELECT ?x
FROM <http://purl.uniprot.org/taxonomy/>
WHERE {?x a :Taxon}
```

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a = rdf:type = <http://www.w3.org/1999/02/22-rdf-syntax-ns#type>

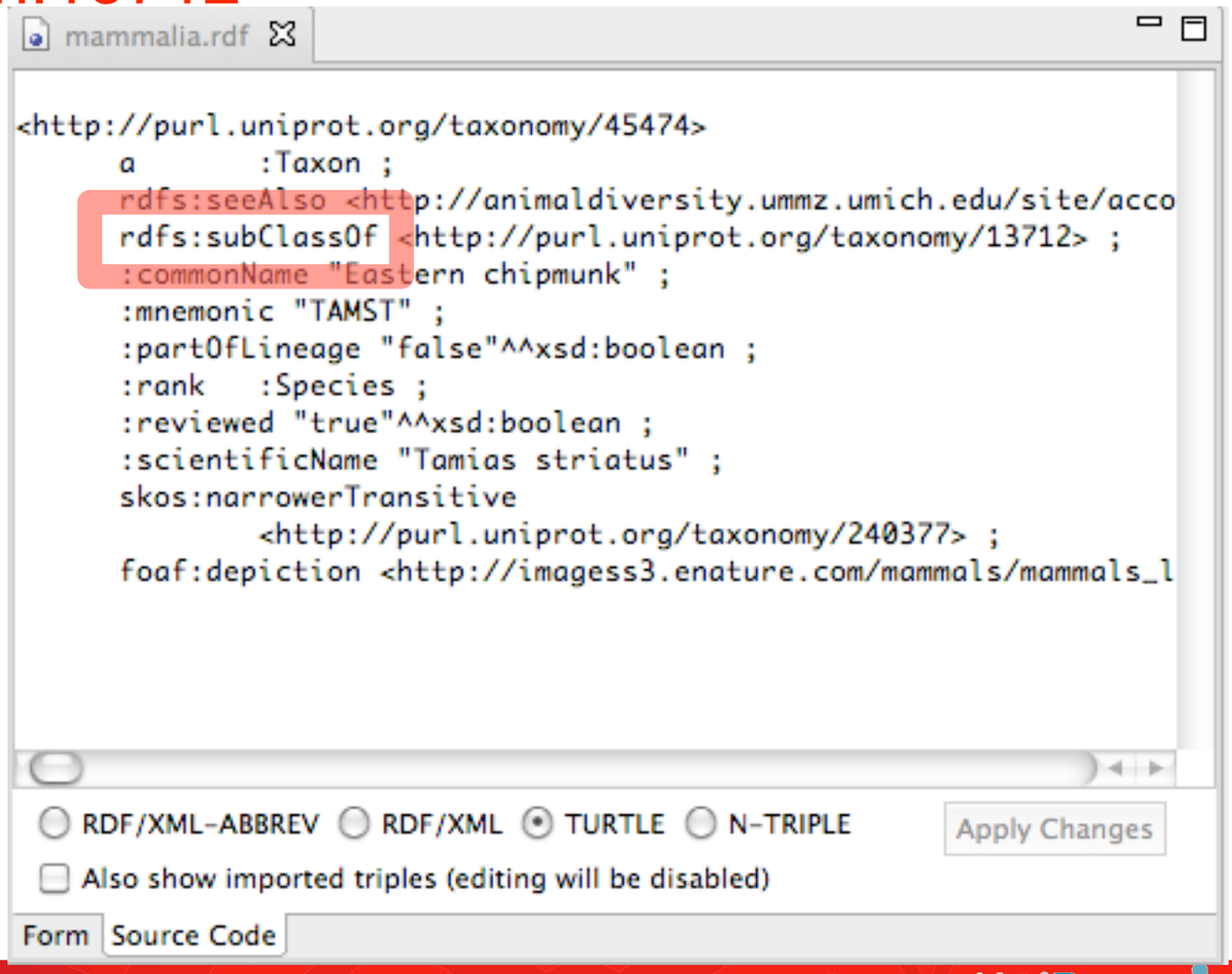


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Useful short hand

rdfs:subClassOf
taxon:45474 is a more specific classification than taxon:13712



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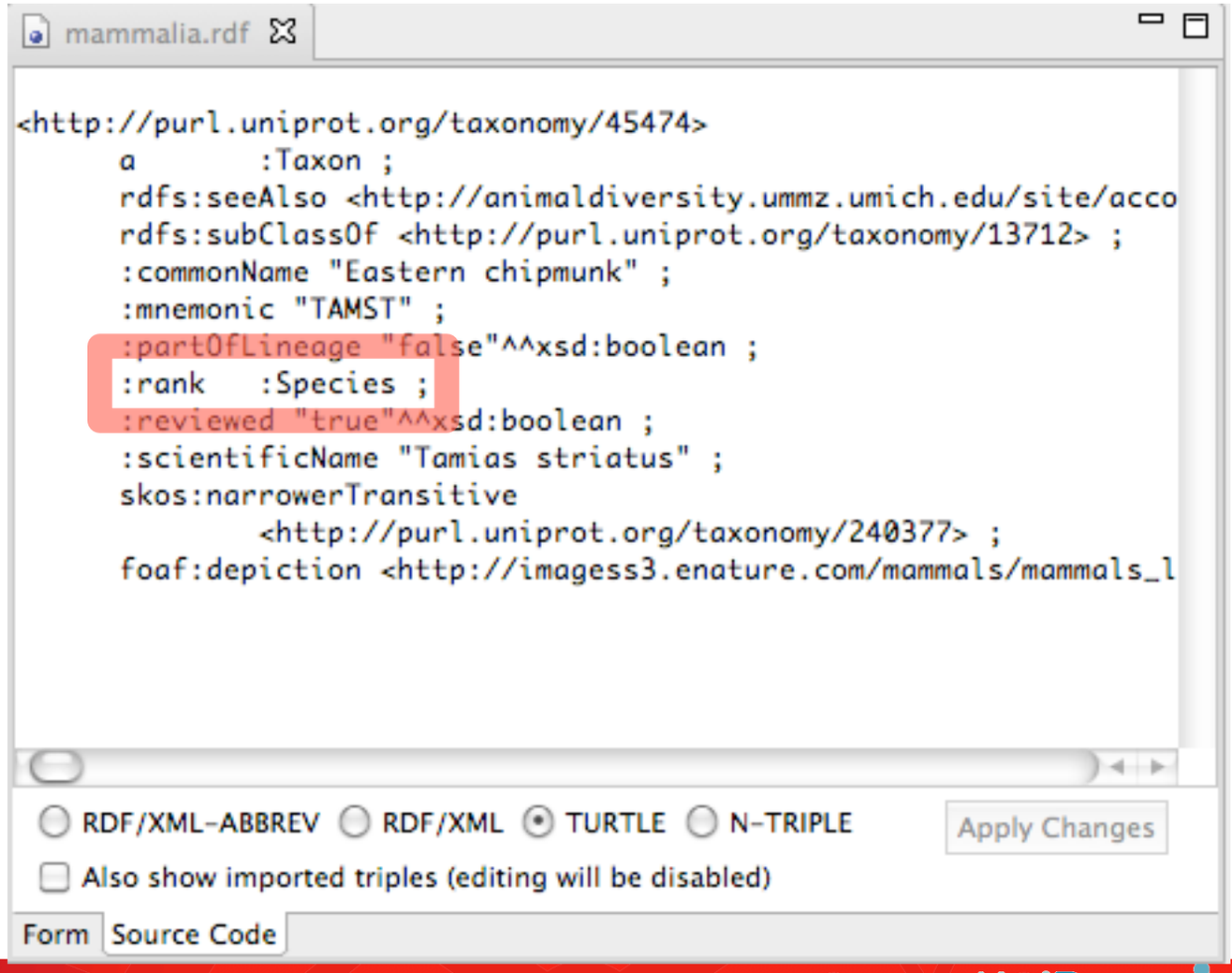
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Useful short hand

rank => “The level, for nomenclatural purposes, of a taxon in a taxonomic hierarchy”



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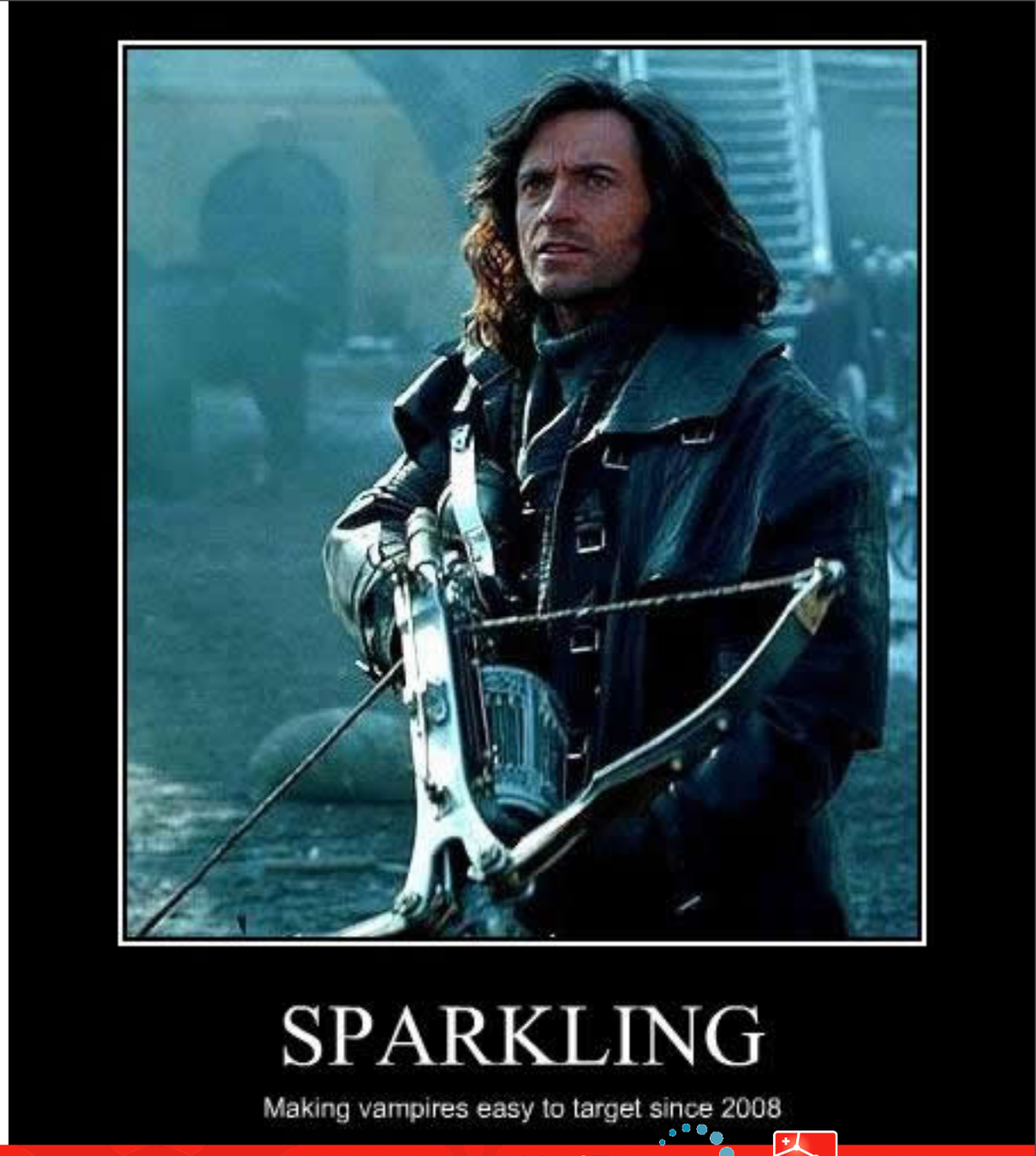
(e.g. all families are for nomenclatural purposes at the same rank, which lies between superfamily and subfamily)

Why learn SPARQL

- Standardized formal query language
  - implementation independent
    - SPARQL ➔ SQL (via R2ML)
    - SPARQL ➔ webservice (via SADI)
    - SPARQL ➔ LDAP (e.g. SquirrelRDF)
    - SPARQL ➔ RDF (triplestore e.g. OWLIM-se)
    - SPARQL ➔ HADOOP/HIVE (e.g. SHARD)
  - How you query independent of how you store!

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Apparently it helps kill vampires !!!



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UniProt

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Lets learn SPARQL

- Queries over RDF data.
  - Four basic types
    - SELECT
      - Returns “tab delimited” results
    - CONSTRUCT
      - Makes new triples
    - DESCRIBE
      - Returns all triples mentioning a resource
    - ASK
      - Return true if anything matches

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In this example session I will only show SELECT and CONSTRUCT

SPARQL:queries triple pattern

```
taxon:9606 rdf:type core:Taxon .
```

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SPARQL:queries triple pattern

```
?anyTaxon rdf:type core:Taxon .
```

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Replace the homo sapiens node with a variable (starts with a ?)

SPARQL:queries triple pattern

```
SELECT ?anyTaxon
WHERE {
   ?anyTaxon rdf:type core:Taxon .
}
```

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Wrap it in a sparql select .

SPARQL:queries triple pattern

taxon:9606 rdf:type core:Taxon .
taxon:9606 core:reviewed “true” .

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Basic triple uniprot record P05067 is rdf:type core:Protein

SPARQL:queries triple pattern

```
?anyTaxon rdf:type core:Taxon .
?anyTaxon core:reviewed “true” .
```

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The two patterns are joined on the shared variable ?anyTaxon.

SPARQL:queries triple pattern

```
SELECT ?anyTaxon
WHERE {
   ?anyTaxon rdf:type core:Taxon .
   ?anyTaxon core:reviewed “true” .
}
```

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Selects all “Swiss-Prot” proteins.

SPARQL:queries triple pattern

```
SELECT ?anyTaxon
WHERE {
   ?anyTaxon rdf:type core:Taxon .
   ?anyTaxin core:reviewed “true” .
}
```

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Not a join due to a typo. Be careful

SPARQL:queries triple pattern

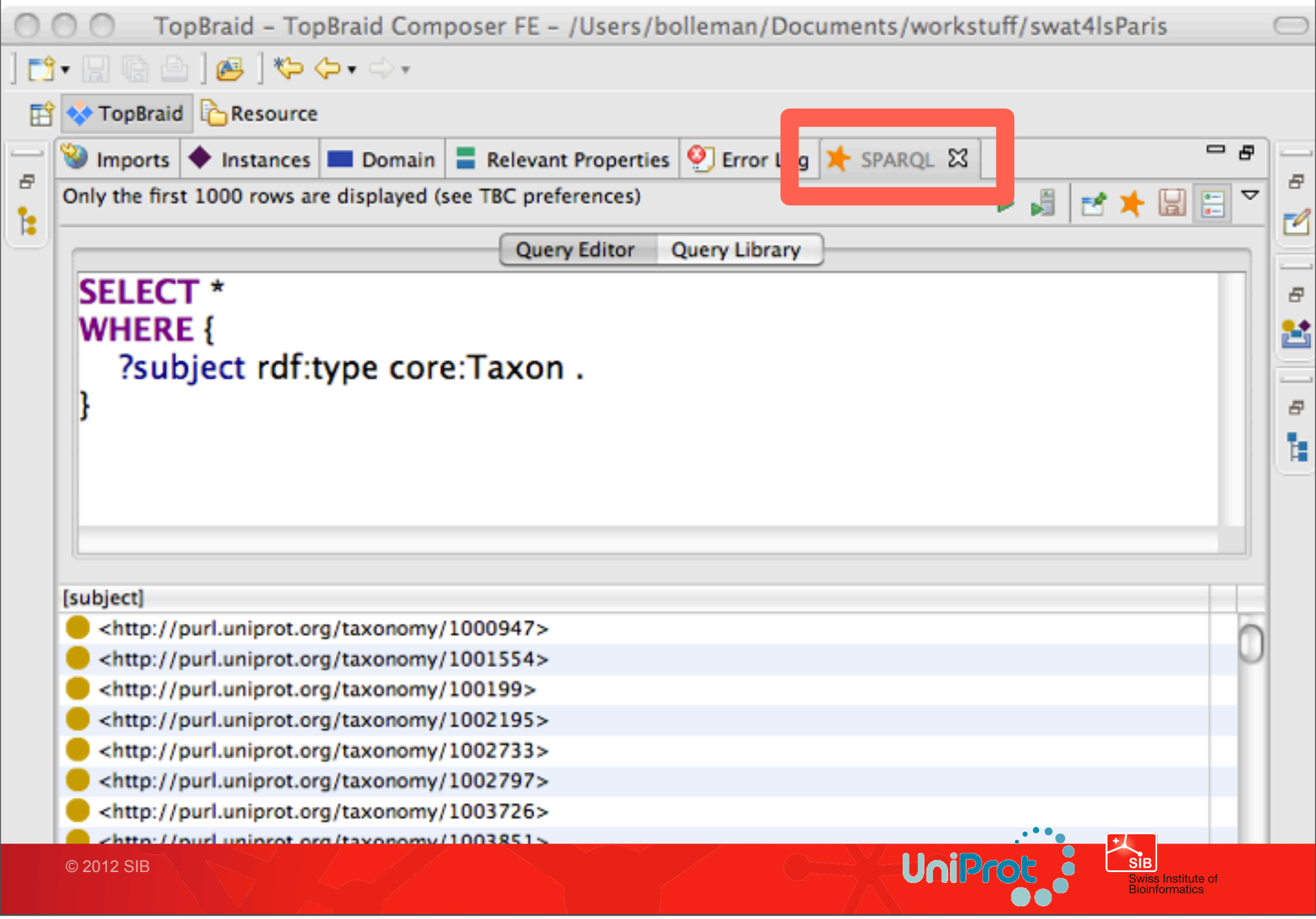
```
SELECT ?anyTaxon
WHERE {
   ?anyTaxon rdf:type core:Taxon .
   $anyTaxon core:reviewed “true” .
}
```

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This is a join $ and ? are interchangeable. I prefer using ?

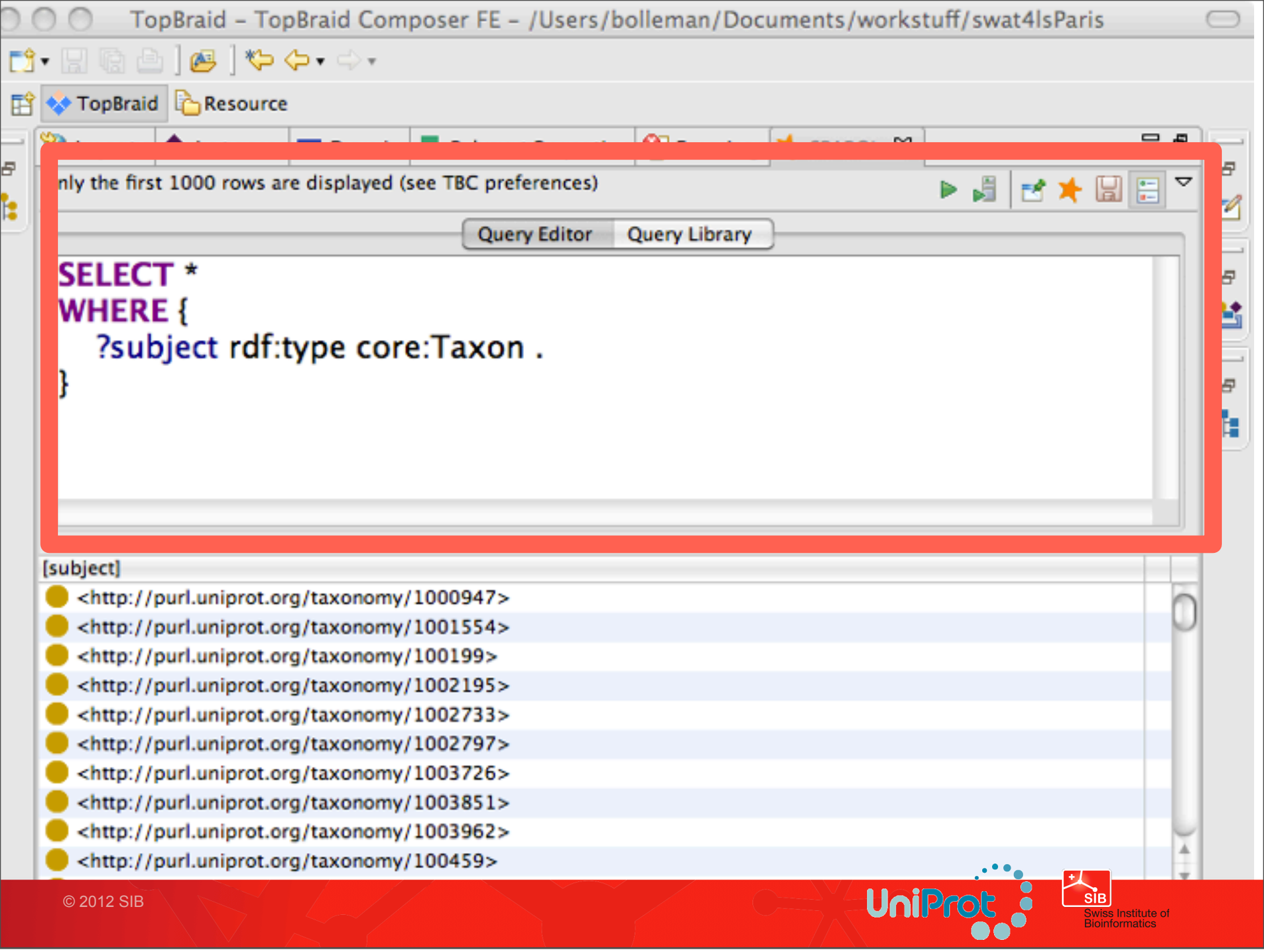
Lets learn SPARQL



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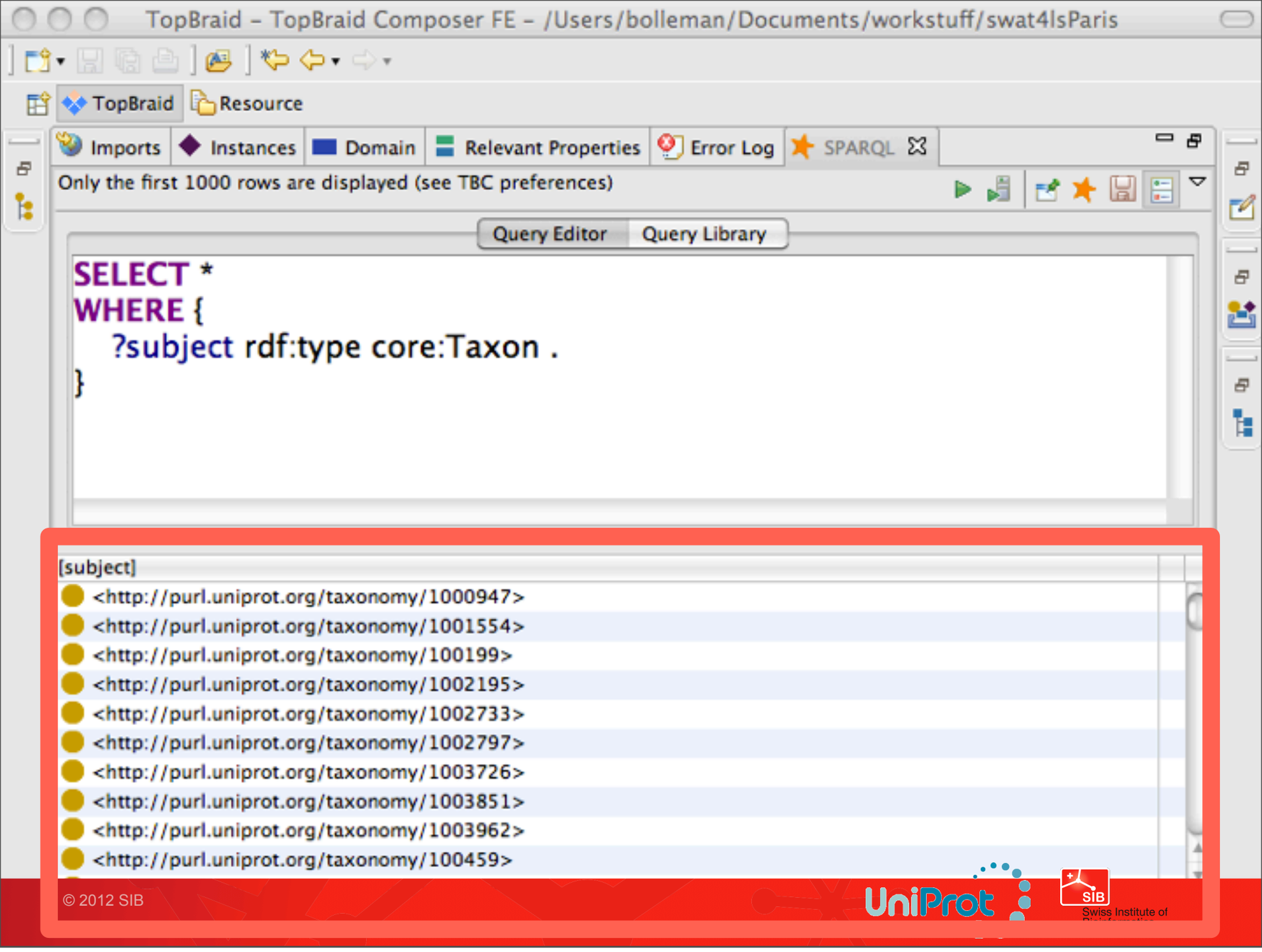
UniProt

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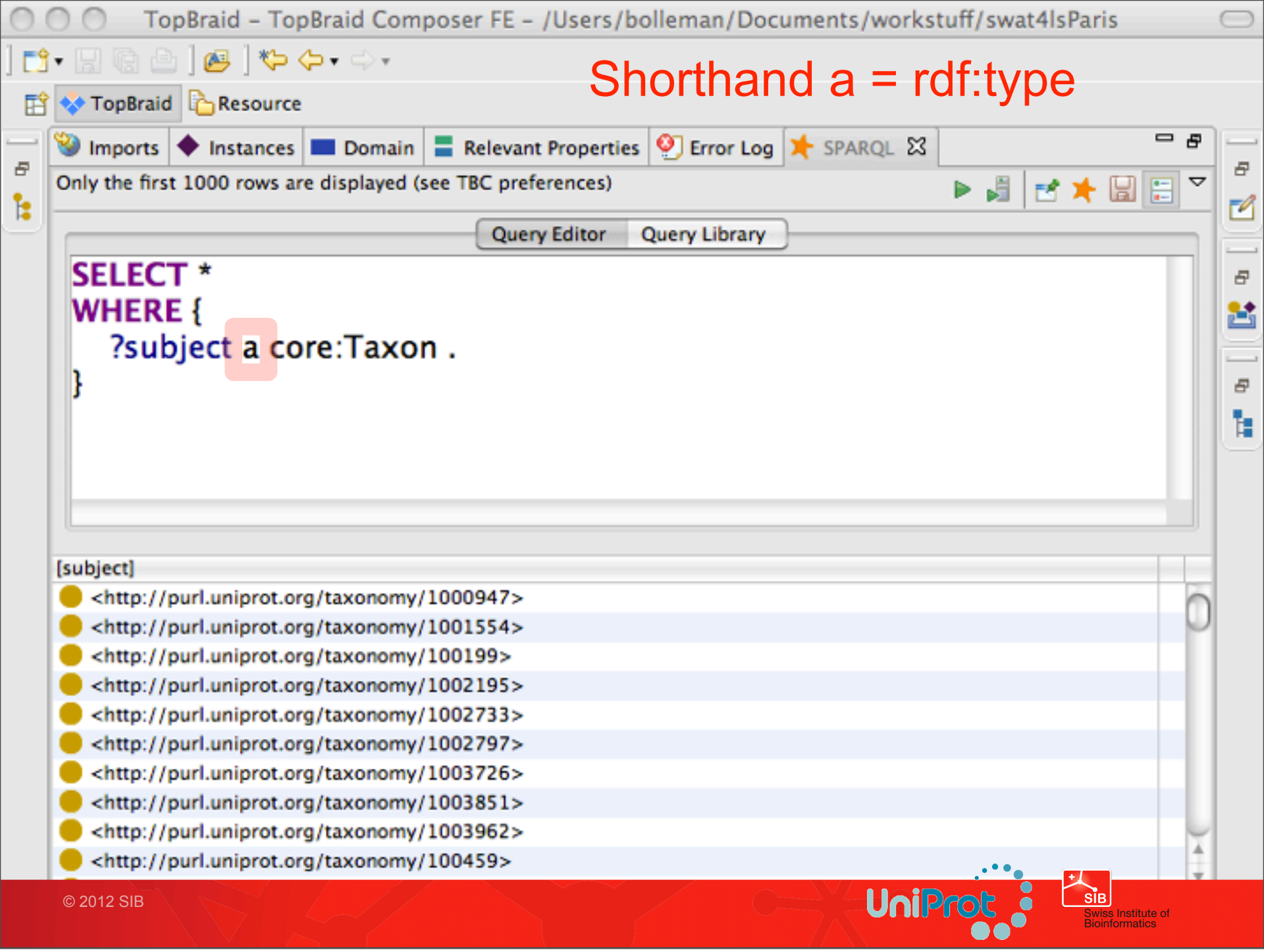
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This is where you type your query.



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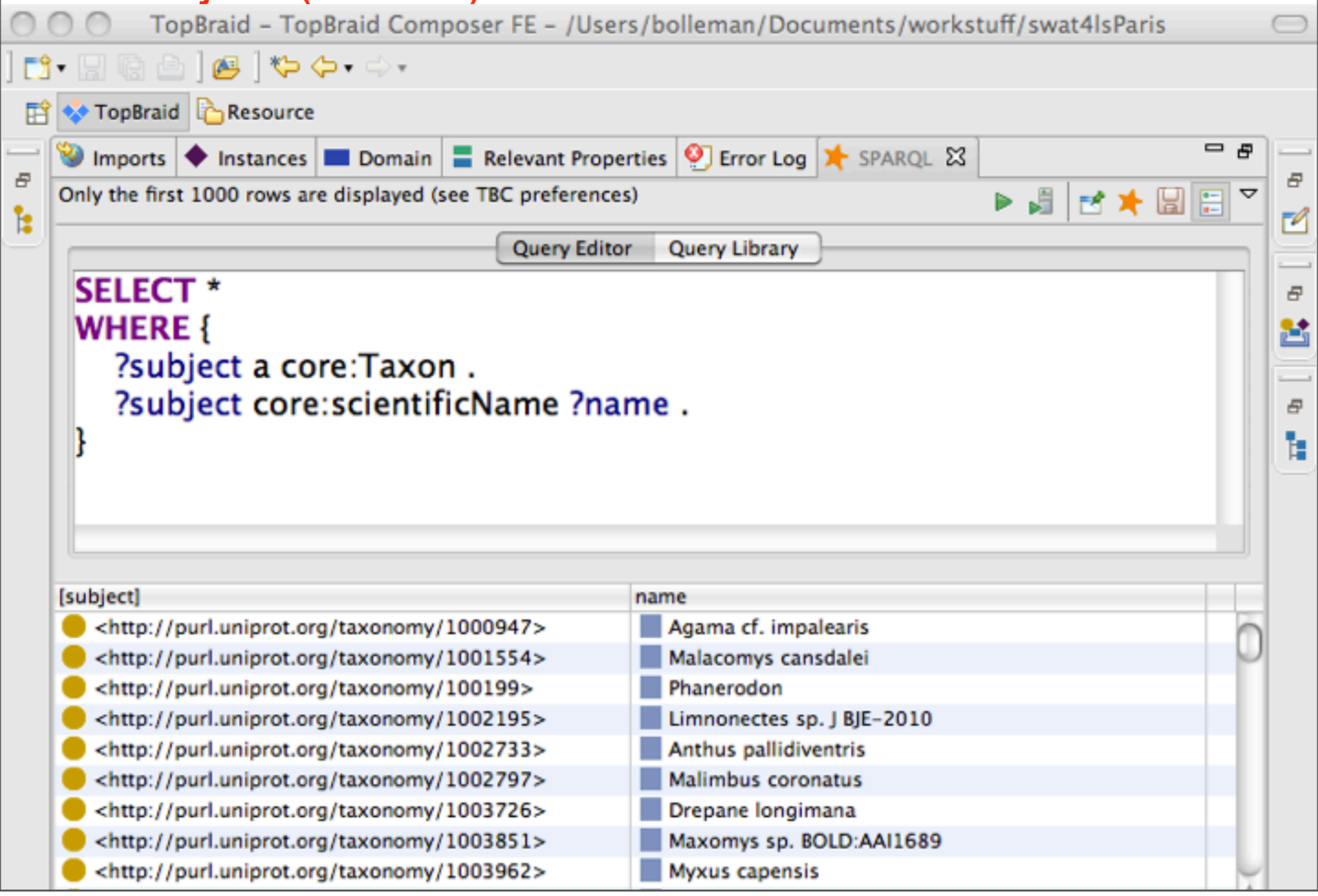
This is the query results.



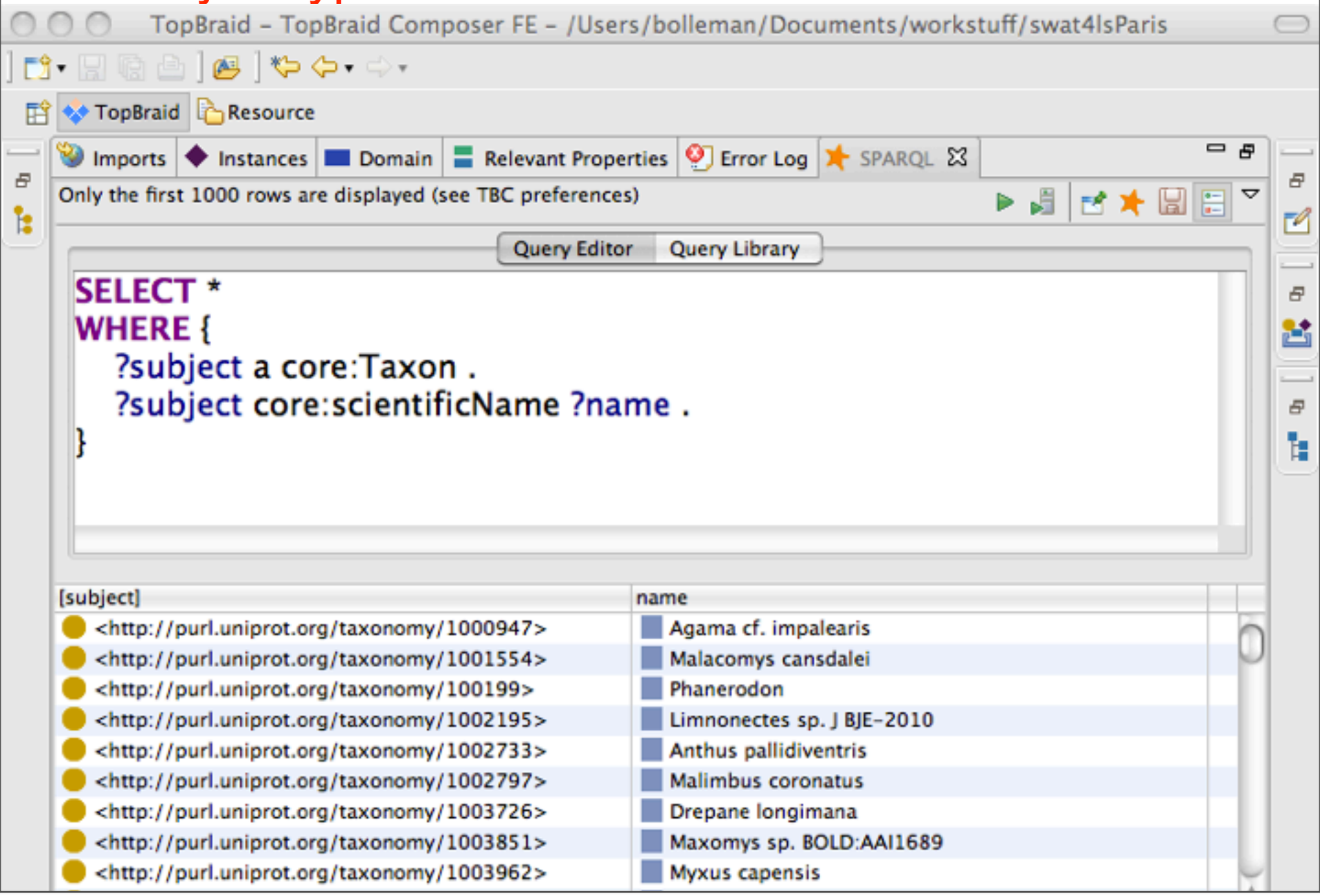
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This is the query results.

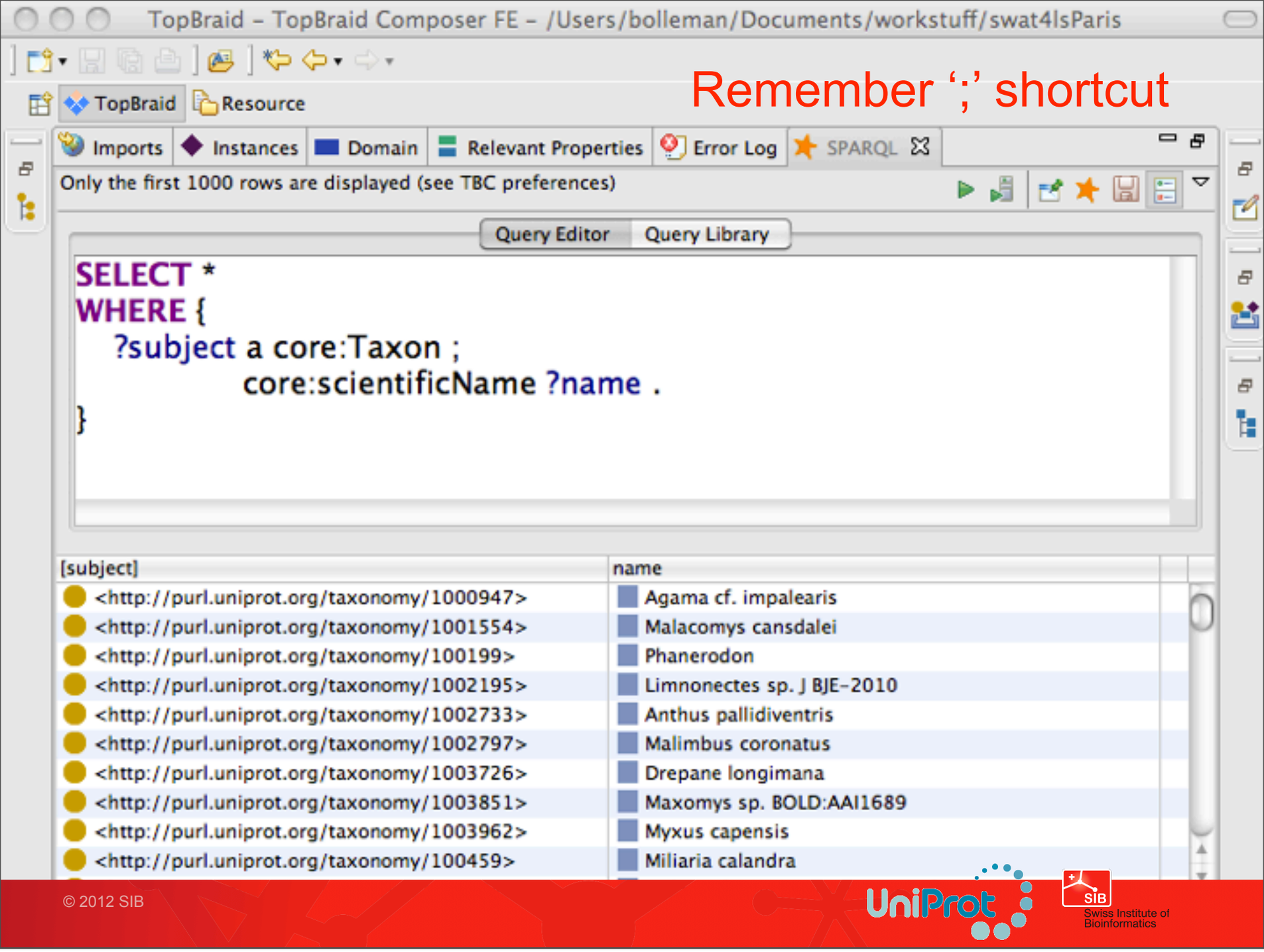
AND join (default)



Now you type

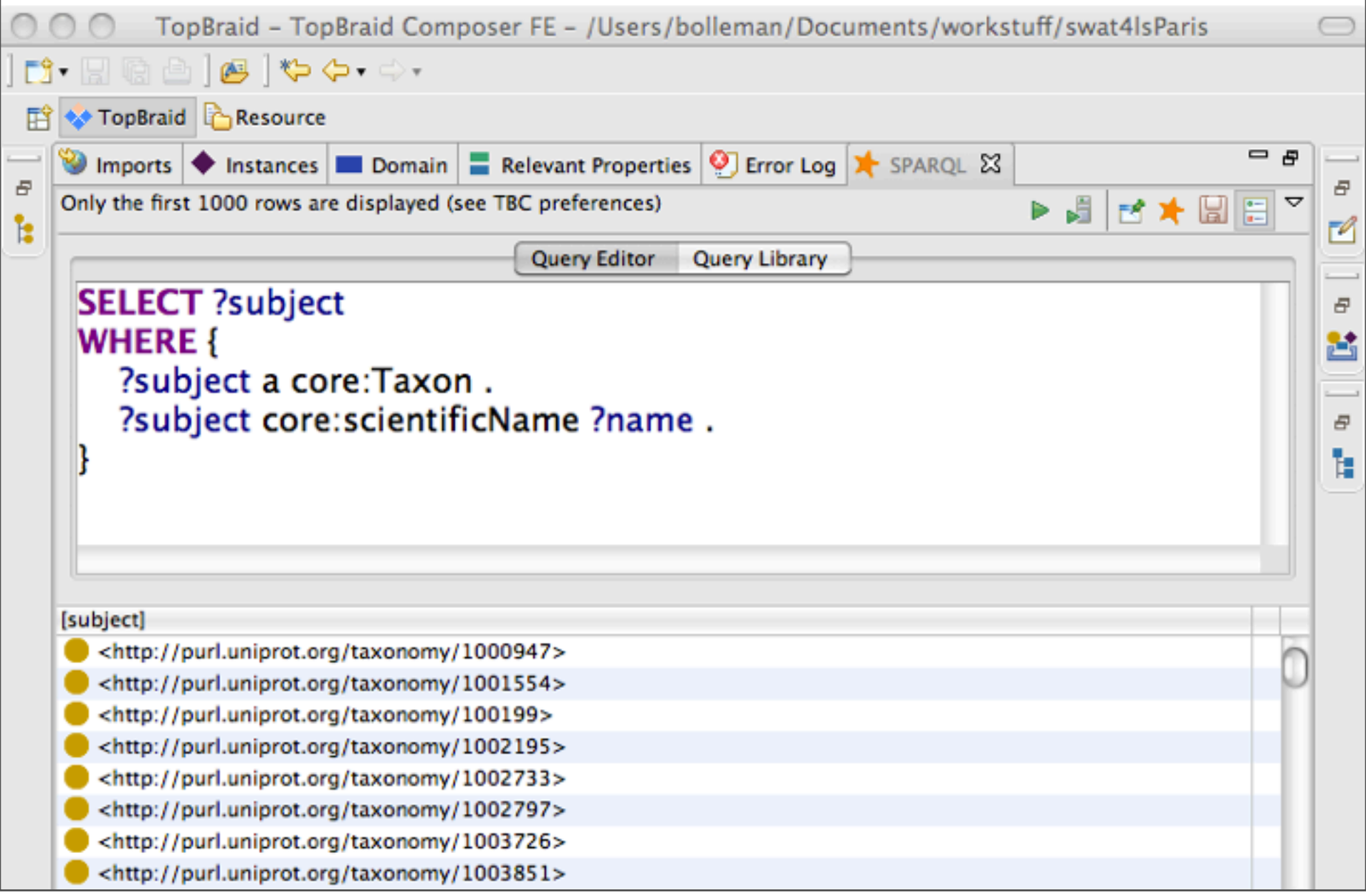



```
SELECT *
WHERE {
   ?subject a core:Taxon .
   ?subject core:scientificName ?name .
}
```




```
SELECT *
WHERE {
   ?subject a core:Taxon ;
            core:scientificName ?name .
}
```

Two variables one output column



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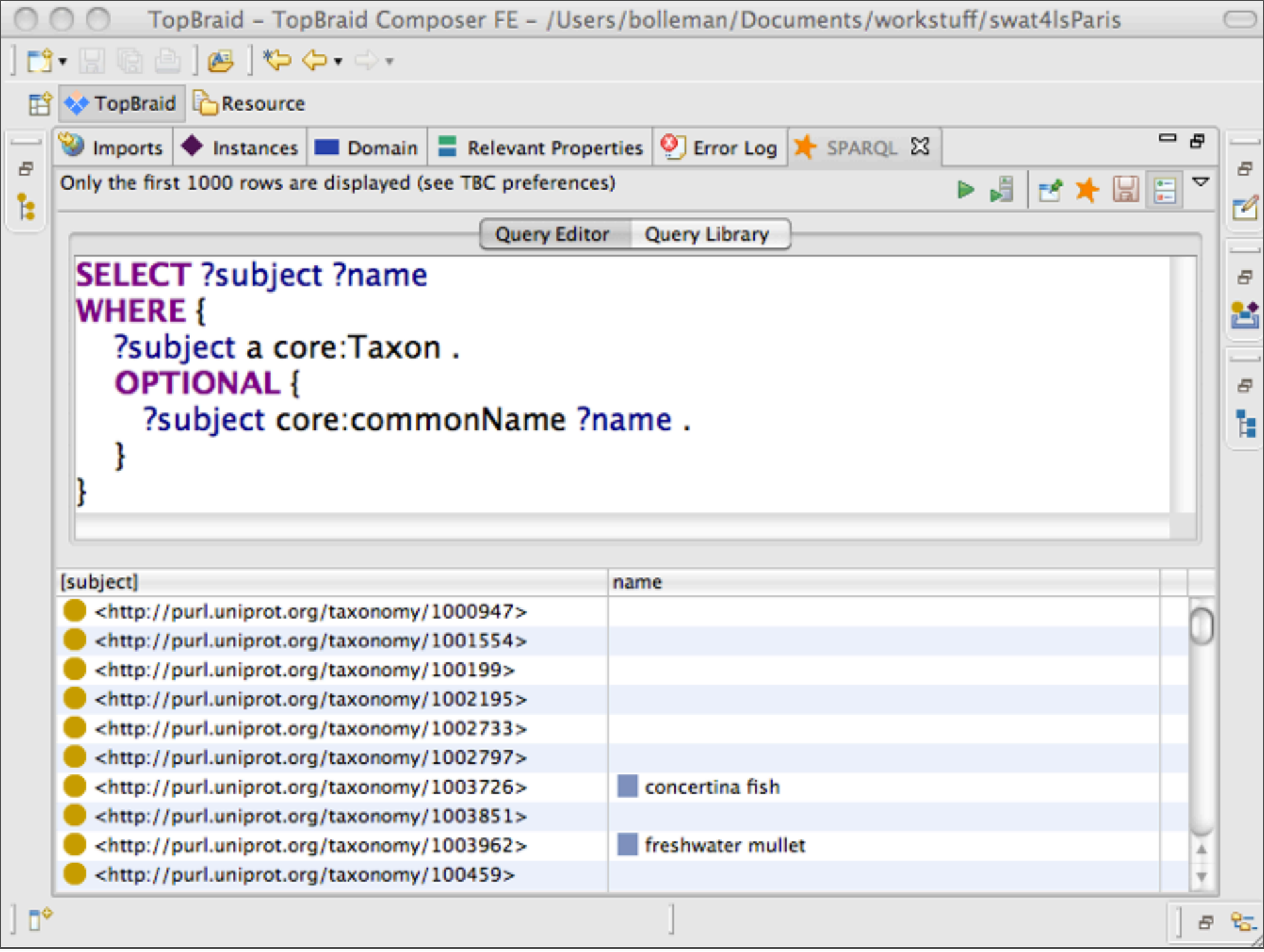
Upto now we have been using * for select all variables.
If you do not need all variables in the output select specific ones.

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name
}
```

Optional

- When values may be missing
  - yet interesting when they are there
- Use as sub query
- bound values from outside stay bound inside
  - ?x ?y?z . OPTIONAL {?x ?b ?c}
    - ?x same variable = same thing

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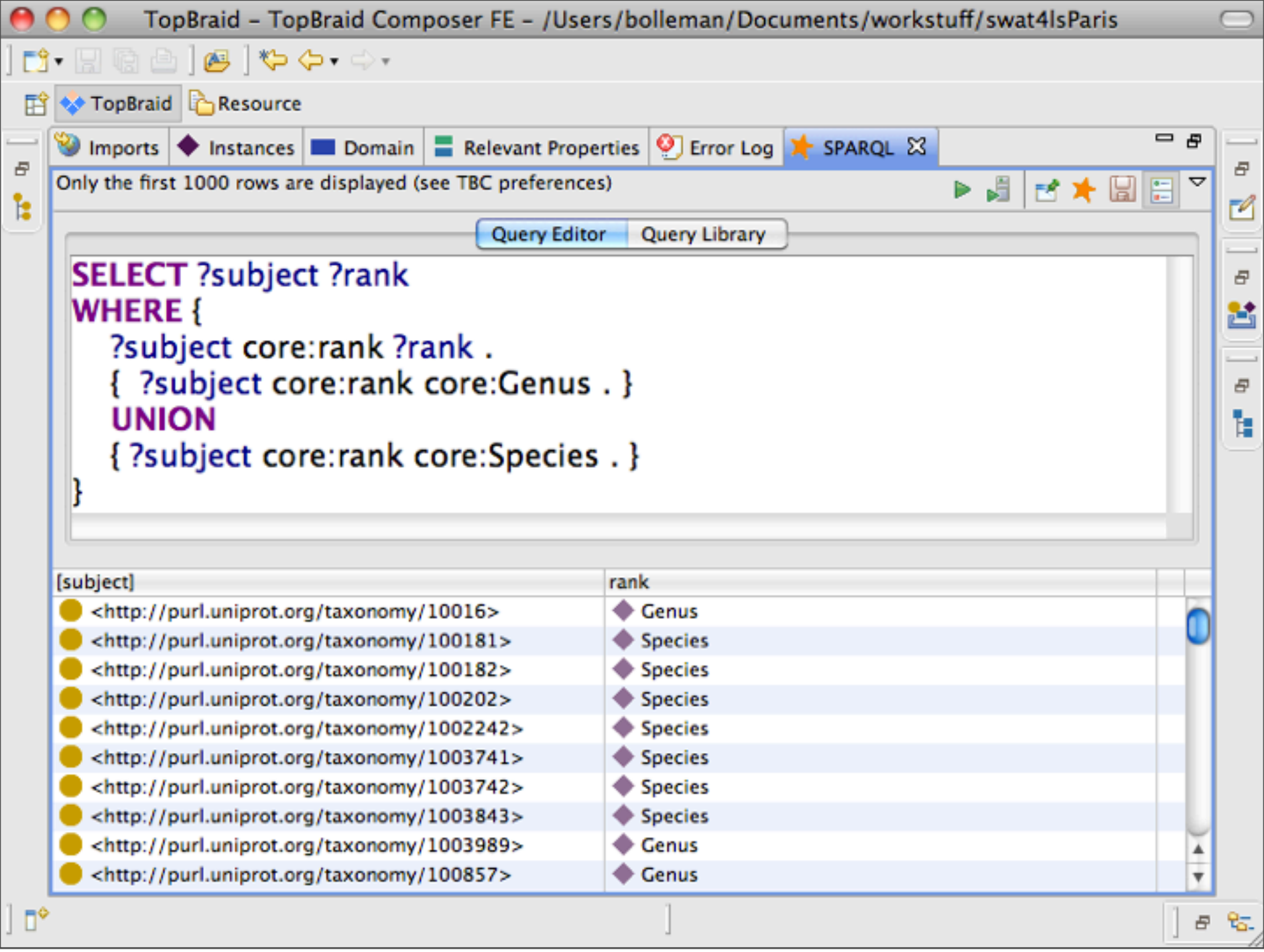
Not every taxon has a “common” name.
Join goes from outside to in.

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  OPTIONAL
  {?subject core:commonName ?name}
}
```

UNION

- Allows you to combine query patterns as an OR operation.
- Joins are still from outer to inner.

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```
SELECT *
WHERE {
  ?subject core:rank ?rank.
  {?subject core:rank core:Genus }
  UNION
  {?subject core:rank core:Species }
}
```

Negation

- When you do not want a certain category of matches.

```
SELECT ?pet
WHERE {
    ?pet a pets:Friendly .
}
```

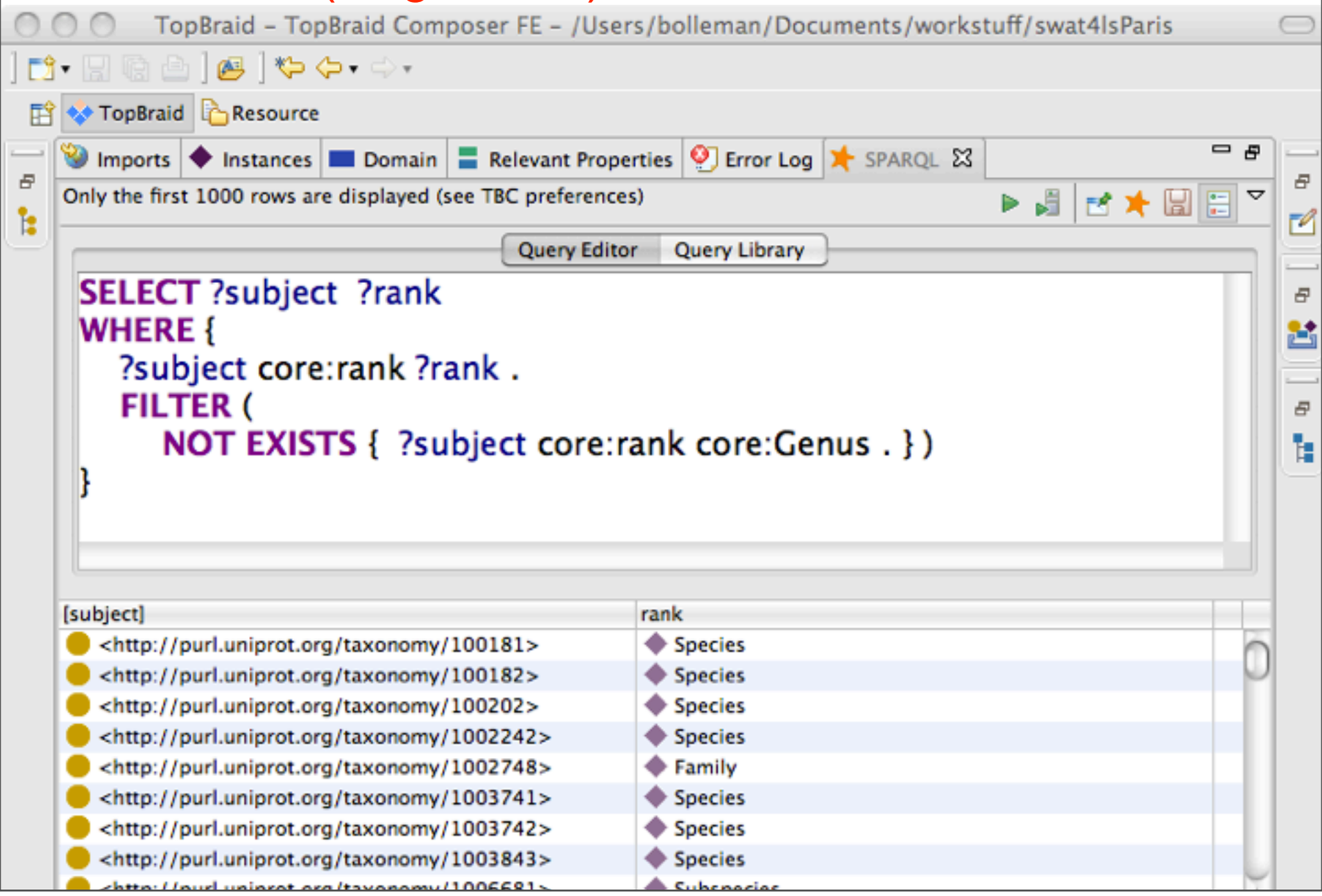
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Oooops



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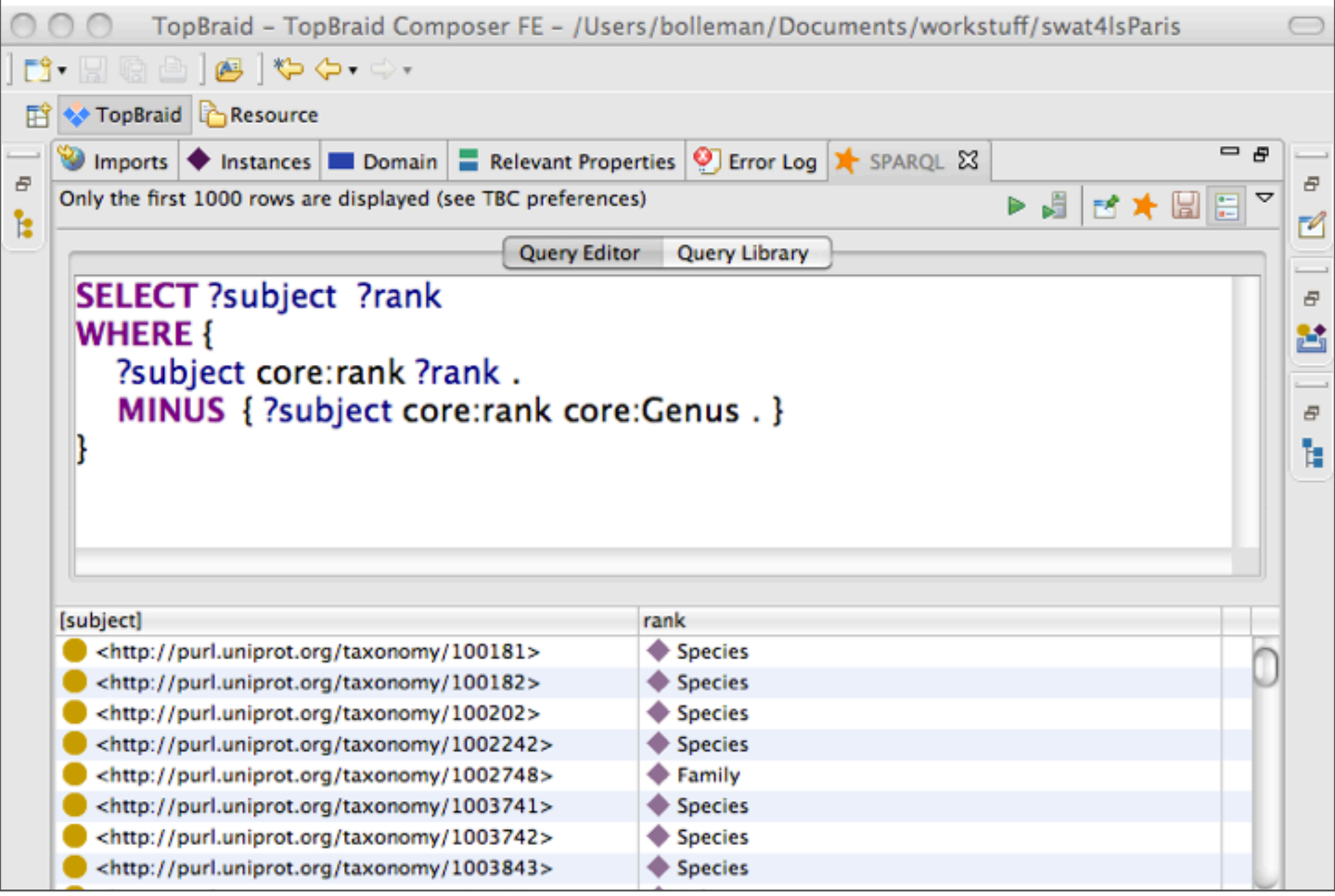
Not exists (Negation 1)



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```
SELECT ?subject  ?rank
WHERE {
   ?subject core:rank ?rank .
   FILTER (
         NOT EXISTS      {  ?subject core:rank core:Genus . } )
}
```

Minus (Negation 2)



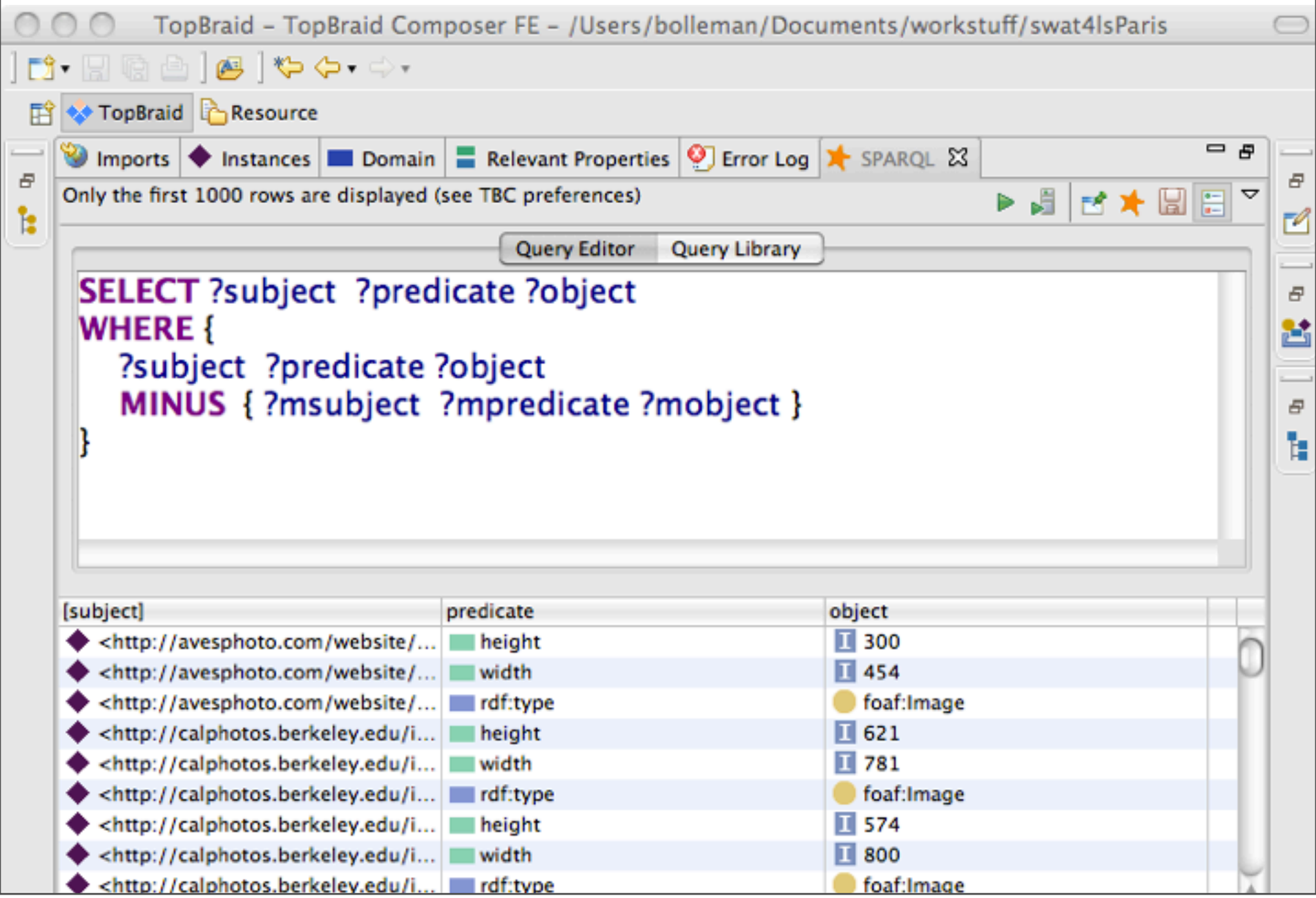

```
SELECT ?subject  ?rank
WHERE {
   ?subject core:rank ?rank .
   MINUS    { ?subject core:rank core:Genus . }
}
```

MINUS{} or FILTER (NOT EXISTS{})

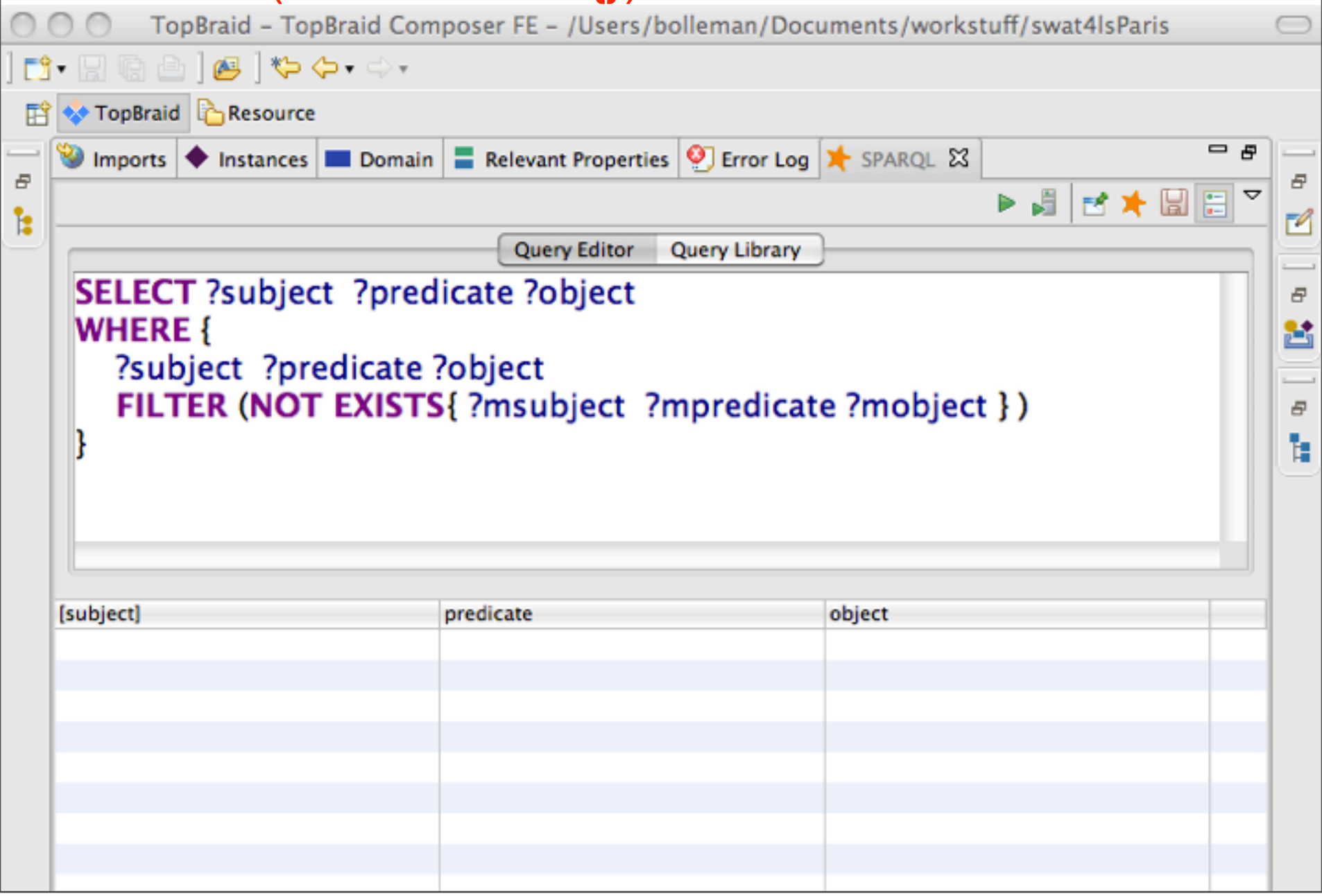
- Whats the difference?
  - MINUS subtracts results
  - NOT EXITS tests if the sub pattern is possible at all.
    - Normally the faster option.

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MINUS all data



FILTER (NOT EXISTS{}) no results



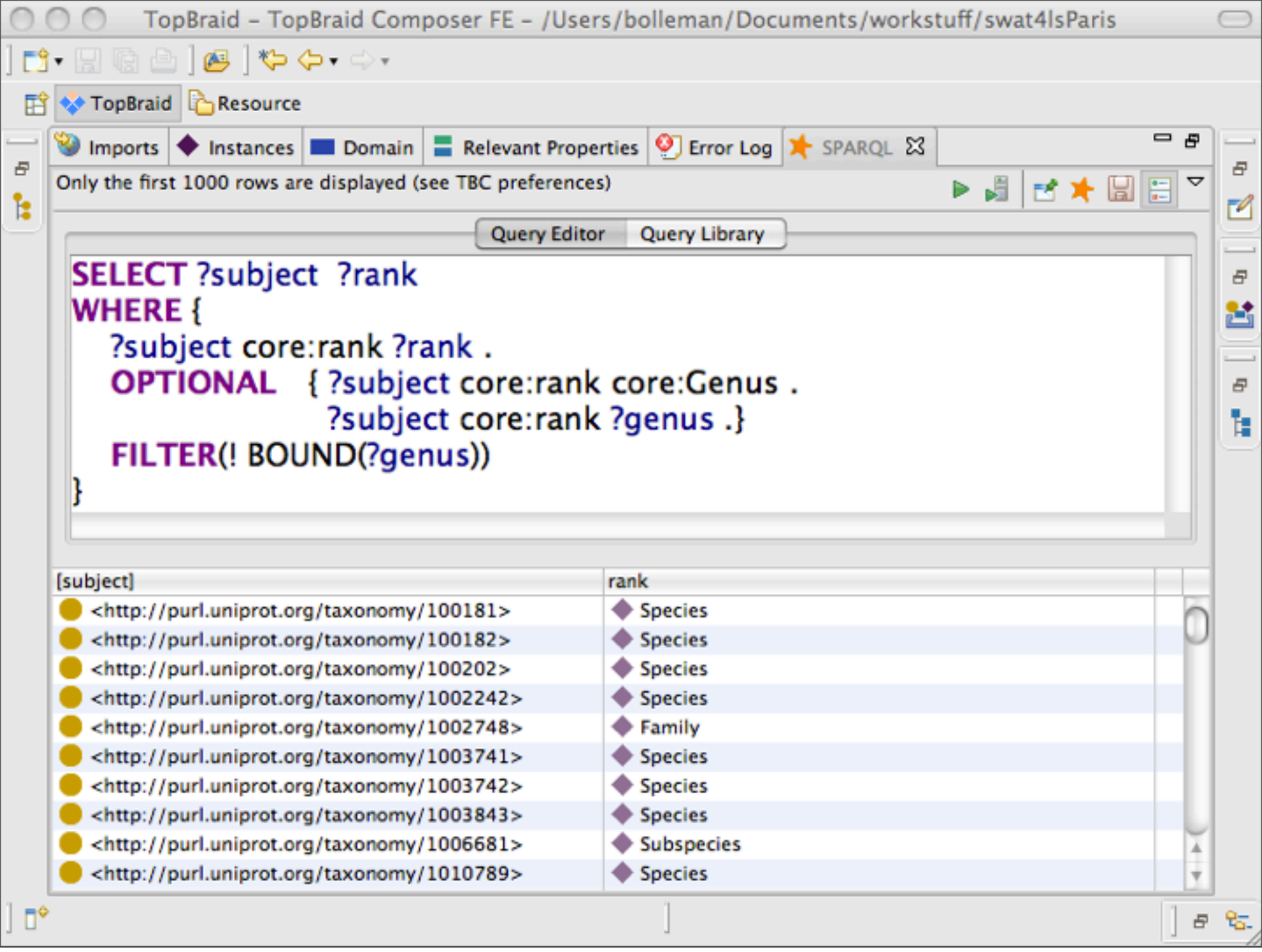
Negation option 3
SPARQL 1.0

```
SELECT ?subject  ?rank
WHERE {
   ?subject core:rank ?rank .
   OPTIONAL    { ?subject core:rank core:Genus .
                          ?subject core:rank ?genus .}
   FILTER(! BOUND(?genus))
}
```

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Uses the special function BOUND
BOUND lets you test if an optional value has been tied to something.



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```
SELECT *
WHERE {
  ?subject core:rank ?rank.
  OPTIONAL {?subject core:rank core:Genus ,
                                  ?genus}
  FILTER(!BOUND(?genus))
}
```

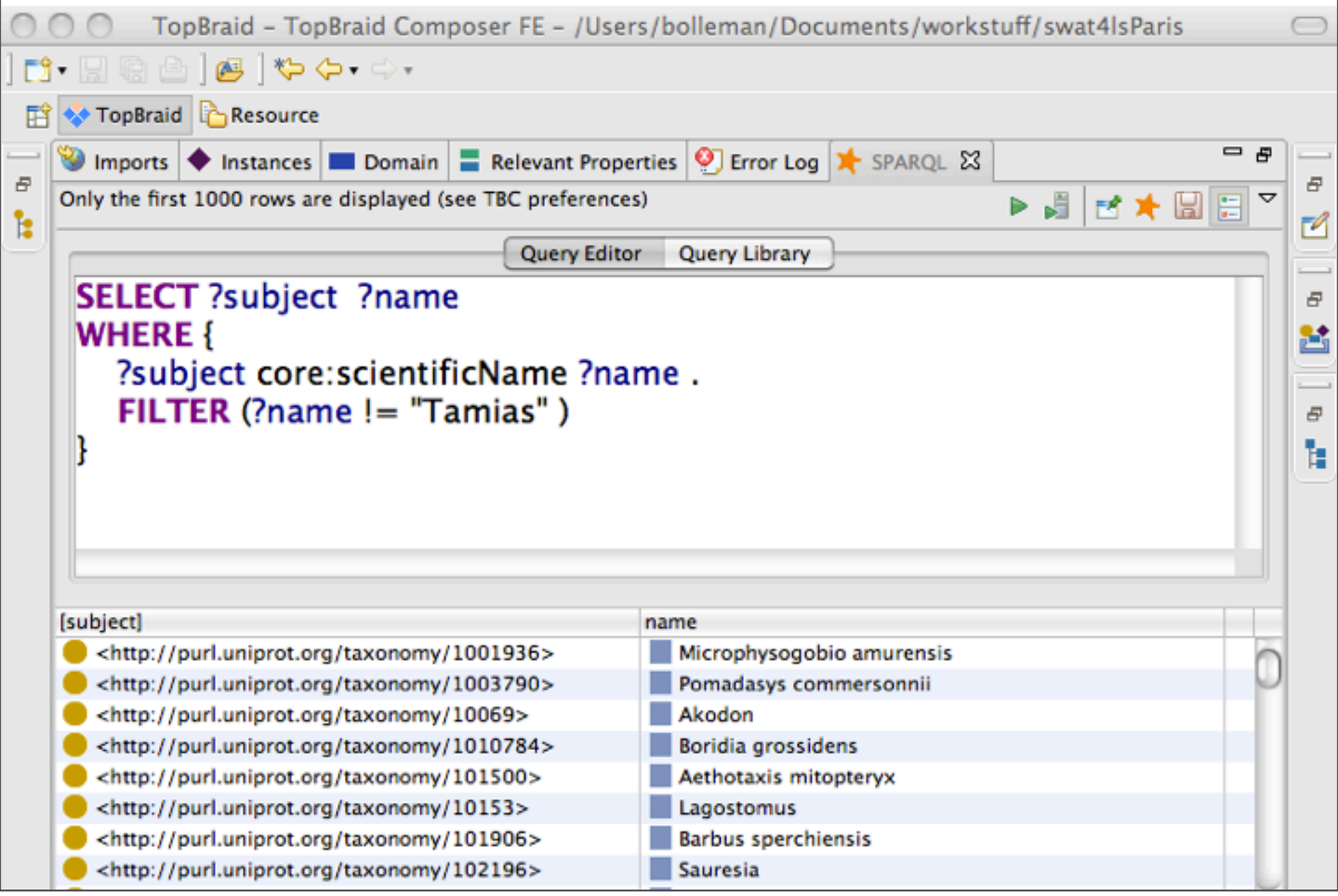
FILTERS

- You just saw it twice
  - Once in the !BOUND
  - Once in the NOT EXISTS

- FILTERS a result set by possibly removing values
  - FILTER do not add a value to the result
- Inside the same graph pattern order independent.

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Filter

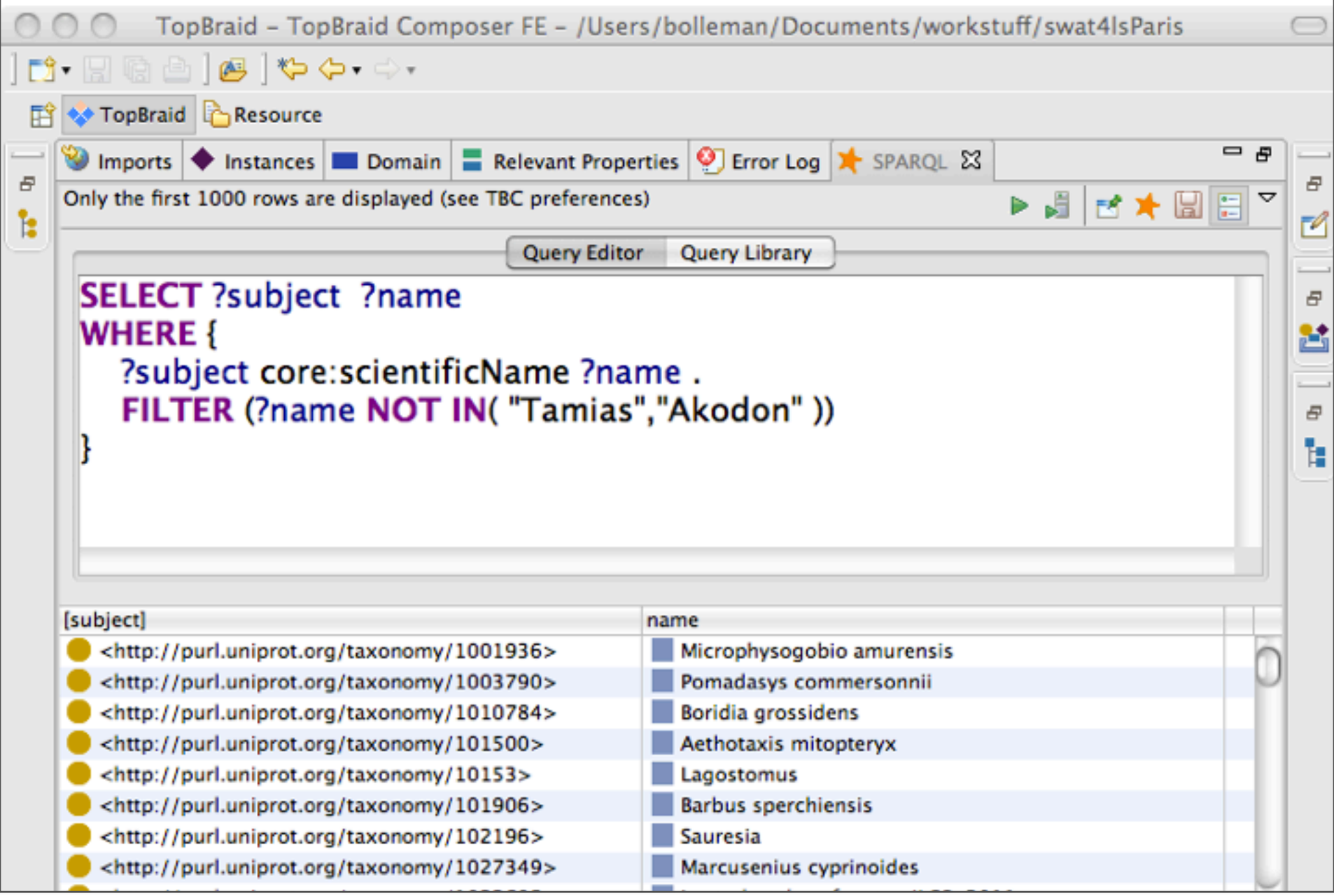


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Finds all taxons where the scientific name does is not “Tamias”

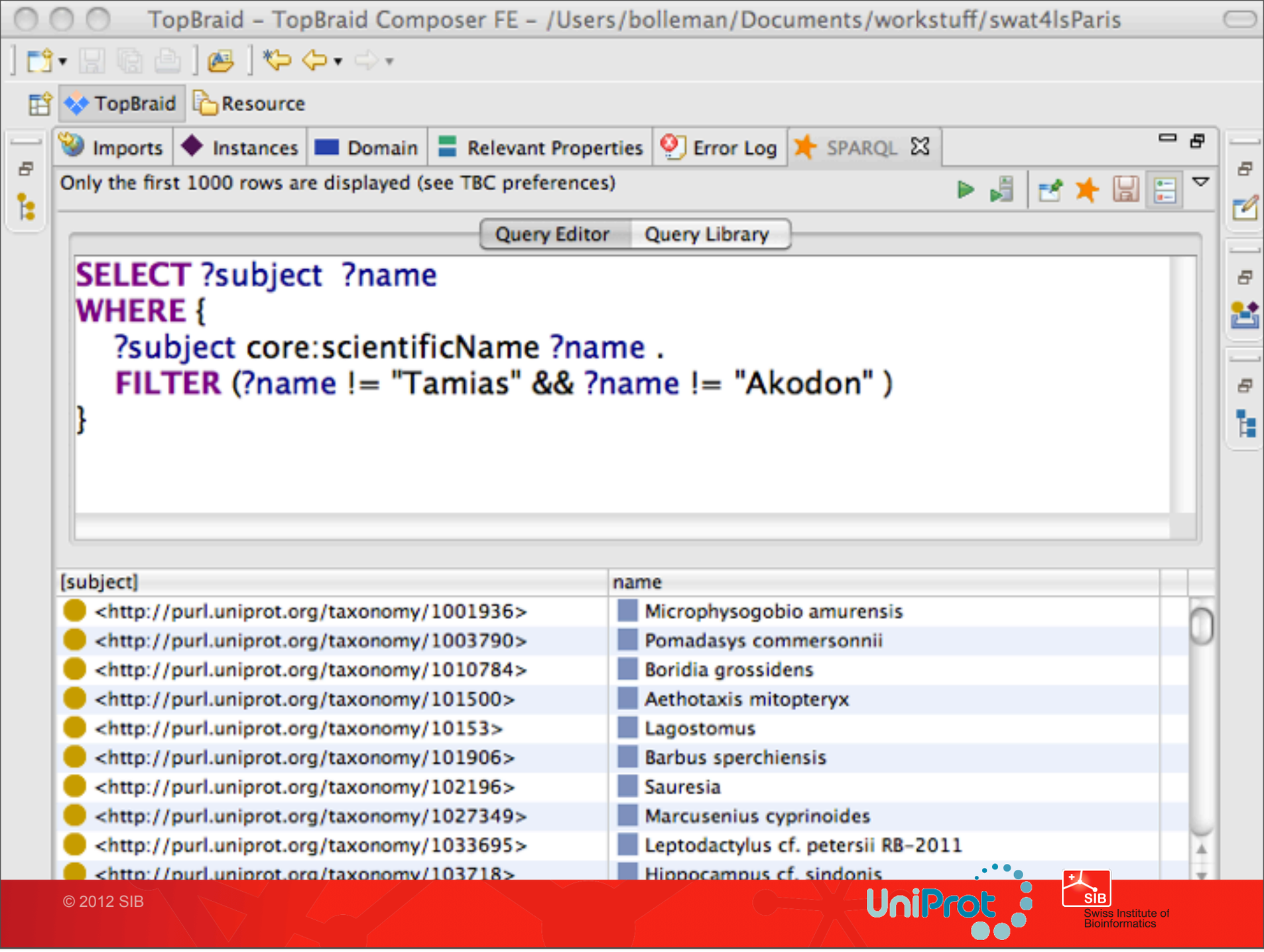
```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
  FILTER(?name != "Tamias")
}
```

Filter on not in



?name is not “Tamias” and name is not “Akodon”

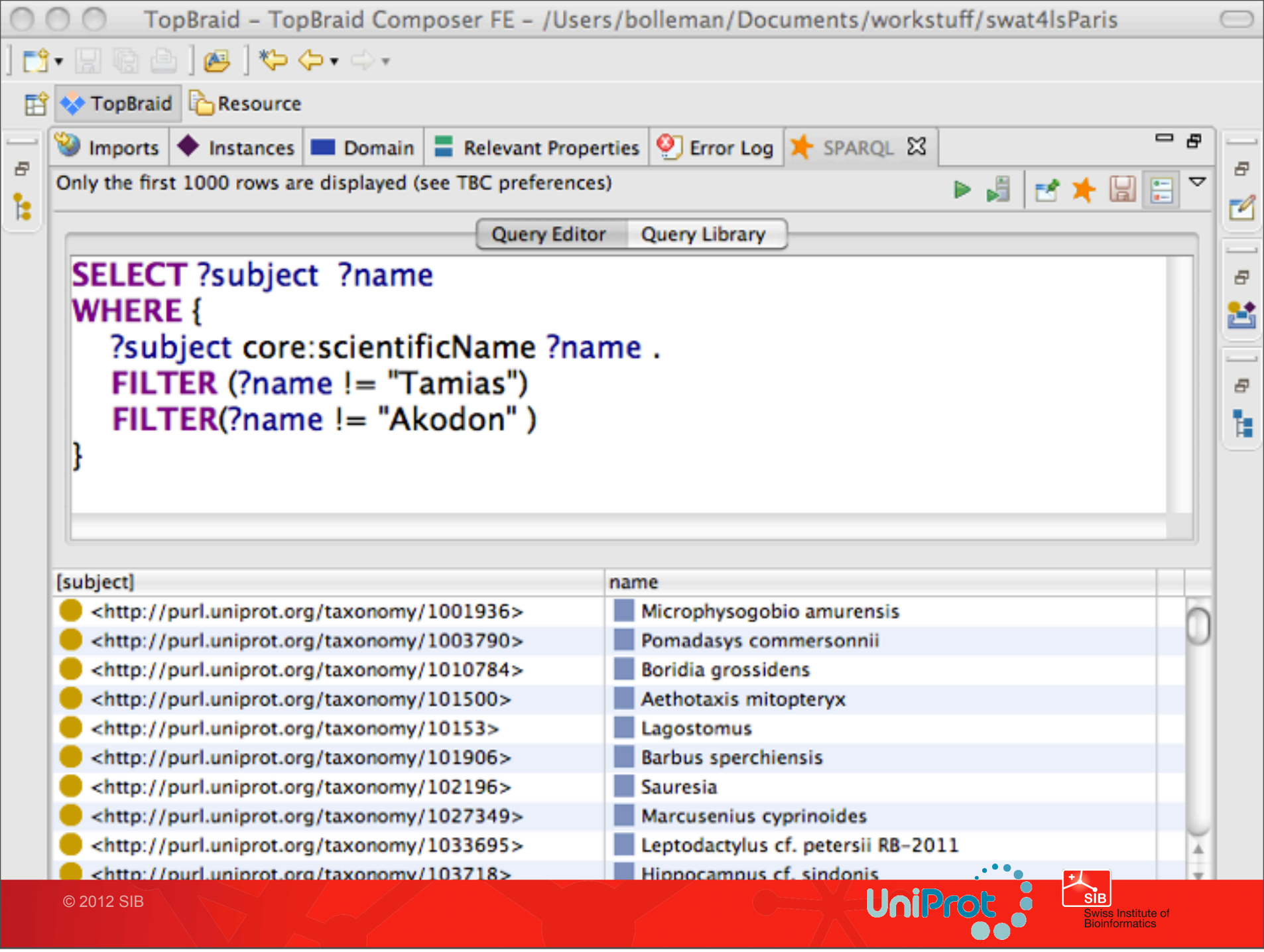
```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
  FILTER(?name NOT IN ("Tamias", "Akodon" ))
}
```



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This is the same but now using two != filters combined with &&

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
  FILTER(?name != "Tamias" && ?name != "Akodon" )
}
```

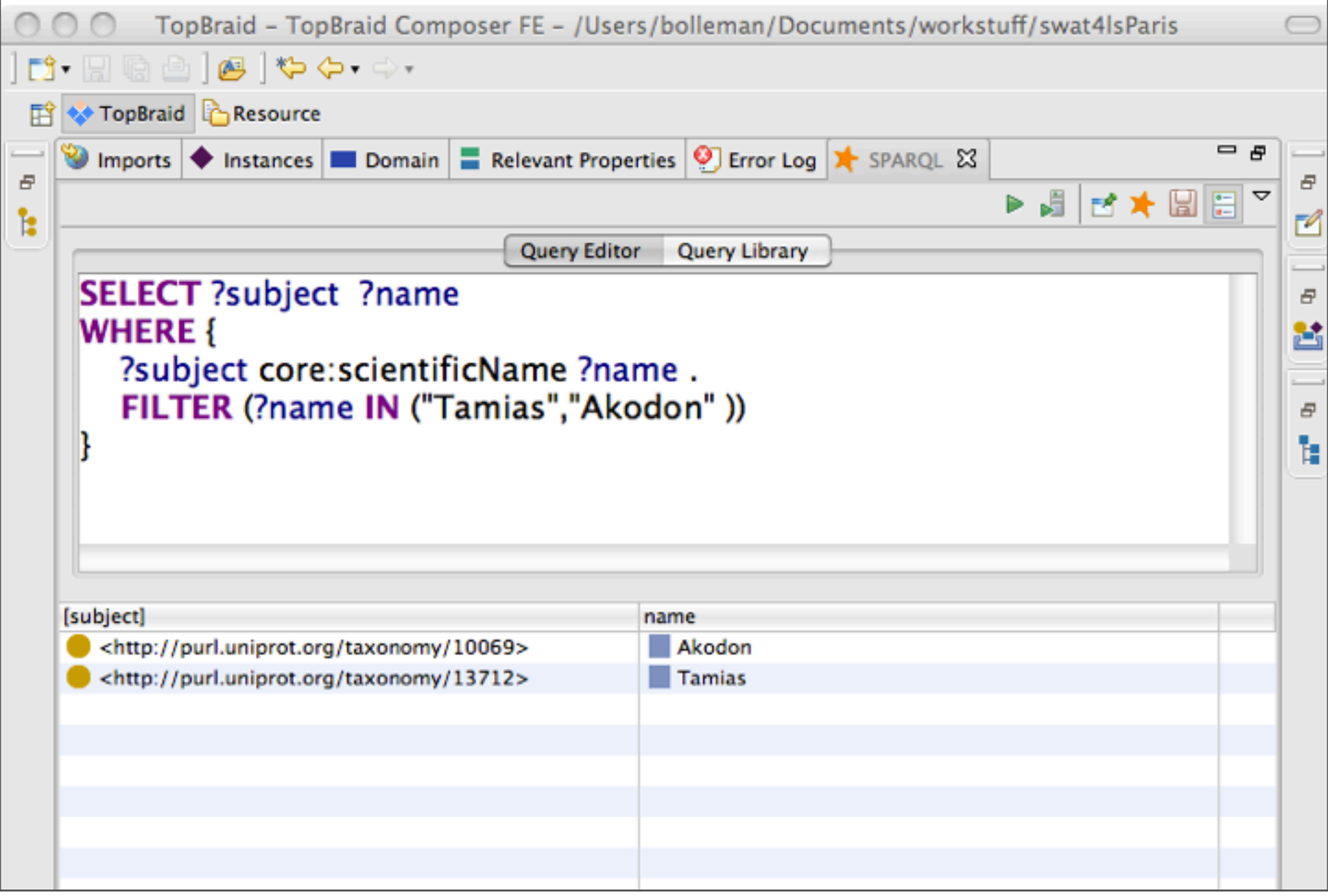


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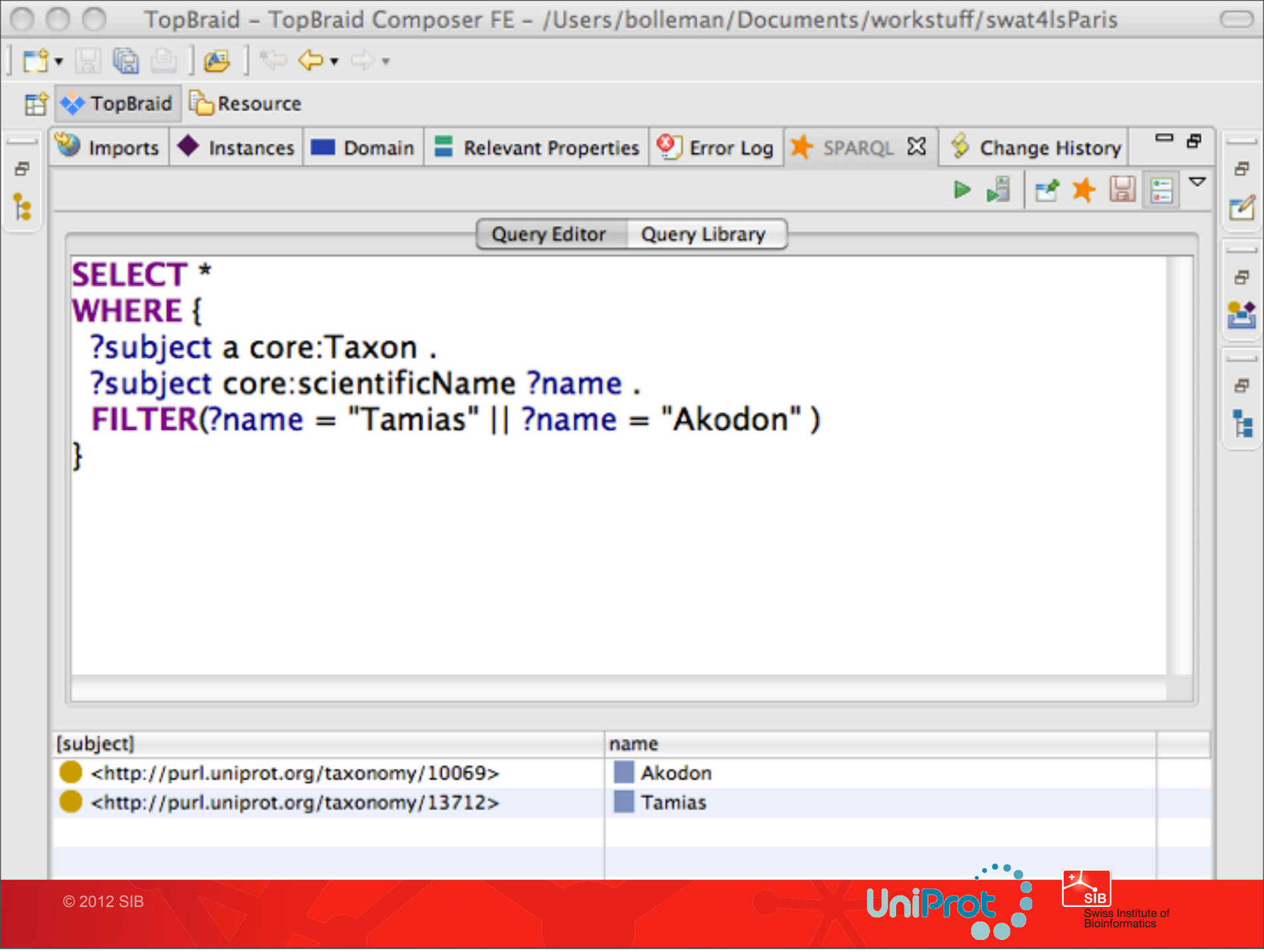
Again the same but now using the implicit AND between two filters.

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
  FILTER(?name != "Tamias")
  FILTER (?name != "Akodon" )
}
```

IN




```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 FILTER(?name IN ("Tamias", "Akodon" ))
}
```




```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
  FILTER(?name = "Tamias" || ?name = "Akodon" )
}
```

FILTER on numbers

- <
  - FILTER (1 < 2)
- >
  - FILTER (2 > 1)
- =
  - FILTER (1 =1)
- !=
  - FILTER(1 != 2)
-

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Filters

- ?x = ?y does casting (value conversions)
  - 1.0^^xsd:float = 1^^xsd:int is true
- sameTerm(?x, ?y) does not
  - sameTerm(1.0^^xsd:float, 1^^xsd:int)

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```
SELECT * WHERE {
BIND(1 AS ?int) .
BIND(1.0 AS ?float) .
FILTER (sameTerm(?int, ?float))
}
or
SELECT * WHERE {
BIND(1 AS ?int) .
BIND(1.0 AS ?float) .
FILTER (?int = ?float)
}
```

FILTER on strings

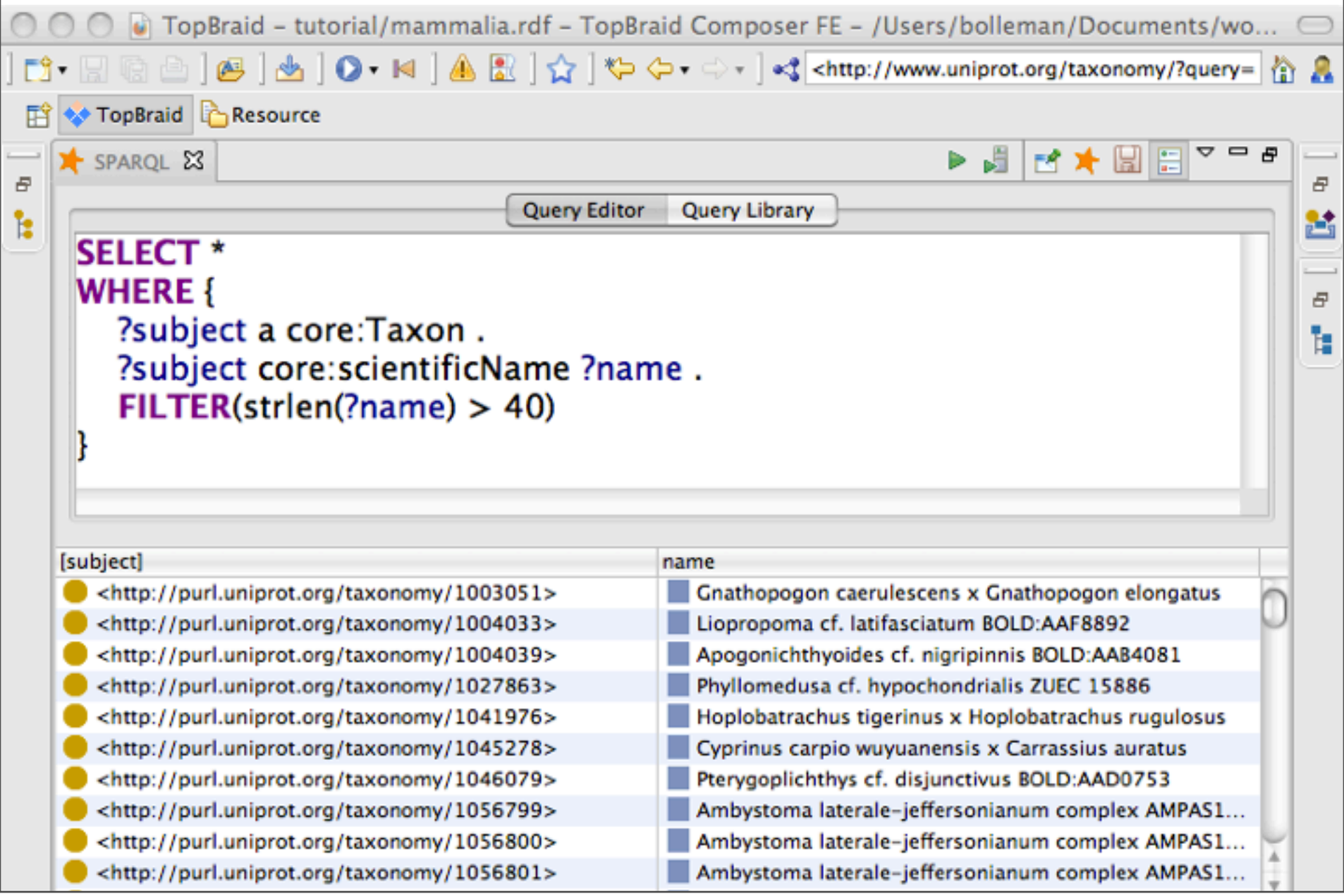
- Functions
  - STRLEN
  - SUBSTR
  - UCASE
  - LCASE
  - STRSTARTS
  - STRENDS
  - CONTAINS
  - STRBEFORE
  - STRAFTER
  - ENCODE_FOR_URI
  - CONCAT
  - langMatches
  - REGEX
  - REPLACE
  - IRI

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IRI we will show at the end of the talk.

STRLEN == String Length

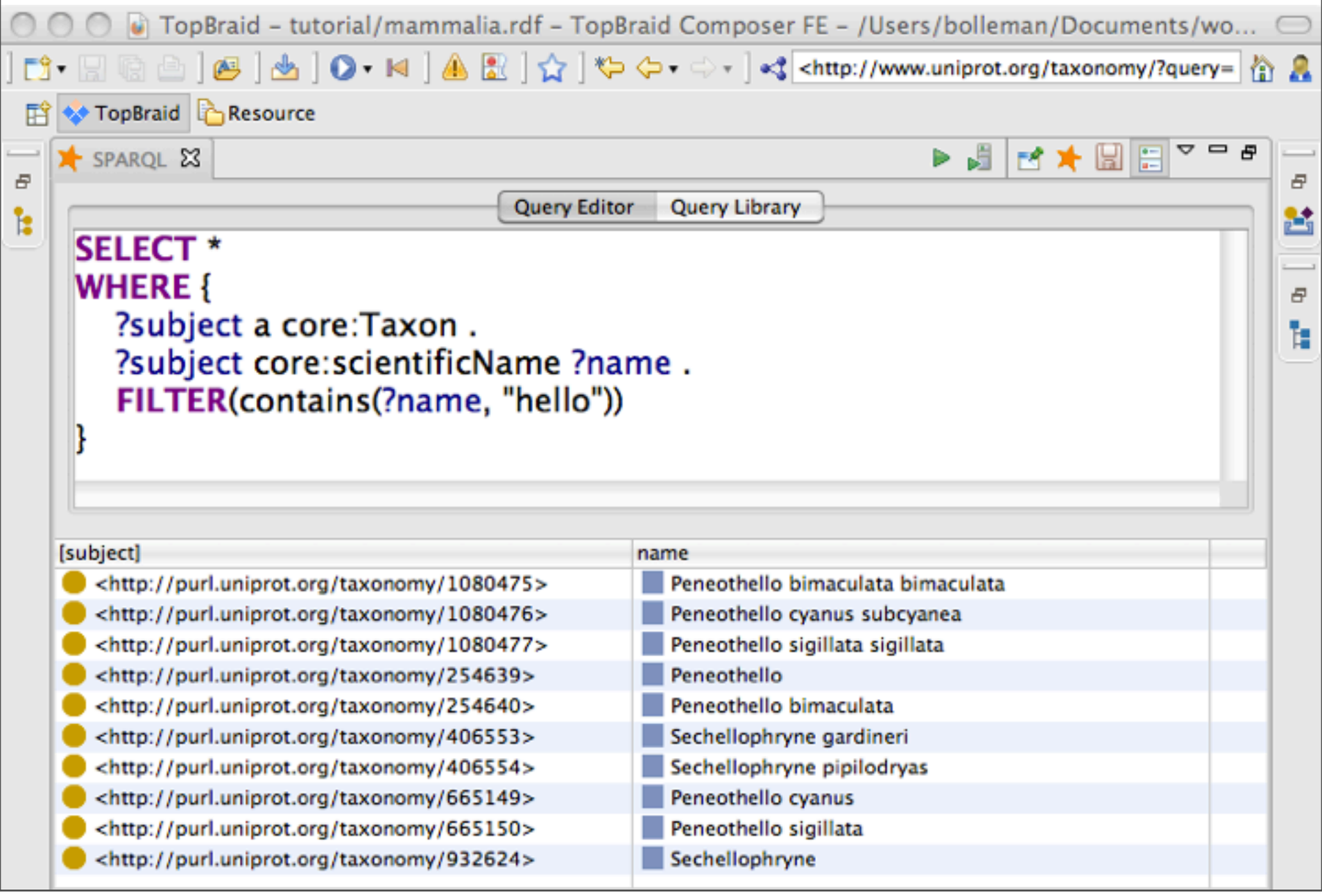


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Returns a number and which is why > works

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 FILTER(strlen(?name) > 40 )
}
```

CONTAINS is case sensitive is it in there

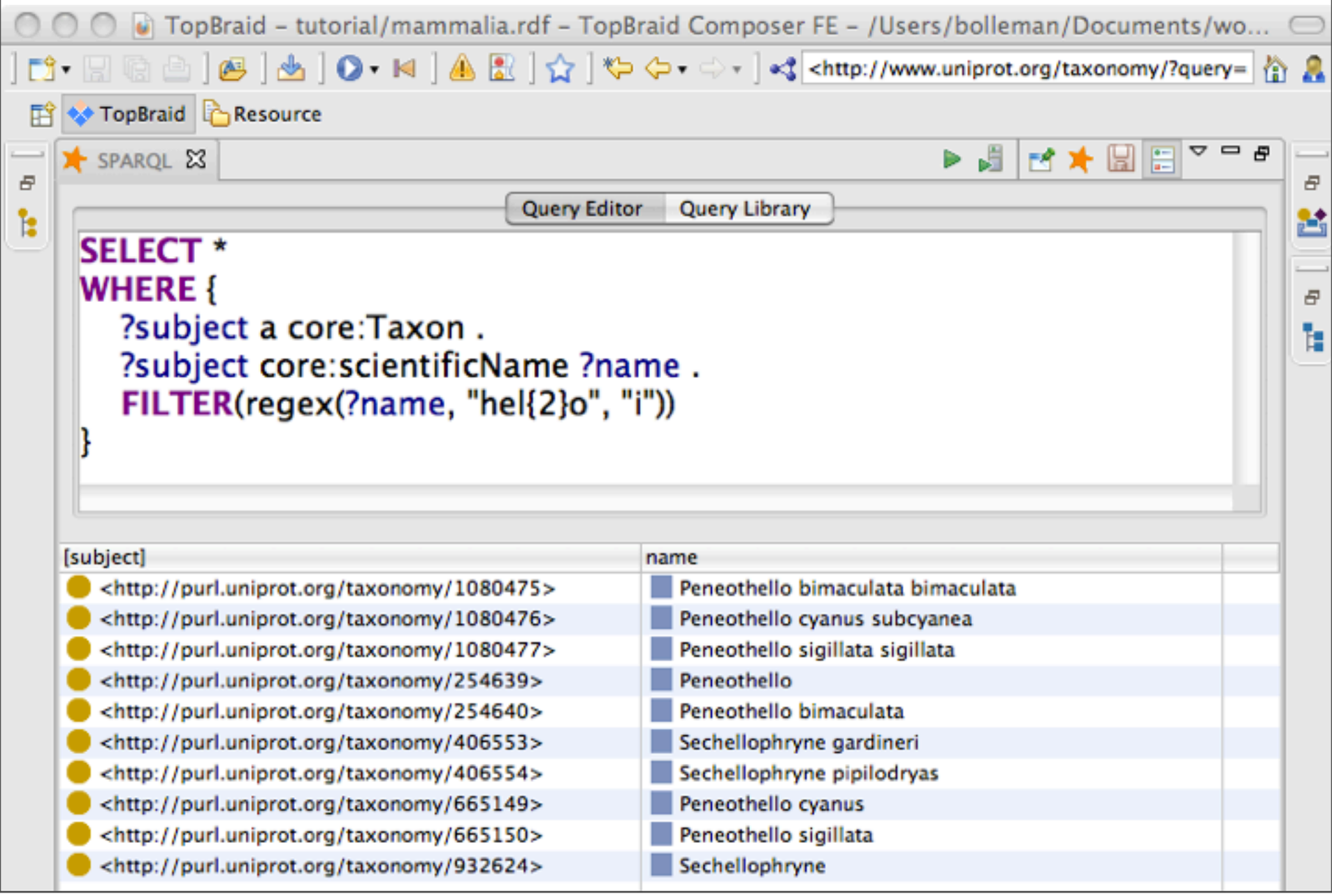


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contains returns a boolean so is directly usable in a FILTER

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 FILTER(contains(?name, "hello"))
}
```

REGEX, just like java regex



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Does not have all the fancy stuff in a PERL regex.
No capturing groups.

```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 FILTER(regex(?name, "hel{2}o", "i"))
}
```

BIND

- Builds new Values
  - Closes the basic graph pattern

```
SELECT ?p WHERE {
    {
        ?taxon a :Taxon .
    }
    BIND (?taxon AS ?p)
}
```

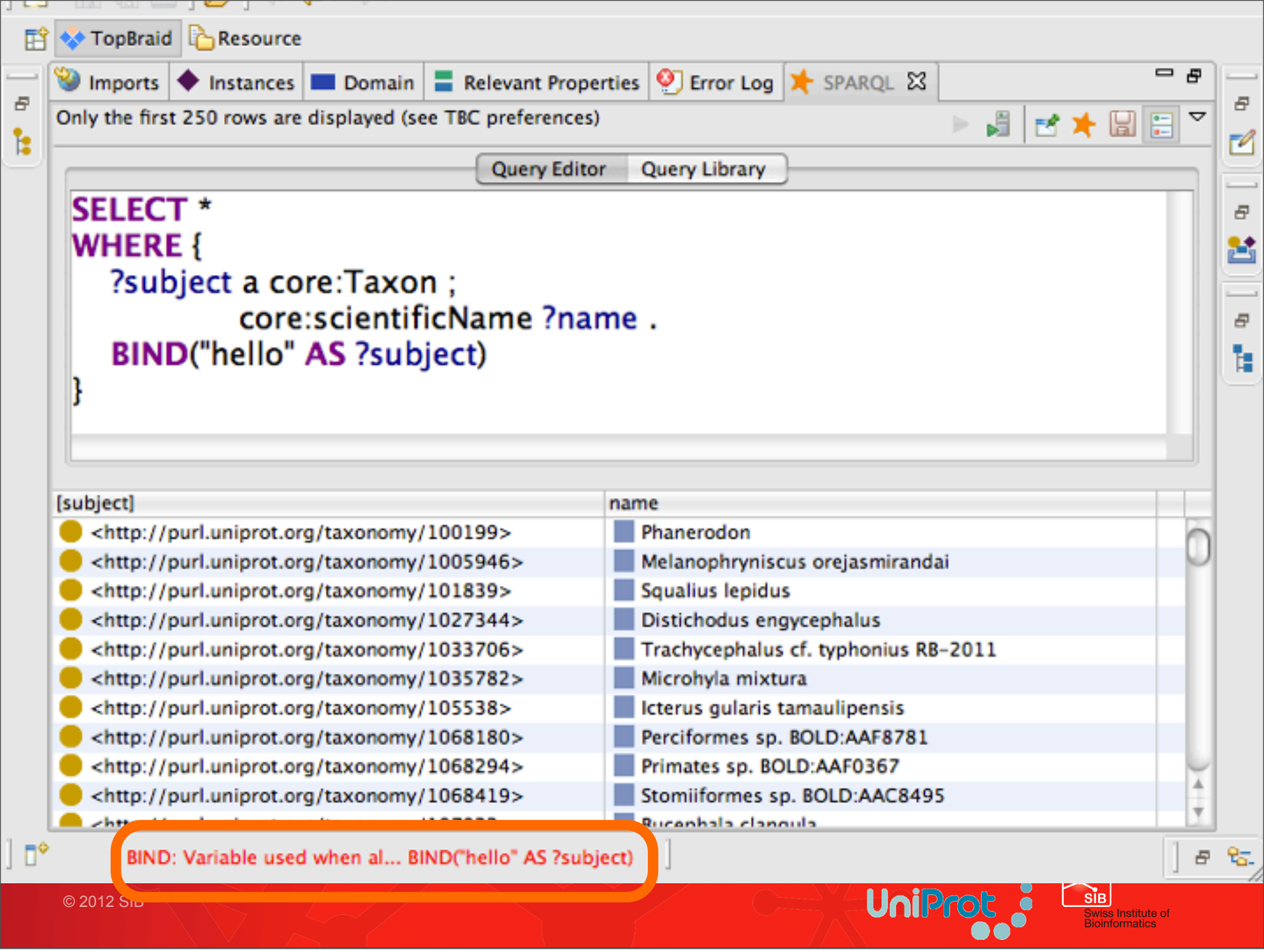
- Always declare before use.

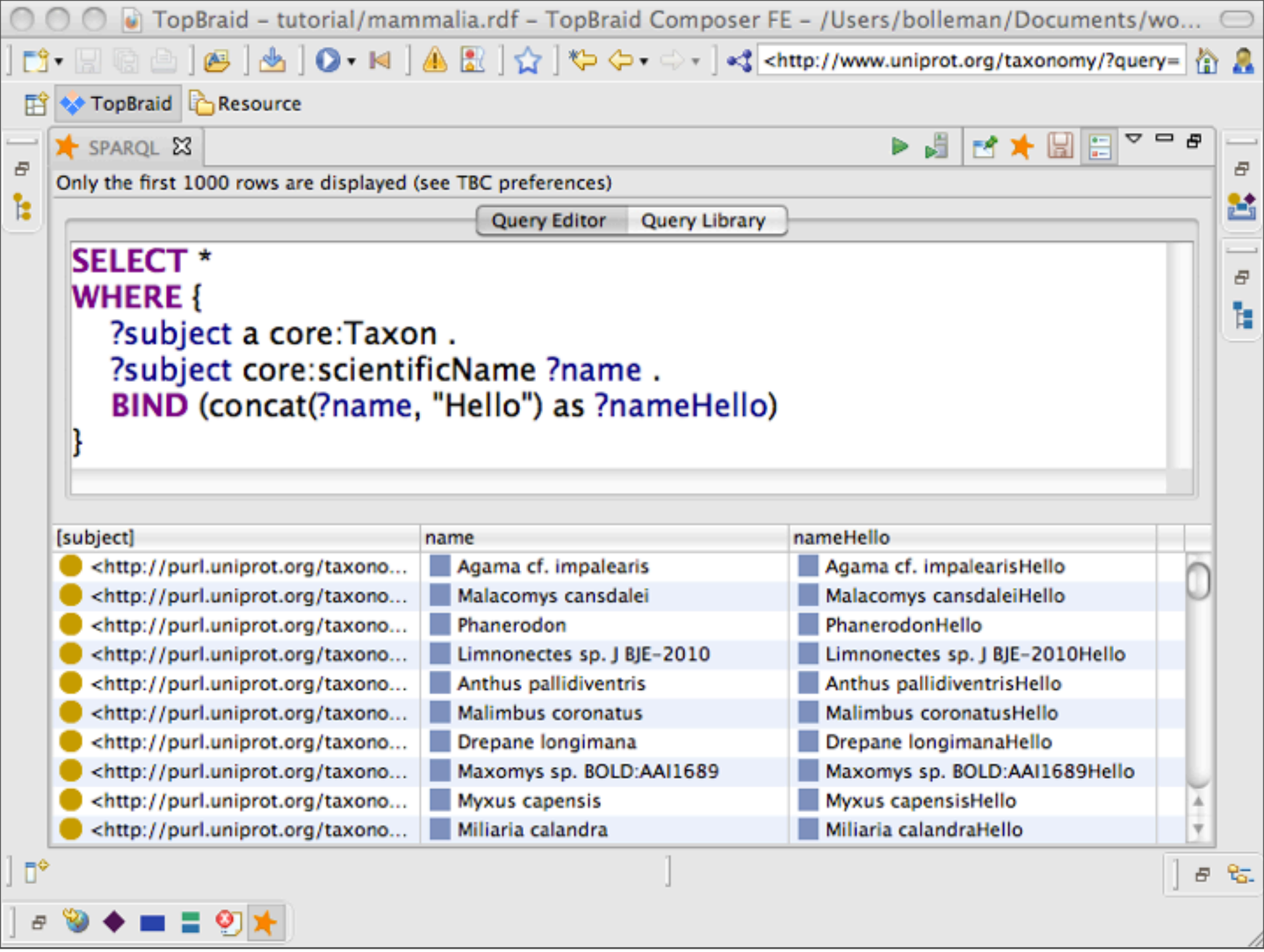
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Triple stores may not warn you if you use before you declare

```
SELECT ?protein WHERE {
{
?protein a :Protein .
}
BIND (<http://purl.uniprot.org/uniprot/P12345> AS ?protein)
}
```

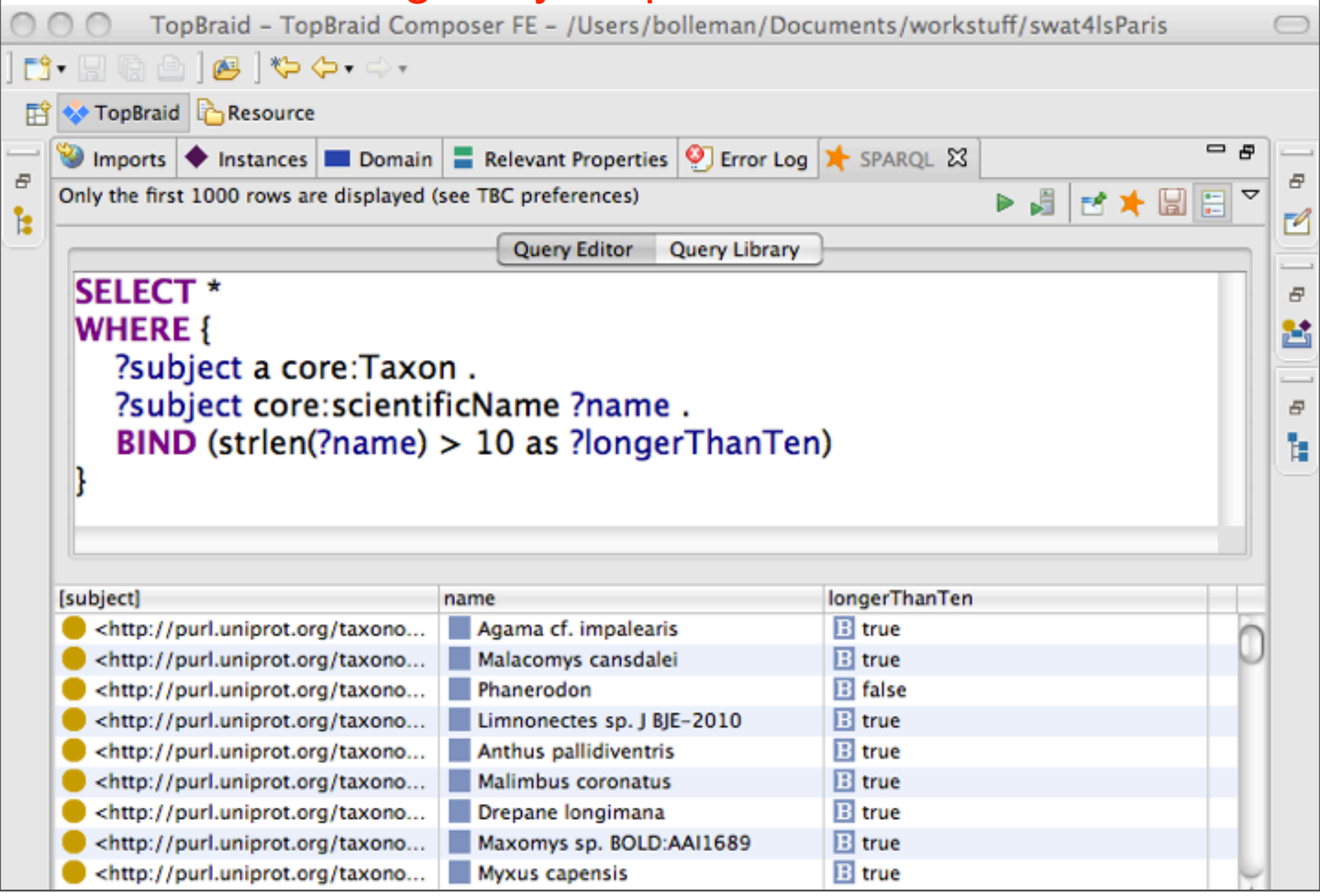

TopBraid
Resource
Imports
Instances
Domain
Relevant Properties
Error Log
SPARQL
Only the first 250 rows are displayed (see TBC preferences)
Query Editor
Query Library
SELECT *
WHERE {
?subject a core:Taxon ;
core:scientificName ?name .
BIND("hello" AS ?subject)
}
[subject]
name
<http://purl.uniprot.org/taxonomy/100199>
Phanerodon
<http://purl.uniprot.org/taxonomy/1005946>
Melanophryniscus orejasmirandai
<http://purl.uniprot.org/taxonomy/101839>
Squalius lepidus
<http://purl.uniprot.org/taxonomy/1027344>
Distichodus engycephalus
<http://purl.uniprot.org/taxonomy/1033706>
Trachycephalus cf. typhonius RB-2011
<http://purl.uniprot.org/taxonomy/1035782>
Microhyla mixtura
<http://purl.uniprot.org/taxonomy/105538>
Icterus gularis tamaulipensis
<http://purl.uniprot.org/taxonomy/1068180>
Perciformes sp. BOLD:AAF8781
<http://purl.uniprot.org/taxonomy/1068294>
Primates sp. BOLD:AAF0367
<http://purl.uniprot.org/taxonomy/1068419>
Stomiiformes sp. BOLD:AAC8495
BIND: Variable used when al... BIND("hello" AS ?subject)
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```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 BIND(concat(?name, "hello") as ?nameHello)
}
```

BIND can assign any output



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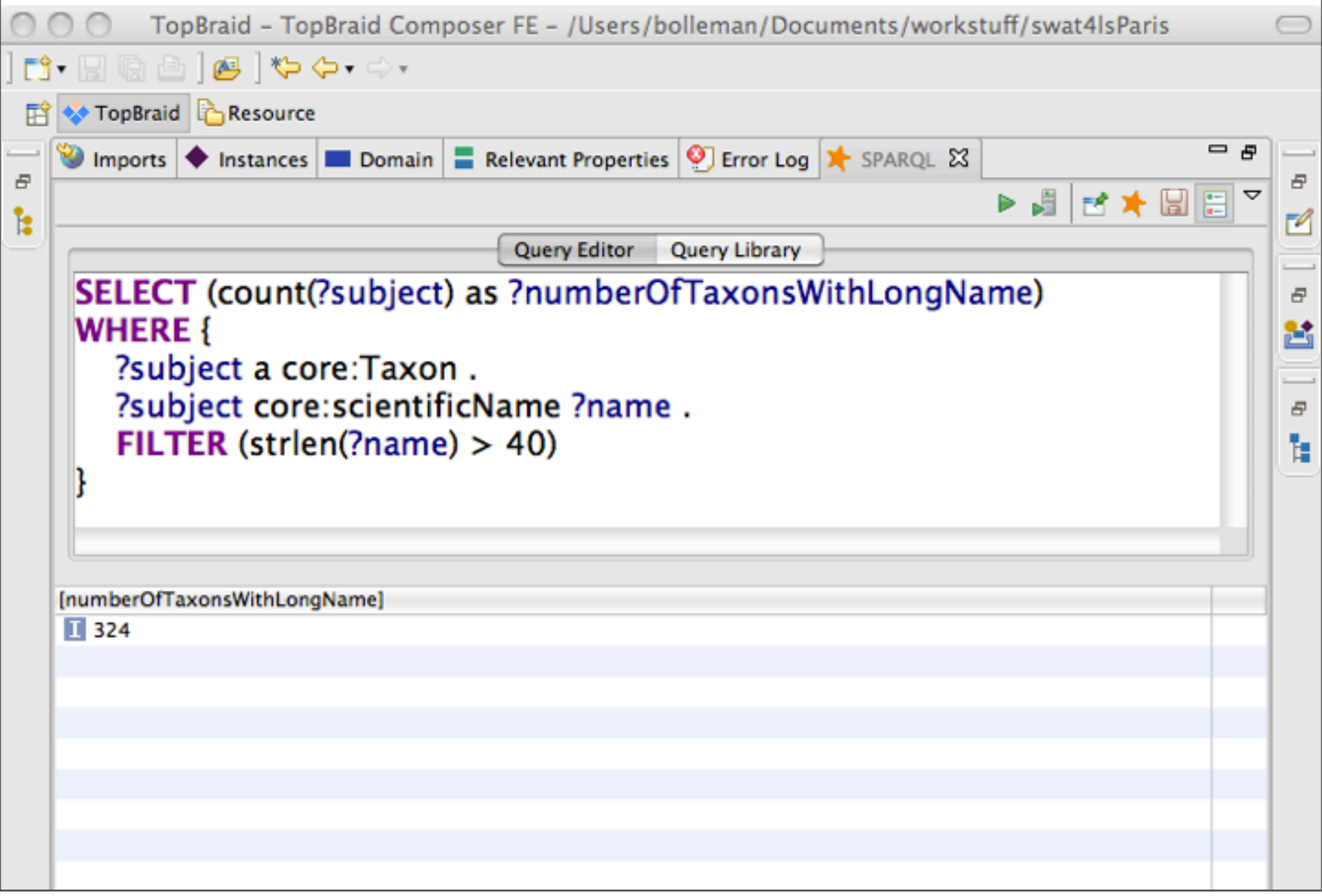
```
SELECT *
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
 BIND(strlen(?name) >10 AS ?longerThanTen)
}
```

Aggregate functions

- on select line
- limited in number
  - count
  - sum
  - avg
  - min
  - max
  - groupConcat
  - sample

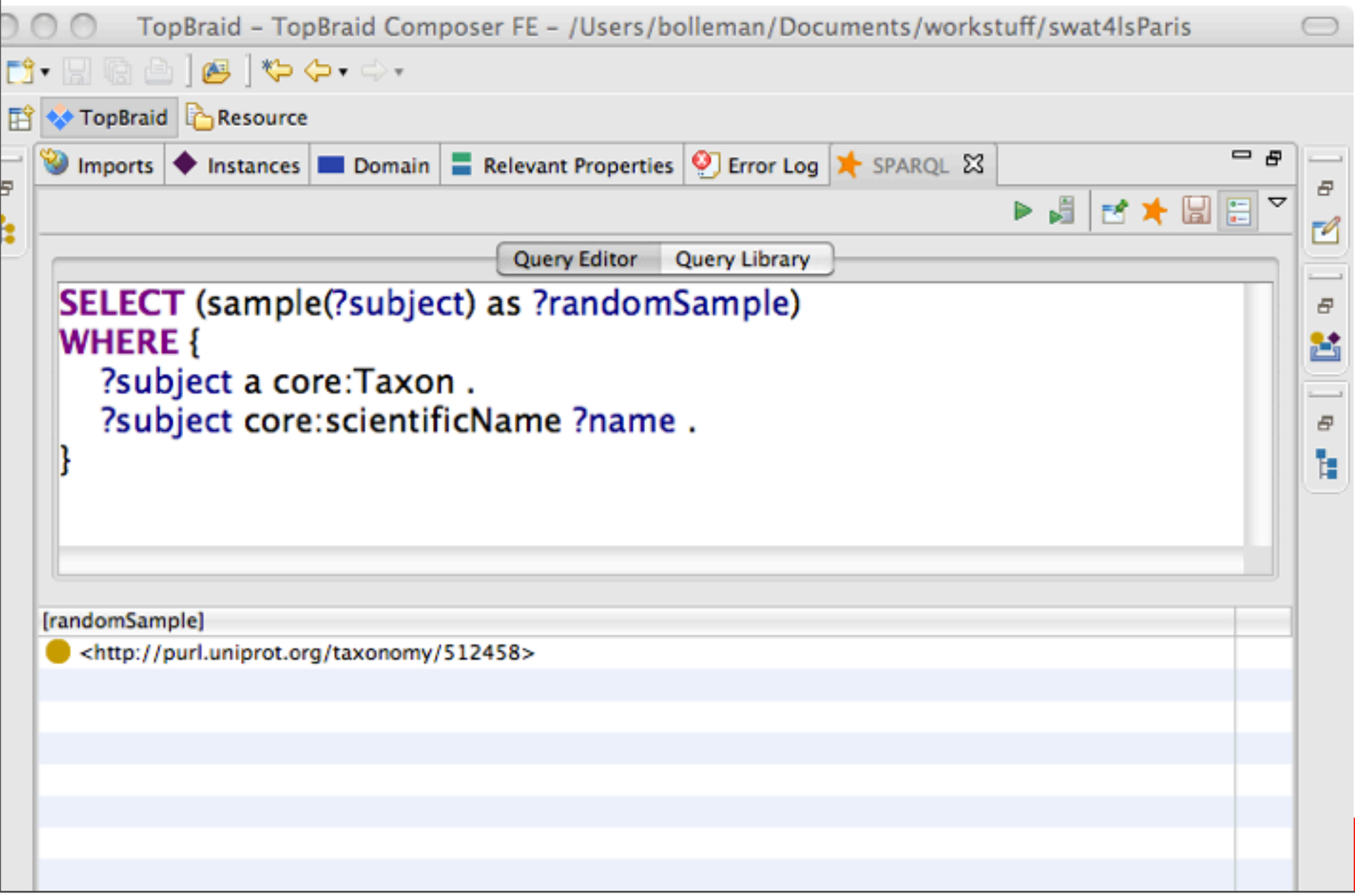
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count




```
SELECT (count(?subject) as ?numberOfTaxonsWithLongName)
WHERE {
   ?subject a core:Taxon .
   ?subject core:scientificName ?name .
   FILTER (strlen(?name) > 40)
}
```

SAMPLE should give a random result back




```
SELECT (sample(?subject) as ?randomSample)
WHERE {
  ?subject a core:Taxon .
  ?subject core:scientificName ?name .
}
```

Follow the path

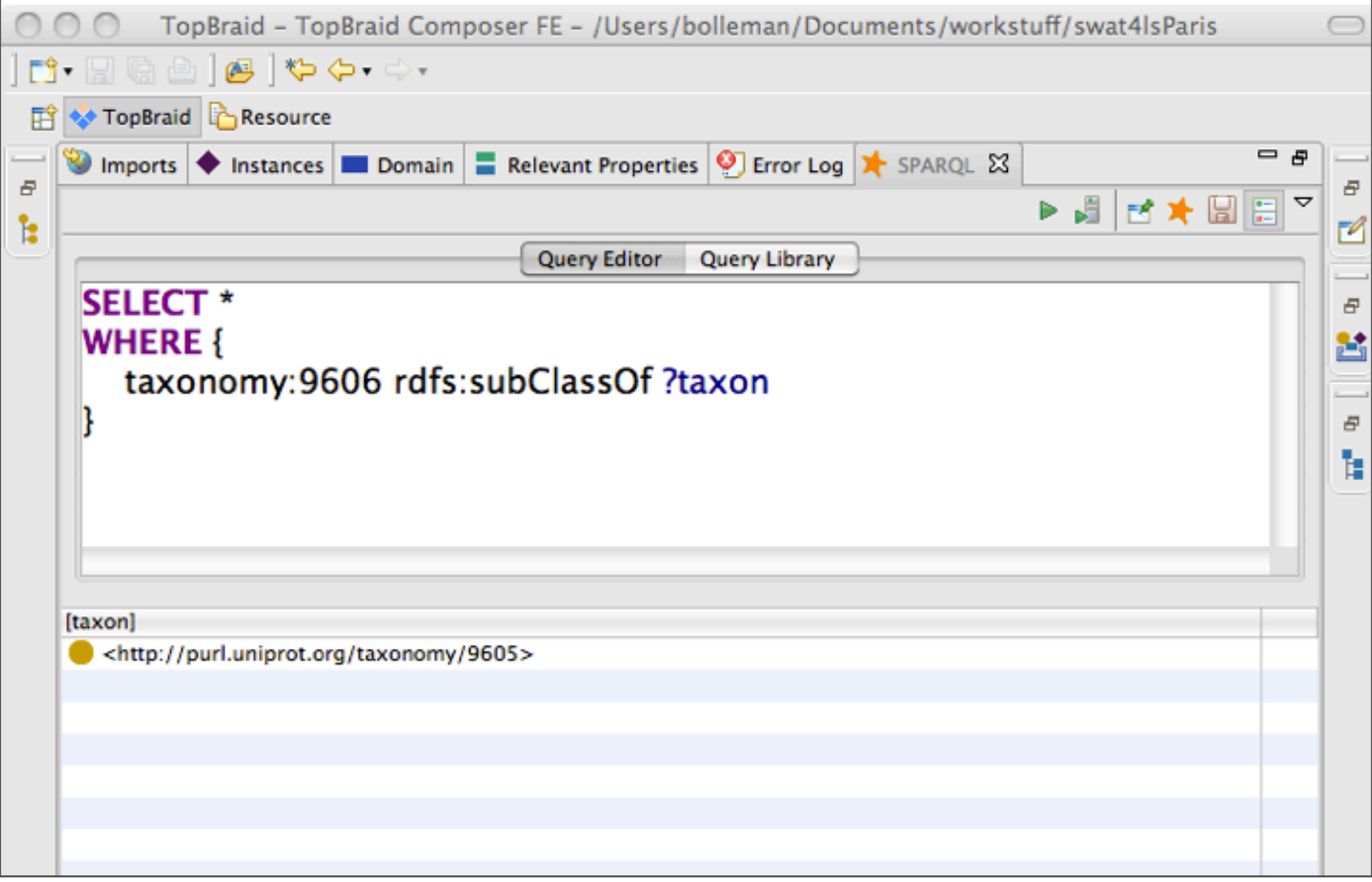


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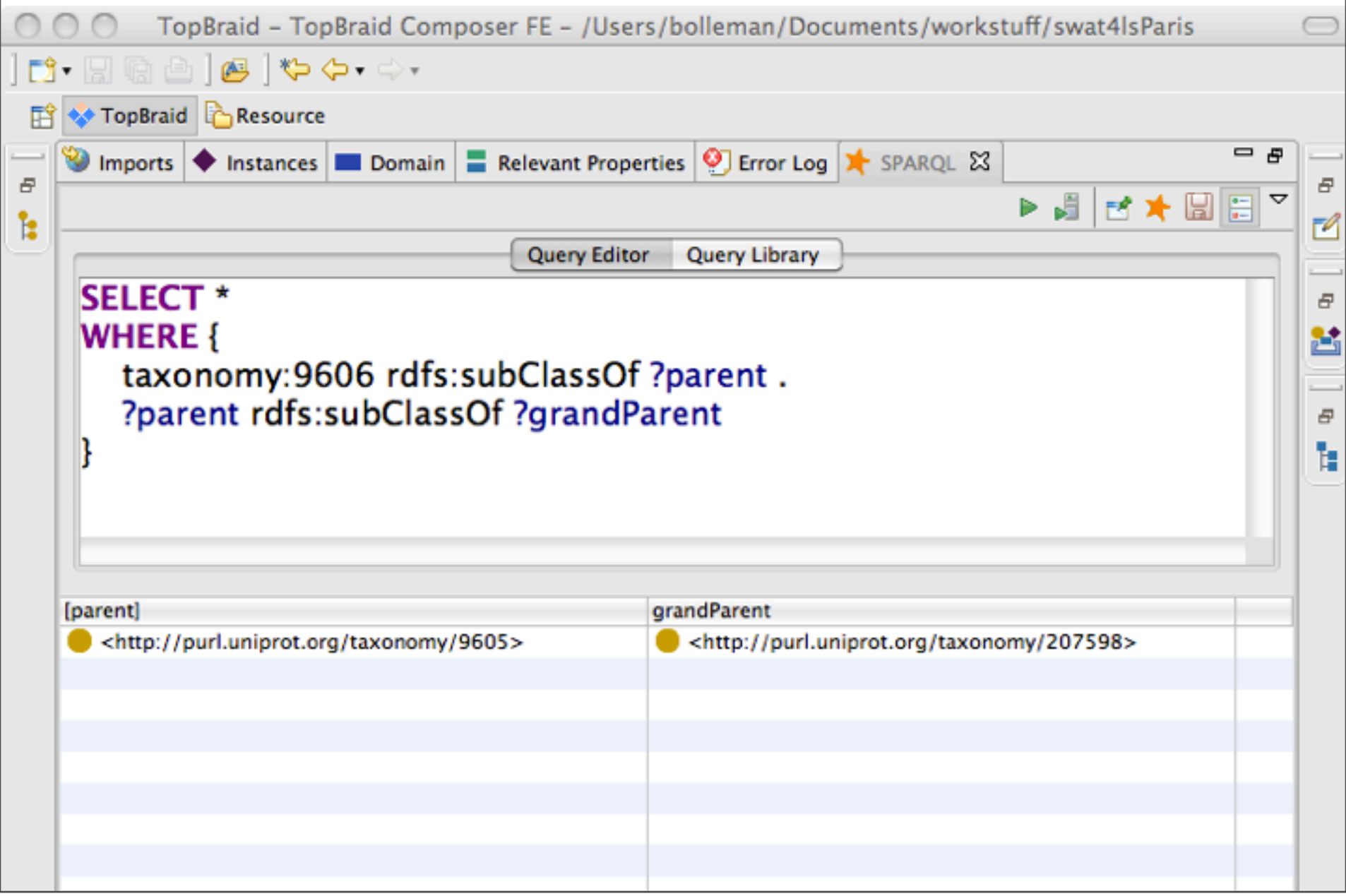
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Path queries



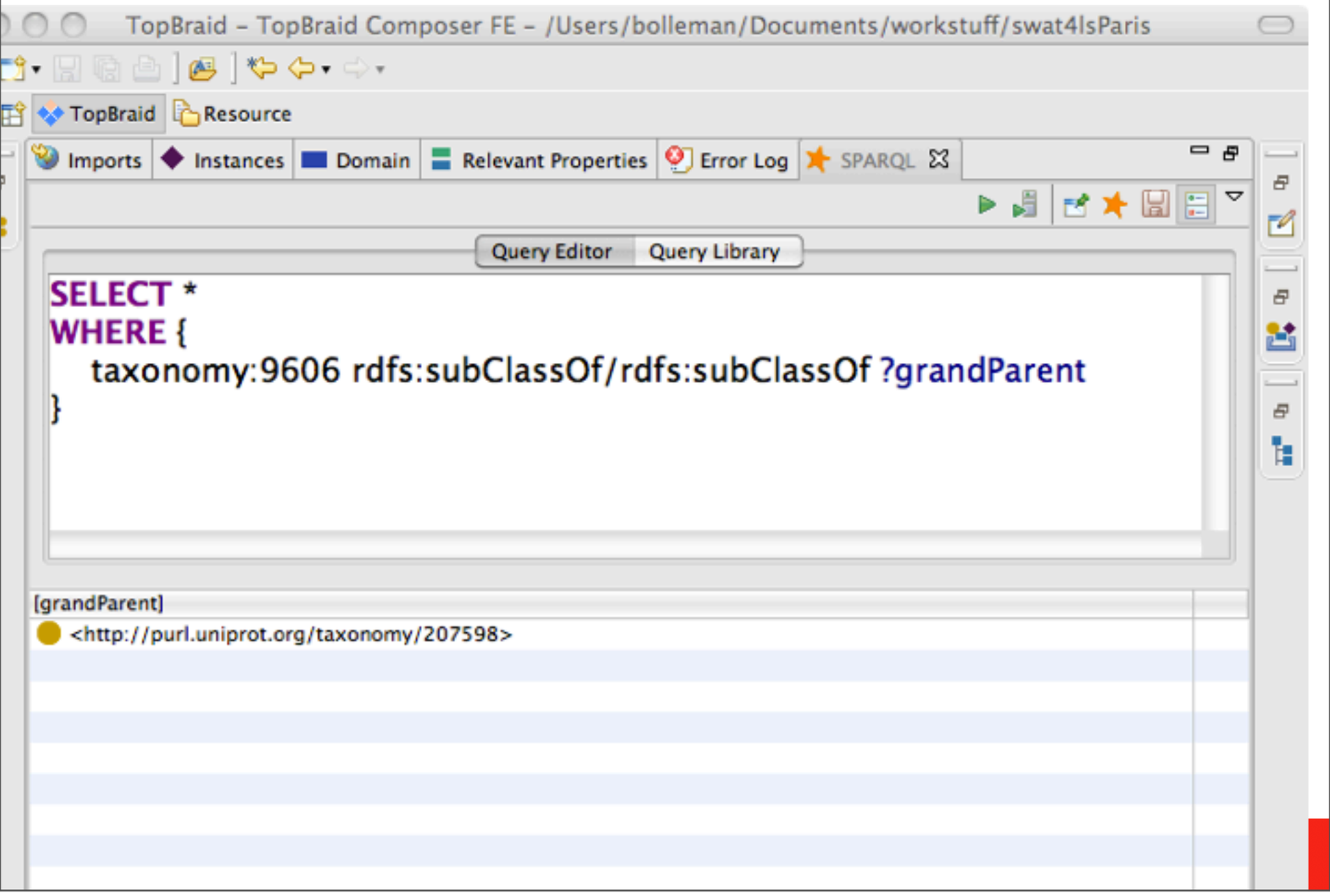

```
SELECT *
WHERE {
  taxonomy:9606 rdfs:subClassOf ?taxon .
}
```

Finding a grand parent using normal joins



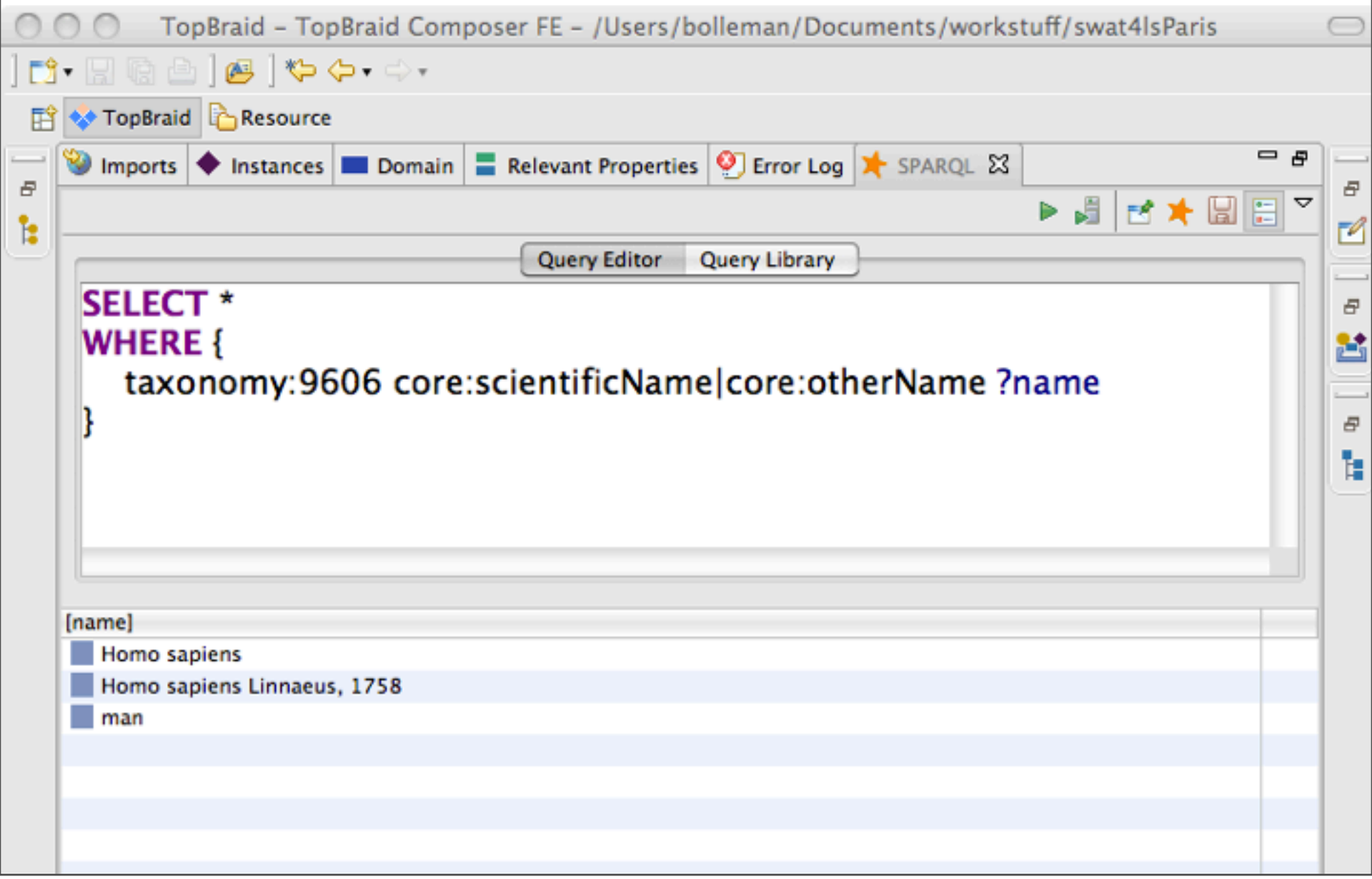

```
SELECT *
WHERE {
  taxonomy:9606 rdfs:subClassOf ?parent .
  ?parent rdfs:subClassOf ?grandParent
}
```

Finding a grandParent using a path query



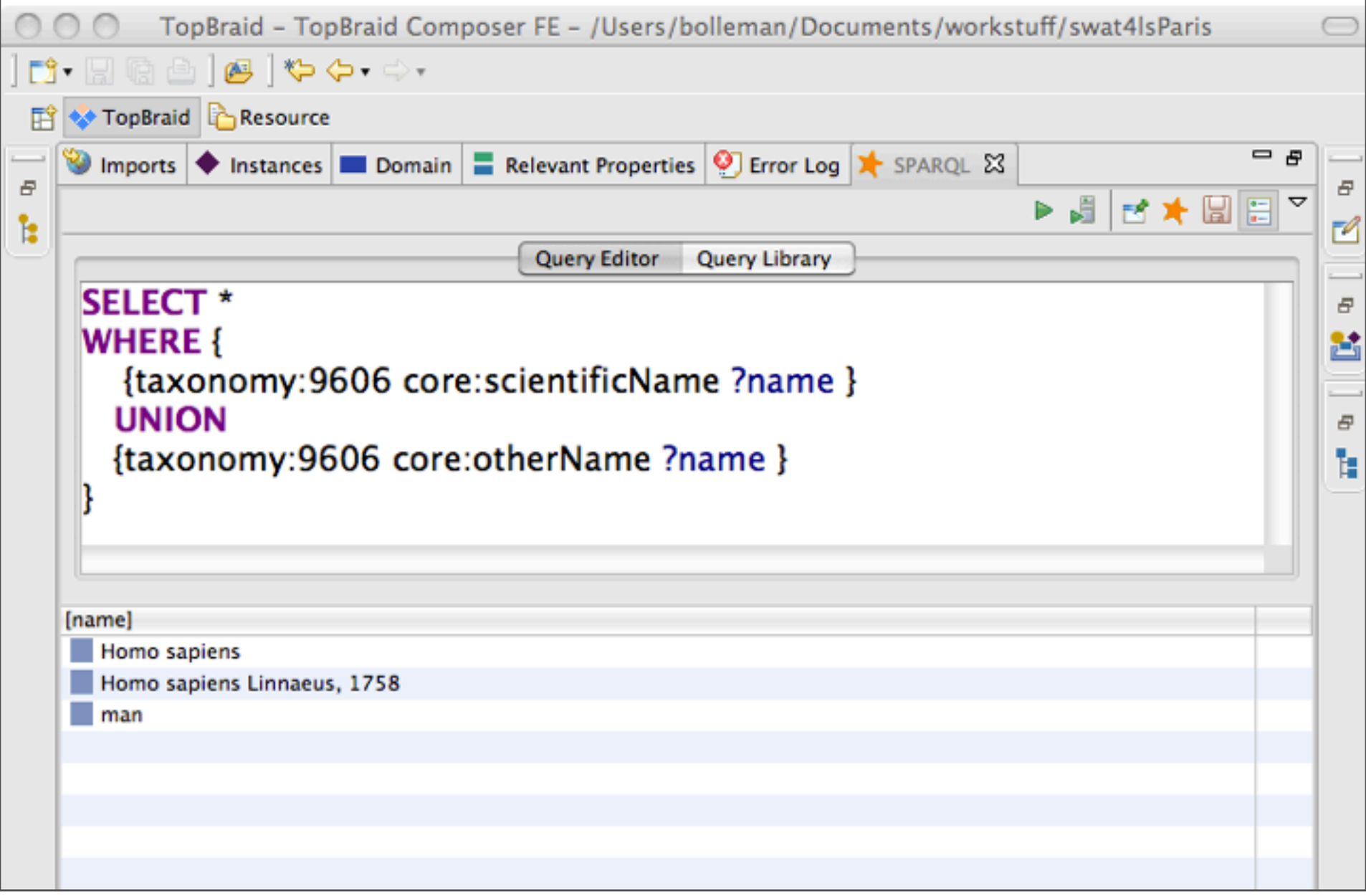

```
SELECT *
WHERE {
  taxonomy:9606 rdfs:subClassOf/rdfs:subClassOf ?grandParent
}
```

| is OR for predicate



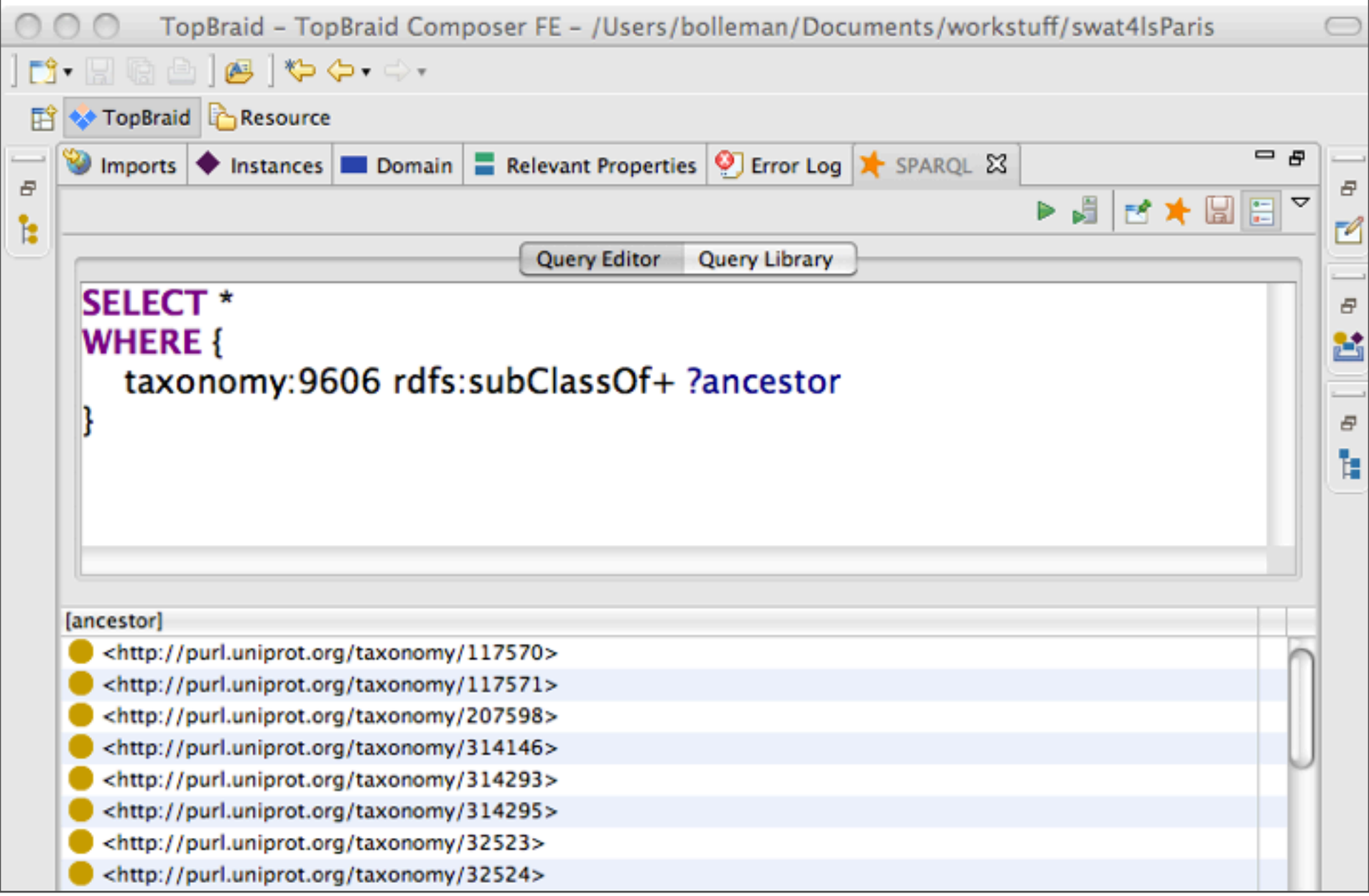

```
SELECT *
WHERE {
  taxonomy:9606 core:scientificName|core:otherName ?name
}
```

Same result with UNION




```
SELECT *
WHERE {
  {taxonomy:9606 core:scientificName ?name}
  UNION
  {taxonomy:9605 core:scientificName ?name}
}
```

Finding any ancestor

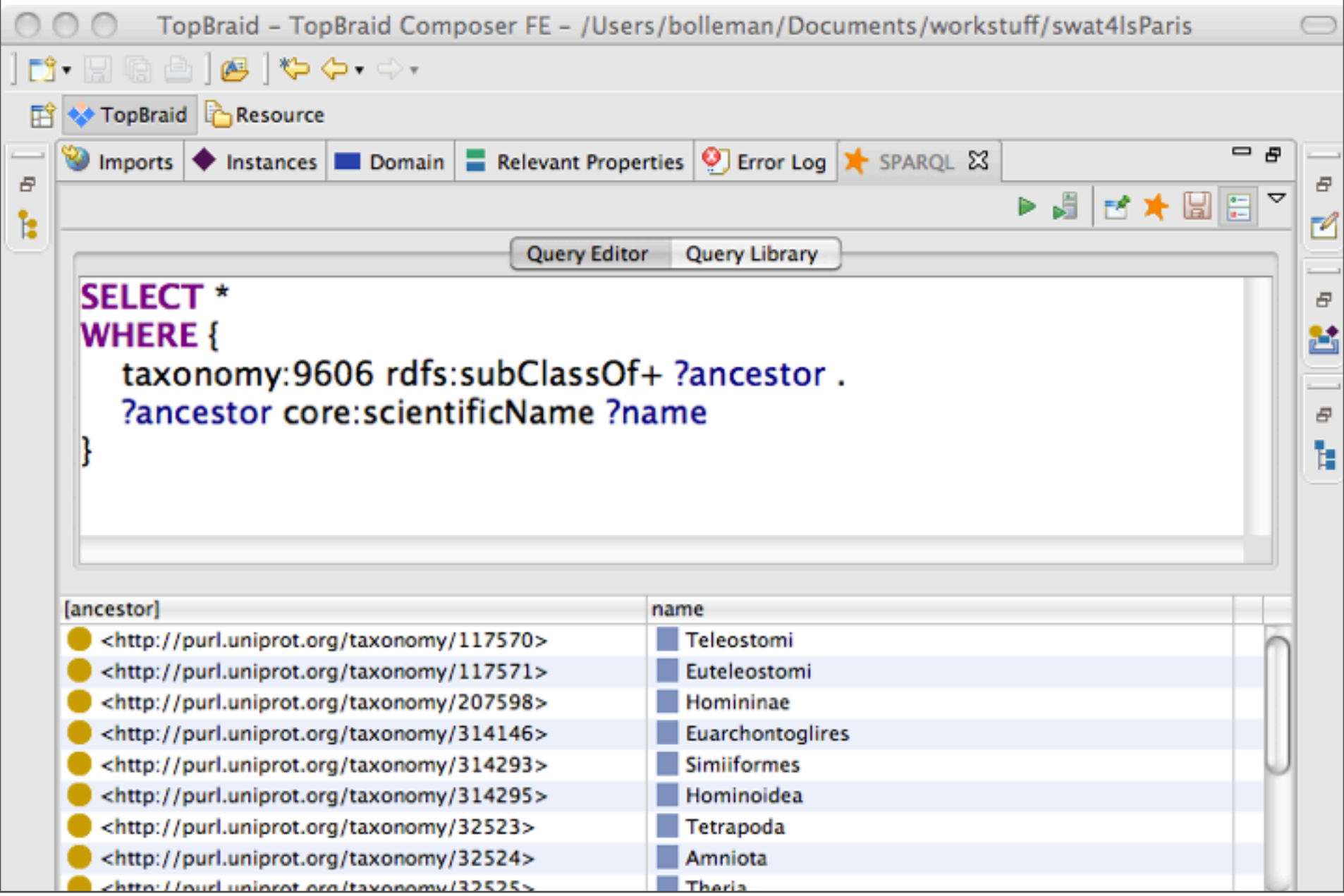


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+ operator means one or more of this predicate

```
SELECT *
WHERE {
  taxonomy:9606 rdfs:subClassOf+ ?ancestor
}
```

Can use the variable in a normal join afterwards



GROUP BY



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GROUP BY

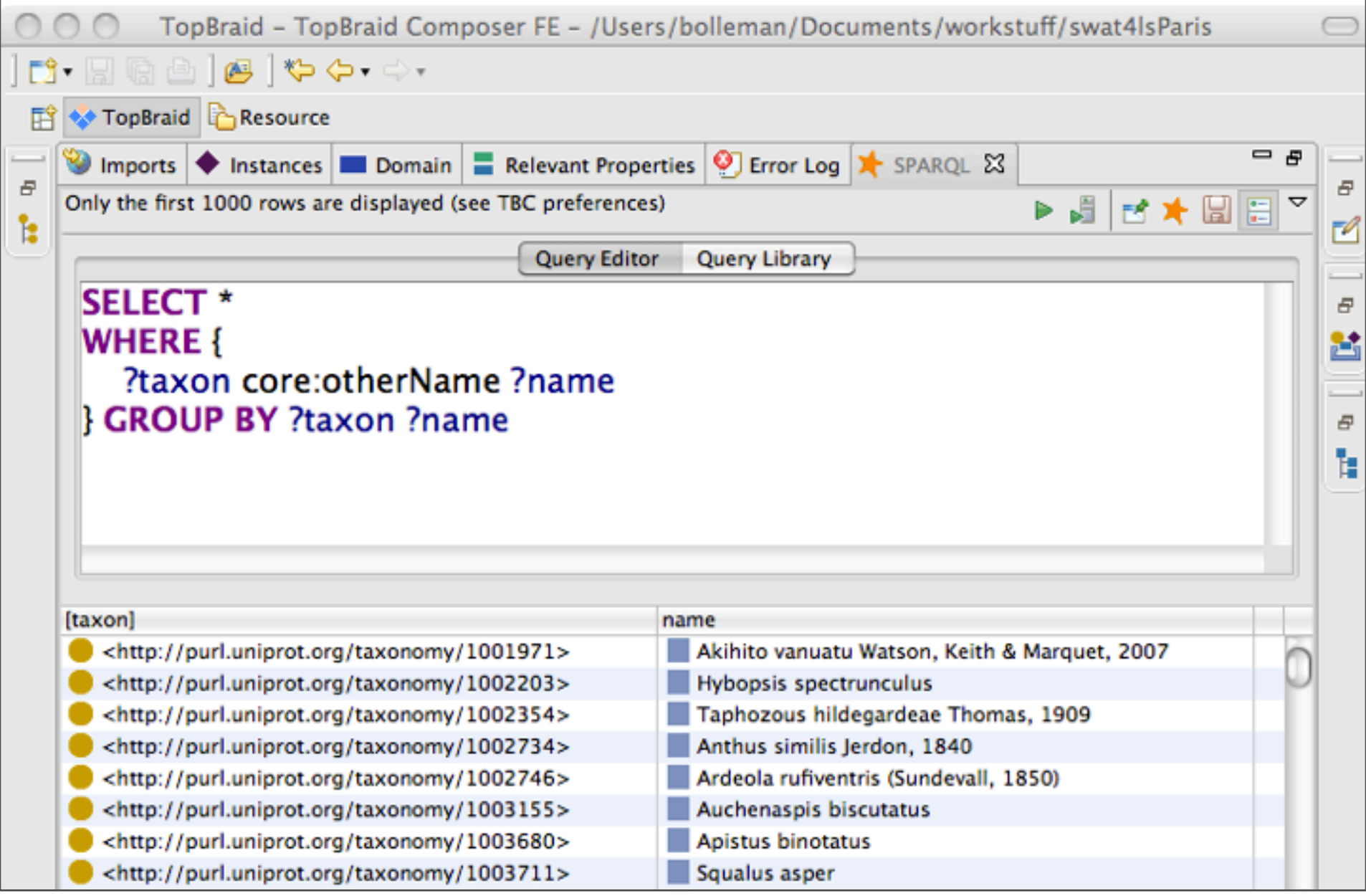
- Needed for aggregate values
- After closing the where clause
  - ... WHERE {?x ?y ?z} GROUP BY ?x

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```
SELECT ?annotation min(?begin) max(?begin)
WHERE {
   ?protein core:annotation ?annotation .
   ?annotation a core:Disulfide_Bond_Annotation .
   ?annotation core:range ?range .
    ?range core:begin ?begin .
} GROUP BY ?annotation
```

GROUP BY



Some taxon nodes have more than one name.

HAVING



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HAVING

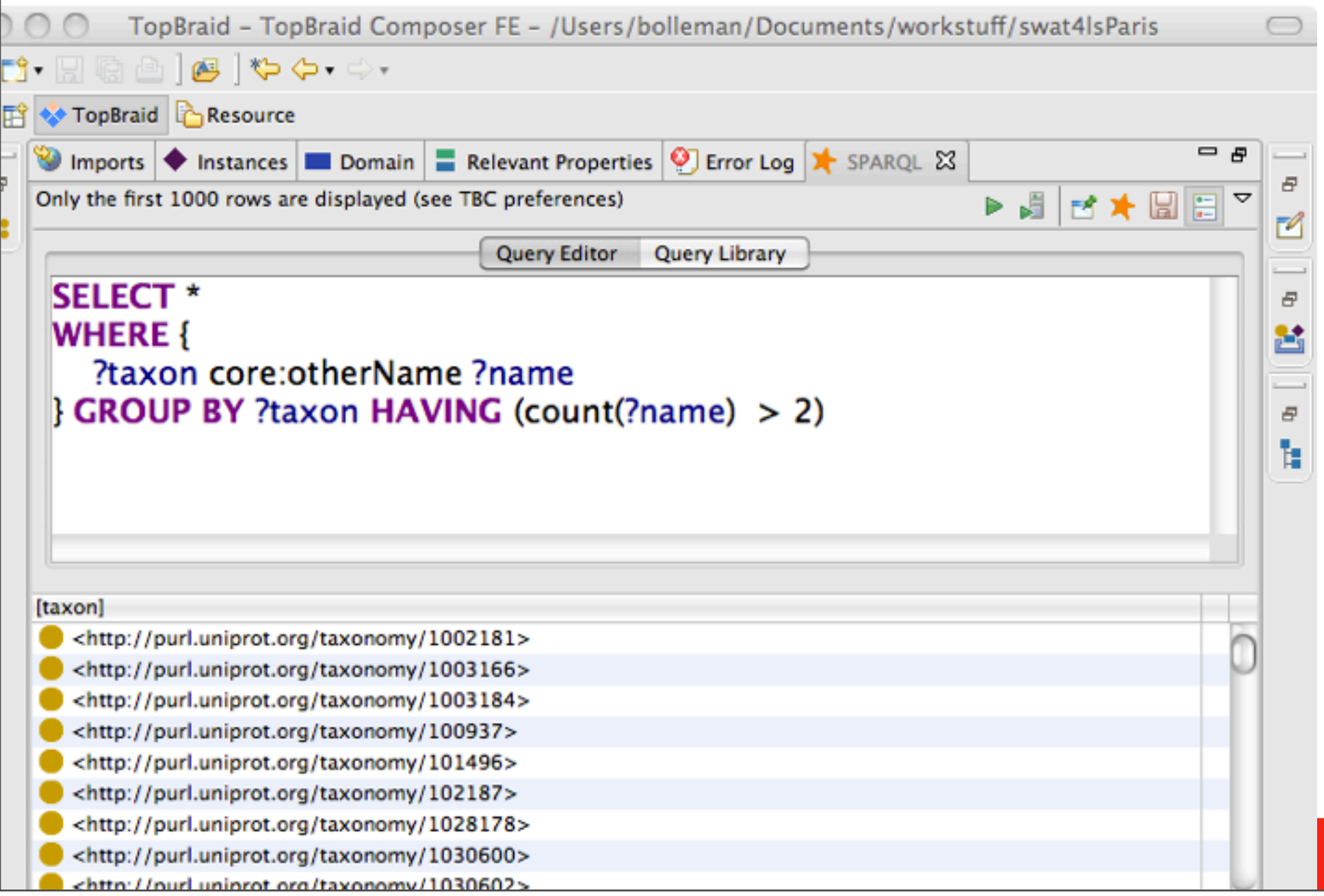
- FILTER for aggregates
- After the GROUP BY clause
  - ... GROUP BY ?x HAVING (count(?y) > 2)
  - ... GROUP BY ?x HAVING (min(?y) = 2)
  - etc...

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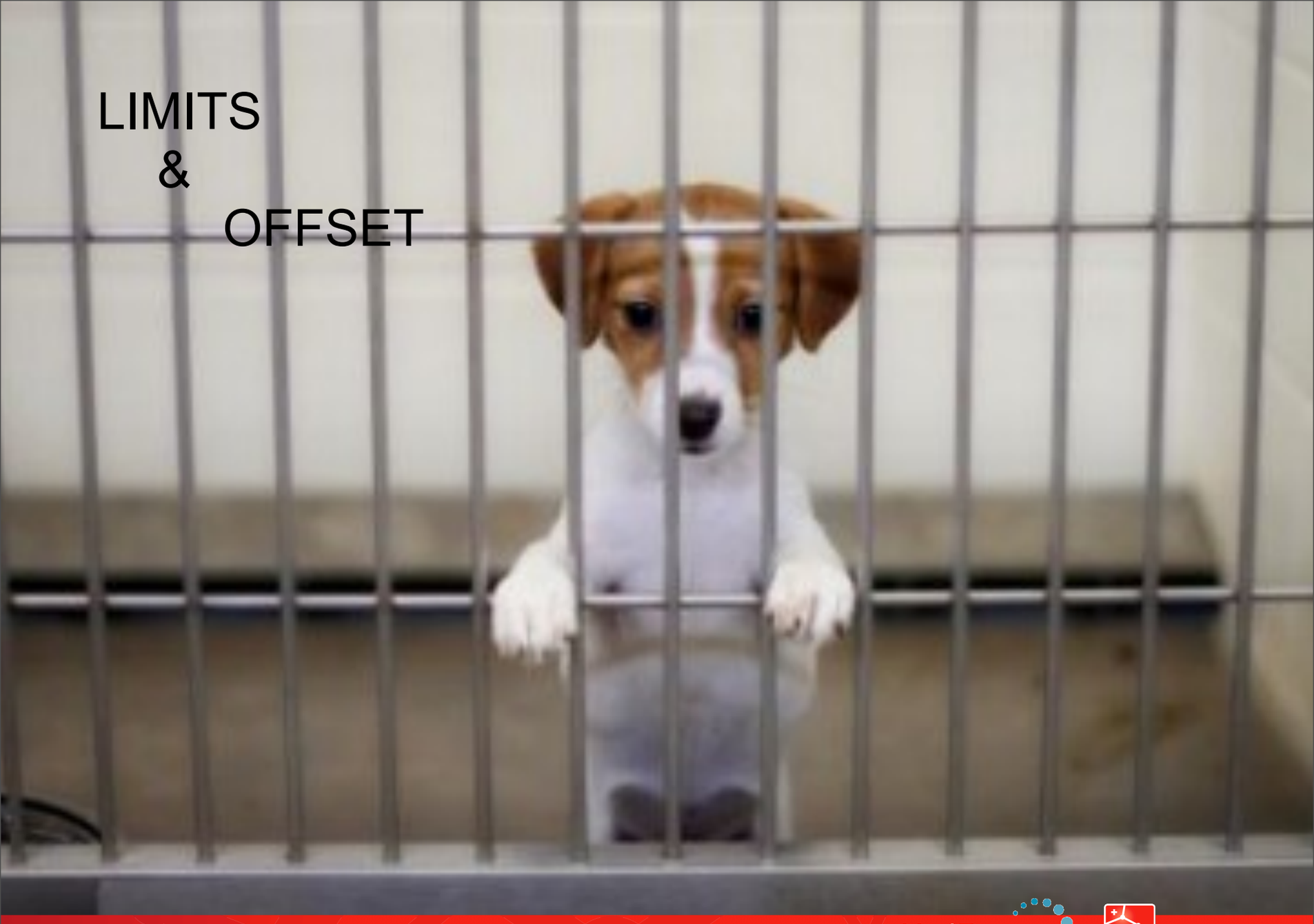
```
SELECT ?annotation min(?begin) max(?begin)
WHERE {
   ?protein core:annotation ?annotation .
   ?annotation core:range ?range .
    ?range core:begin ?begin .
} GROUP BY ?annotation HAVING (count(?begin) > 1)
```

HAVING




```
SELECT *
WHERE {
    ?taxon core:otherName ?name
} GROUP BY ?taxon HAVING (count(?name)  > 2)
```

LIMITS & OFFSET



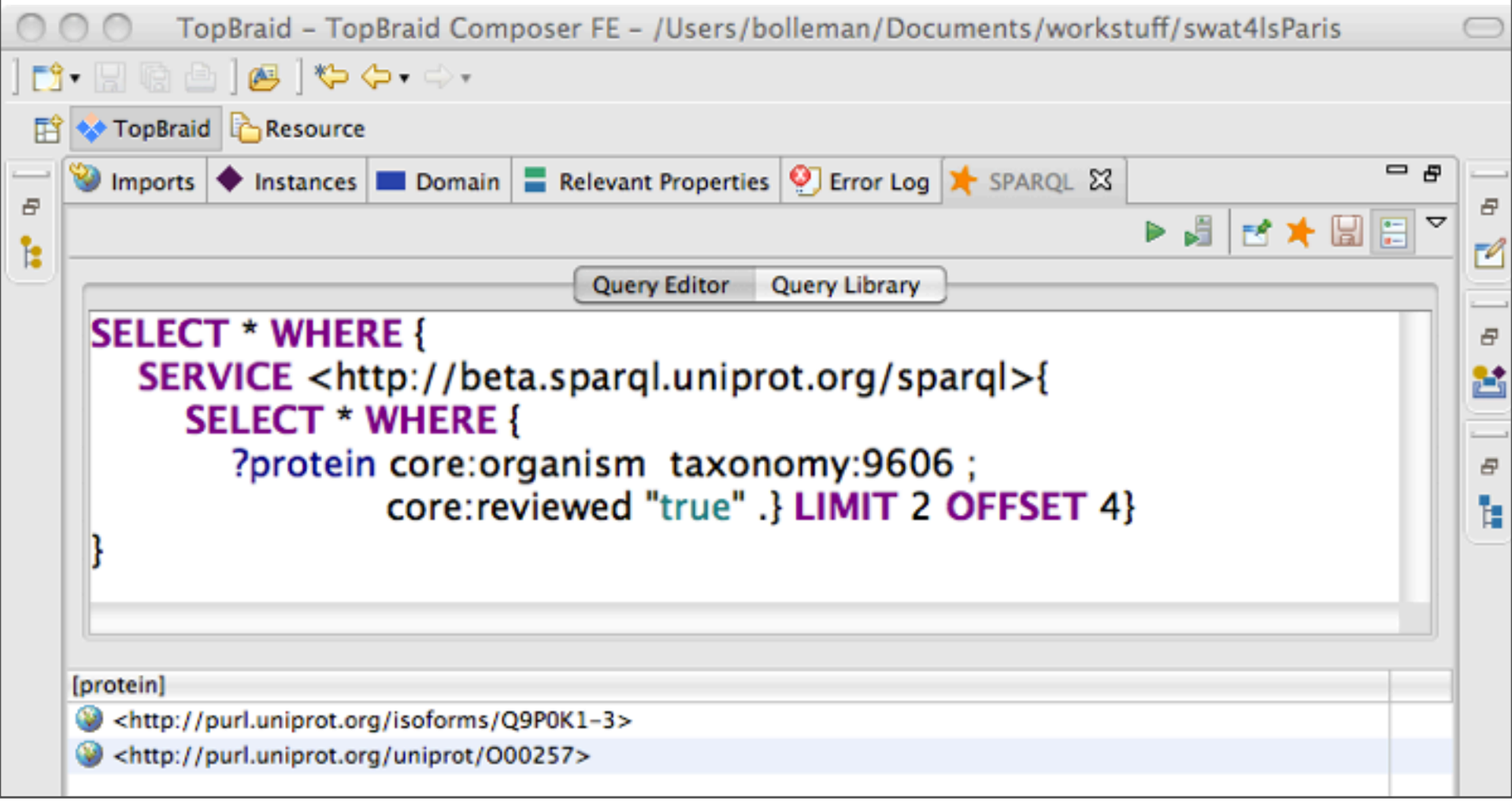
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LIMIT and OFFSET

- OFFSET is skip first results
- LIMIT return no more than x results

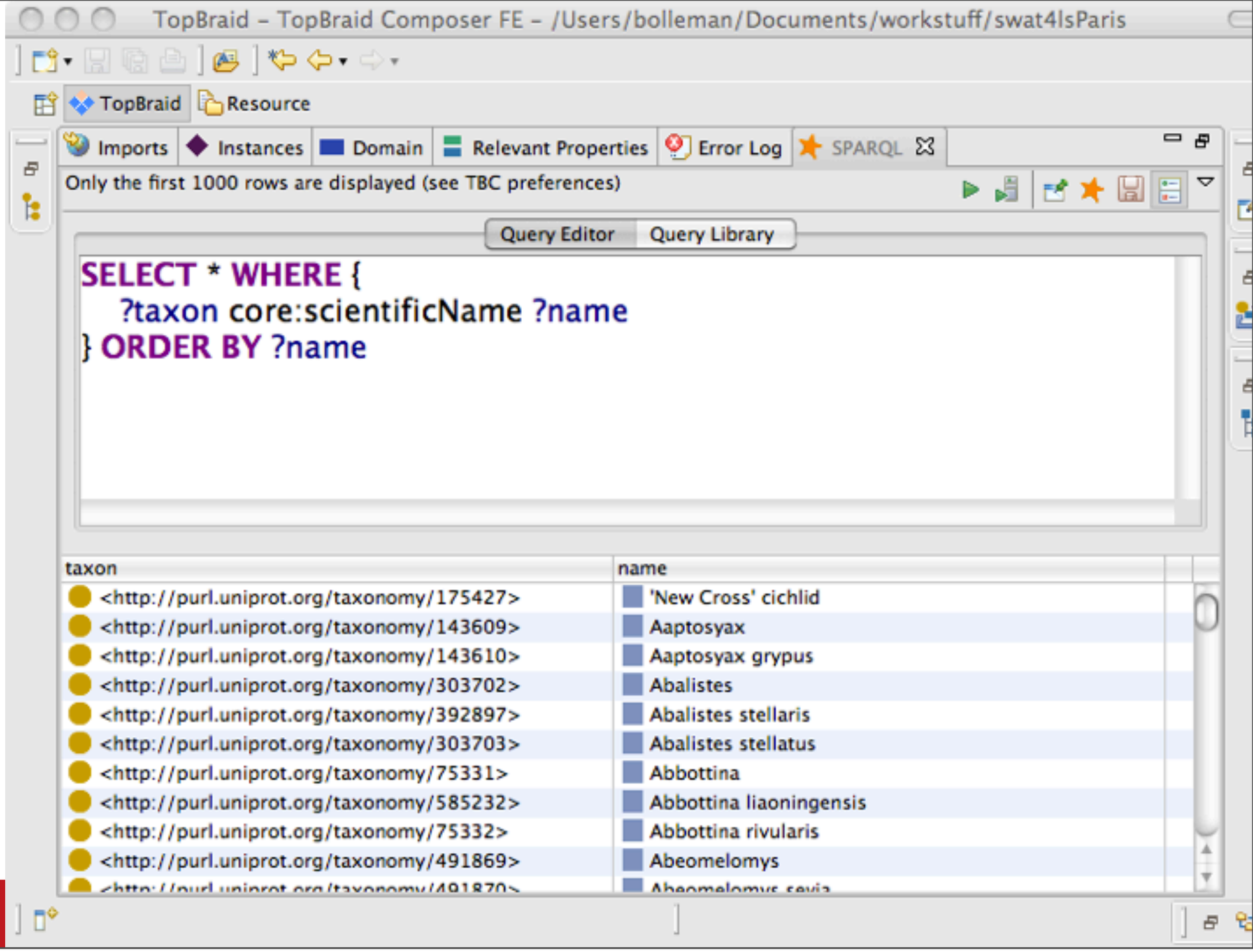


Useful for paging
Per standard only works as you expect when order by is present. In practice works as expected.

ORDER



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```
SELECT * WHERE {
    ?taxon core:scientificName ?name
} ORDER BY ?name
```

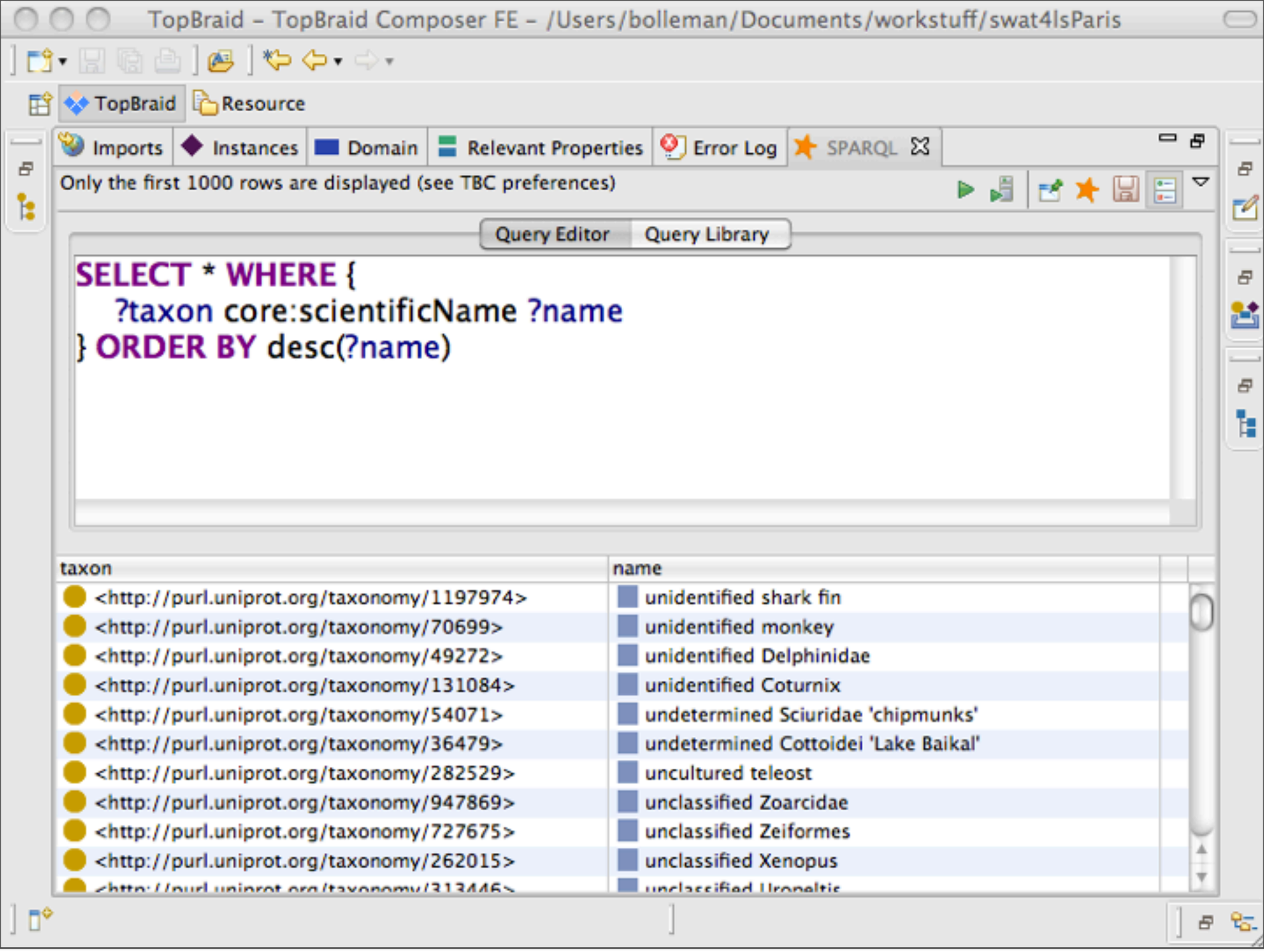


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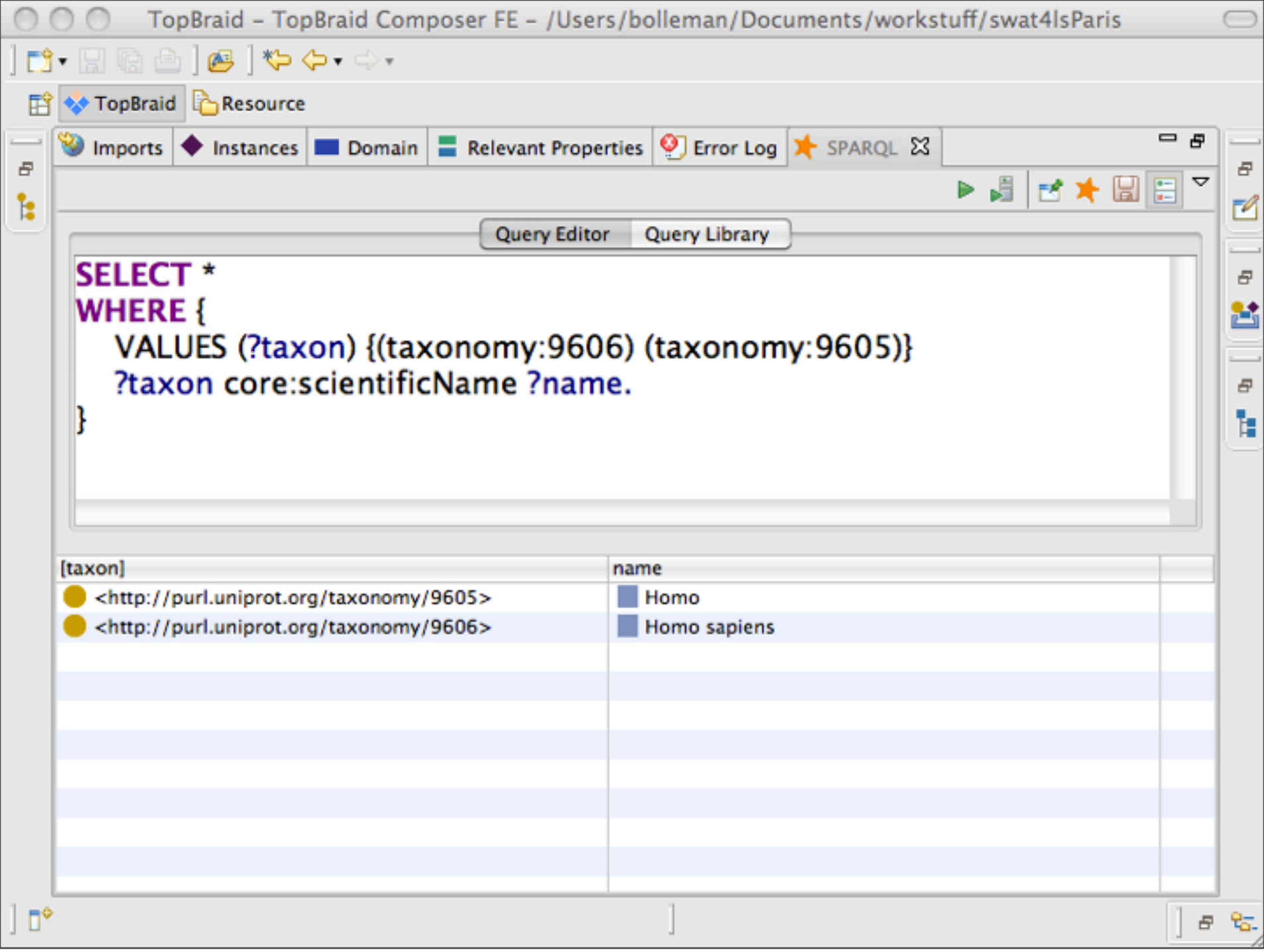
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```
SELECT * WHERE {
    ?taxon core:scientificName ?name
} ORDER BY desc(?name)
```

VALUES

- Super BIND
- Provide inline data

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TopBraid – TopBraid Composer FE – /Users/bolleman/Documents/workstuff/swat4lsParis
TopBraid
Resource
Imports
Instances
Domain
Relevant Properties
Error Log
SPARQL
Query Editor
Query Library
SELECT *
WHERE {
VALUES (?taxon) {(taxonomy:9606) (taxonomy:9605)}
?taxon core:scientificName ?name.
}
[taxon]
name
<http://purl.uniprot.org/taxonomy/9605>
Homo
<http://purl.uniprot.org/taxonomy/9606>
Homo sapiens

Examples

- Parameter lists are between ()

```
VALUES (?annotation) {
  (core:Disease_Annotation)
  (core:Disulfide_Bond_Annotation)
}
```

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Examples

- Undef means no value at
  - all not bound

```
VALUES (?annotation ?begin) {
  (core:Disease_Annotation UNDEF)
  (core:Disulfide_Bond_Annotation 2)
}
```

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?begin maps to UNDEF for disease and 2 for the disulfide bond.

VALUES

- After declaring a set of values you can use them in your query.

```
SELECT ?comment  WHERE {
  VALUES (?annotation ?begin) {
    (core:Disease_Annotation UNDEF)
    (core:Disulfide_Bond_Annotation 2)
  }
  ?annotation rdfs:comment ?comment .
}
```

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SERVICE: Using other sparql endpoints

- SERVICE<URL of other endpoint>
  - Runs a sub query on the other endpoint and merges it back into your query.

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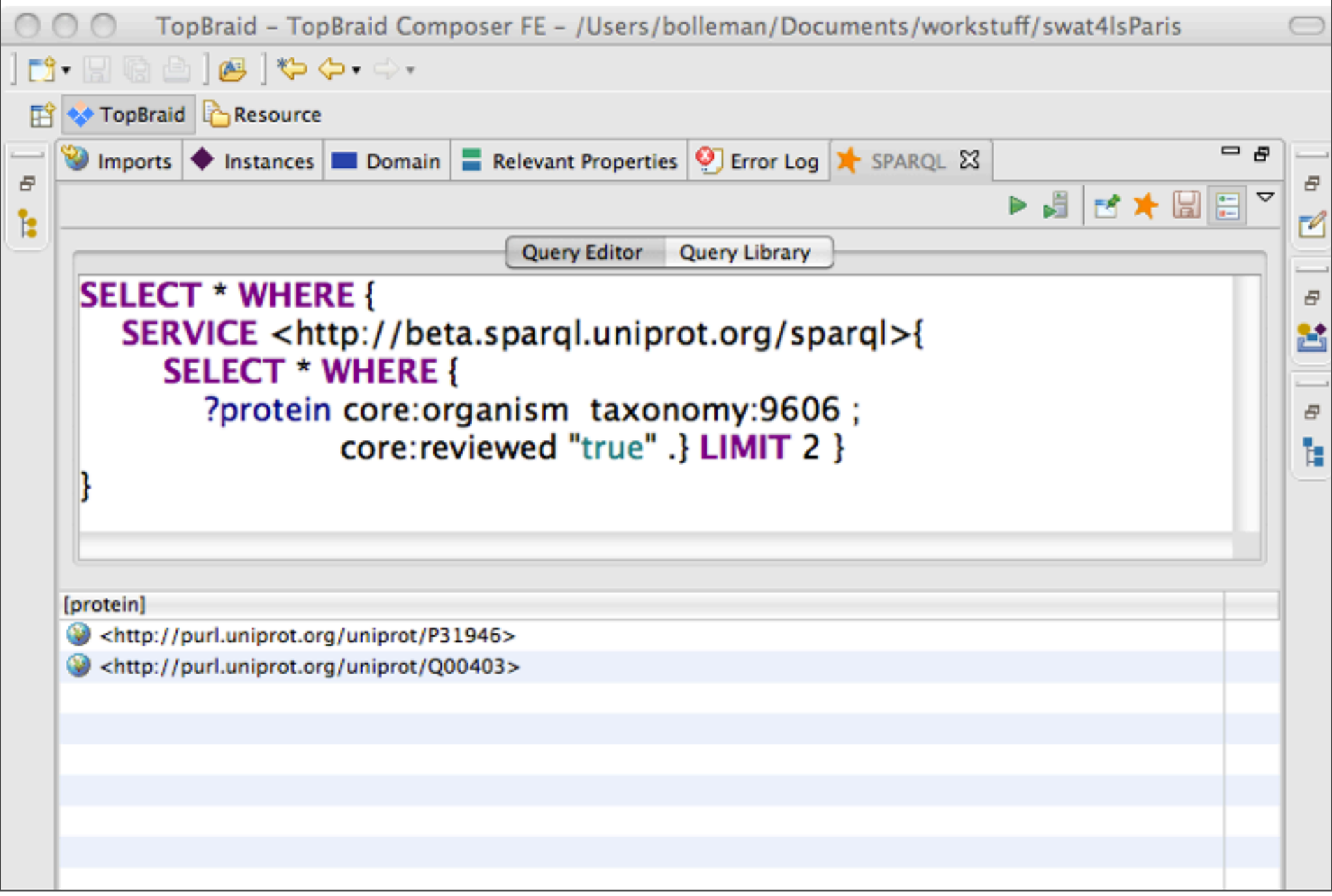


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SERVICE




```
SELECT * WHERE {
    SERVICE <http://beta.sparql.uniprot.org/sparql>{
        SELECT * WHERE {
            ?protein core:organism  taxonomy:9606 ;
                        core:reviewed "true" .} LIMIT 2 }
}
```

SERVICE

- Useful
  - Quick experimenting with combing multiple datasources
  - Quick for queries where not to much data is send to the remote point
- Slow
  - When you ask for to much data
  - Remote endpoint not resourced for your questions

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Want to select one book to read from the library but first move all books into your own home and backagain.



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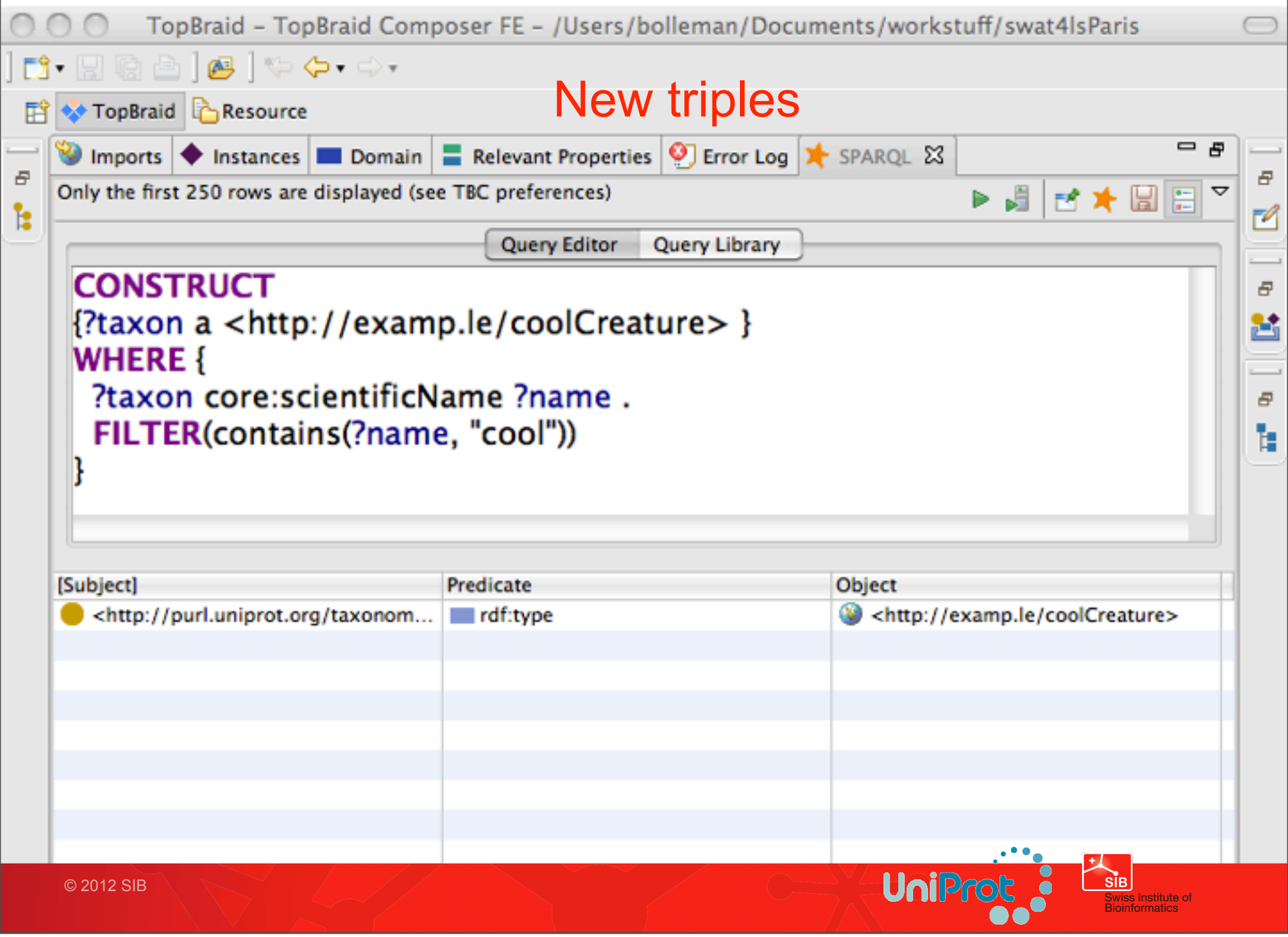
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Construction

- CONSTRUCT
  - New triples
    - downloads RDF
  - Does not update store

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New triples

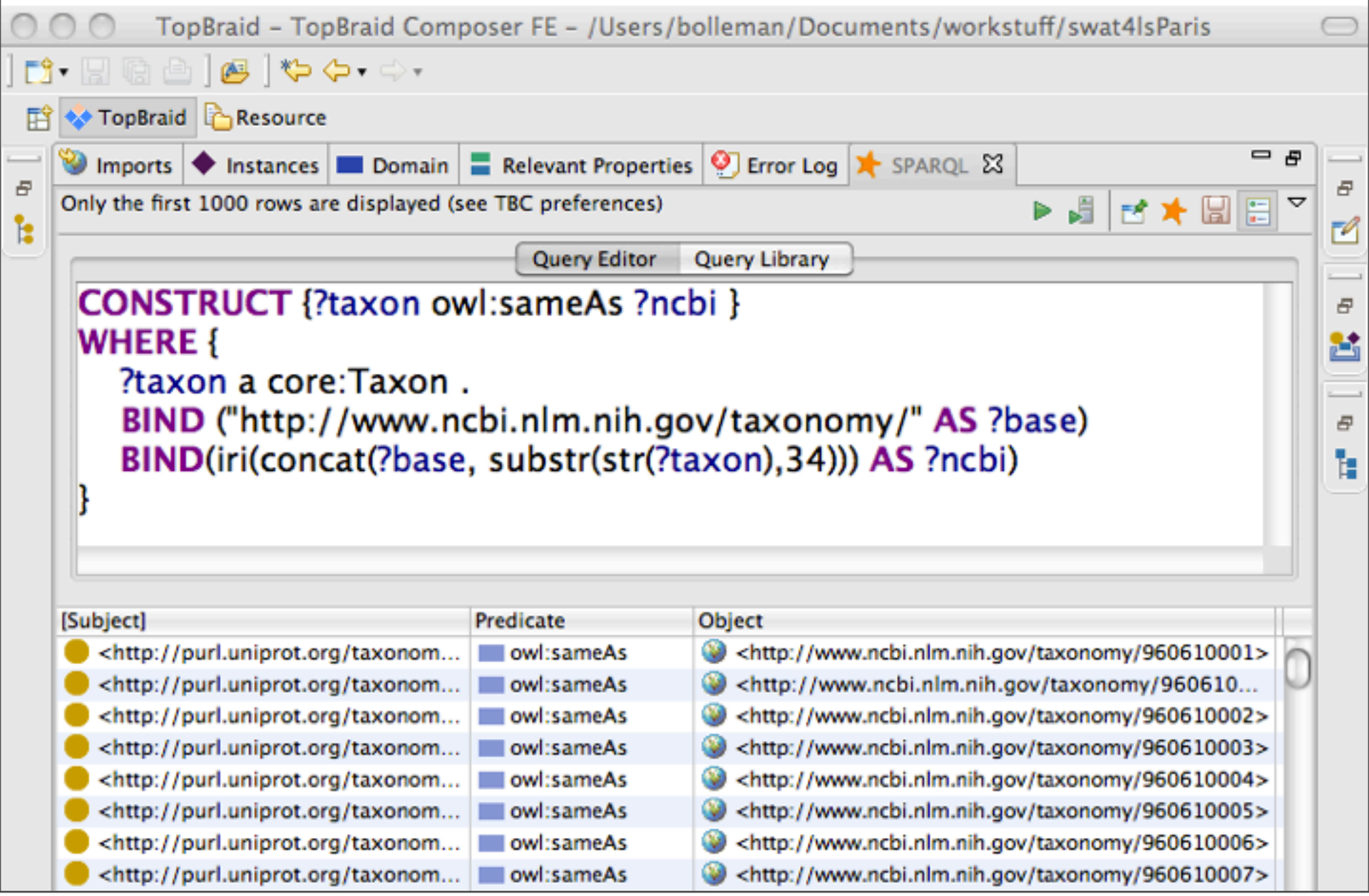


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Constructing an owl:sameAs between two URI



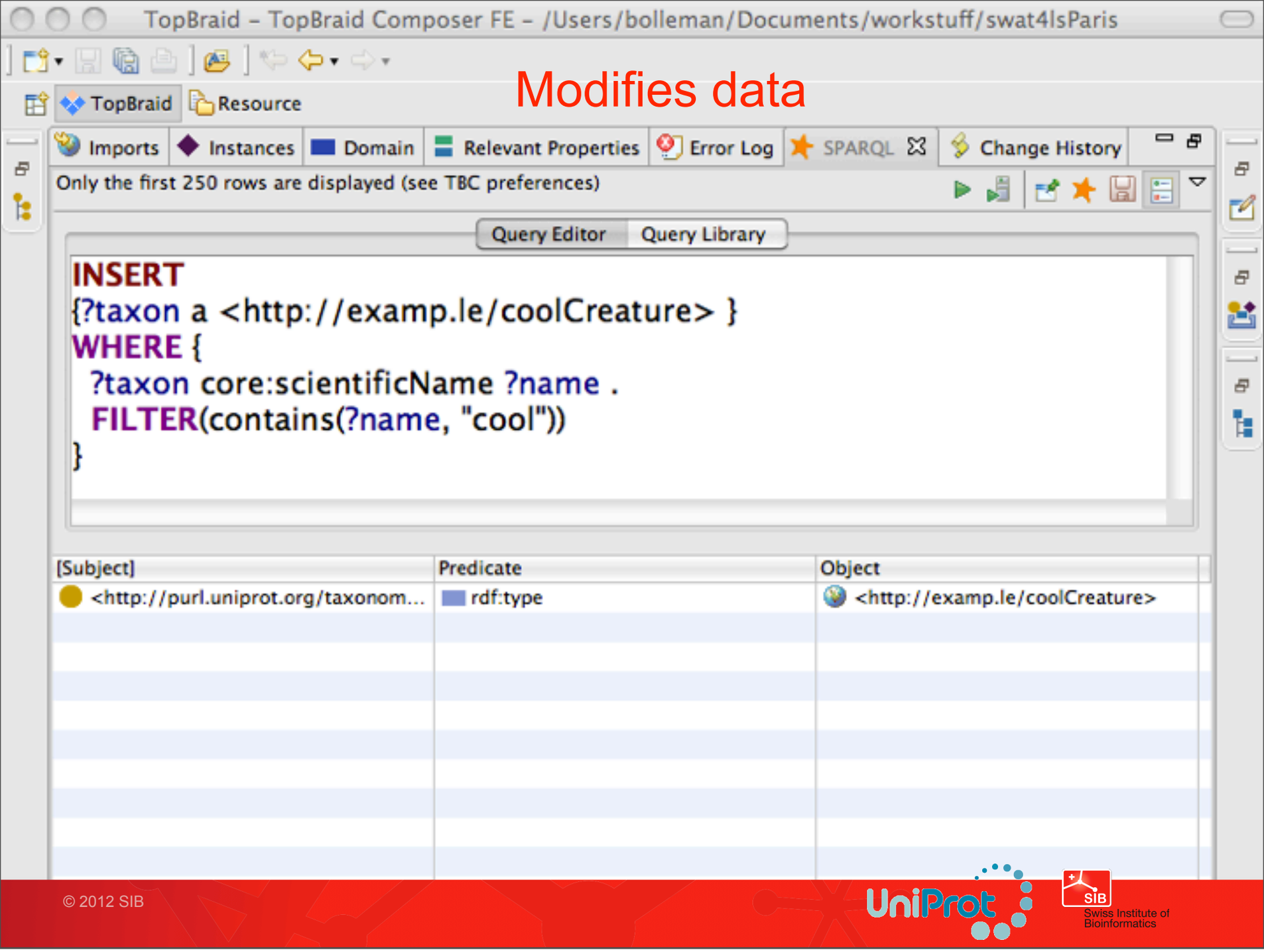
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We use two lines otherwise it does not fit on the beamer screen.

INSERT

- Adds data
  - like construct

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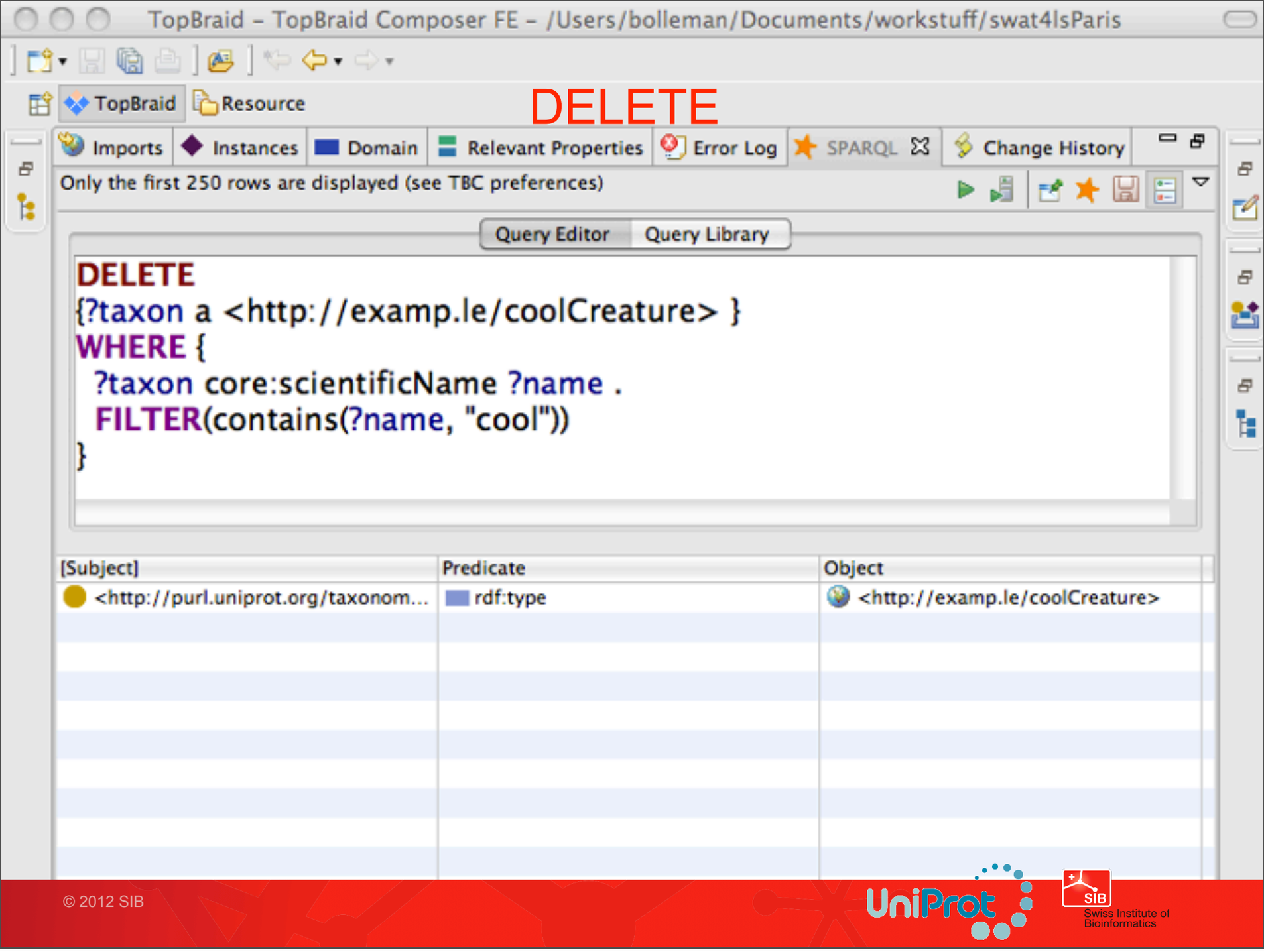



```
INSERT
{?taxon a     <http://examp.le/coolCreature> }
WHERE {
  ?taxon core:scientificName ?name .
  FILTER(contains(?name, "cool"))
}
```

DELETE

- Removes data
  - Triples matching are removed from the data
  - Triples can be bound using where clause

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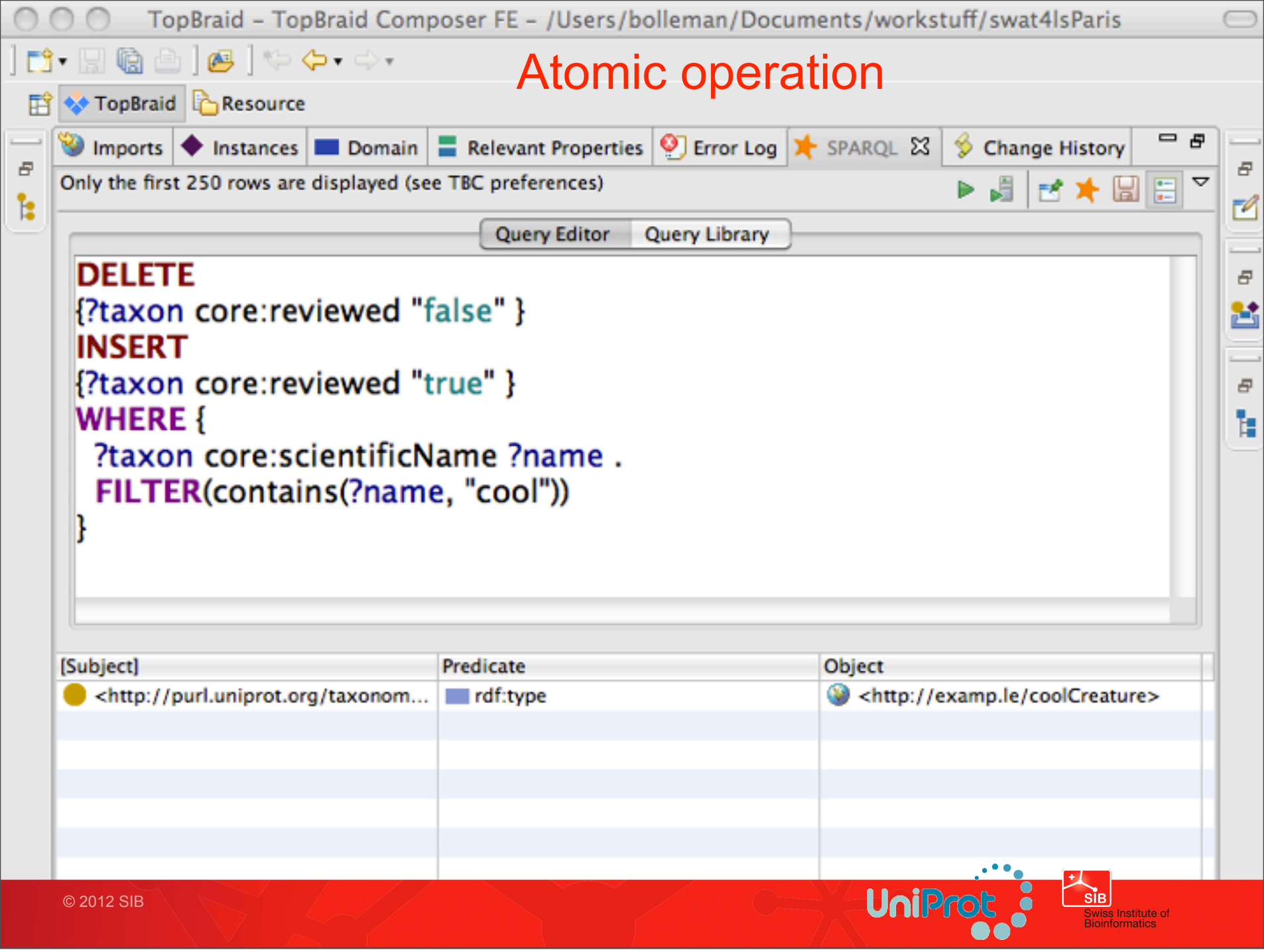
```
DELETE
{?taxon a     <http://examp.le/coolCreature> }
WHERE {
  ?taxon core:scientificName ?name .
  FILTER(contains(?name, "cool"))
}
```

DELETE
INSERT

- Single atomic operation.

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Atomic operation




```
DELETE
{?taxon core:reviewed "false" }
INSERT
{?taxon core:reviewed "true" }
WHERE {
  ?taxon core:scientificName ?name .
  FILTER(contains(?name, "cool"))
}
```



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Questions





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